Chapter 9

Transportation Problem

"We want these assets to be productive. We buy them. We own them. To say we care only about the short term is wrong. What I care about is seeing these assets in the best hands."

– Carl Icahn

PREVIEW

The structure of transportation problem involves a large number of shipping routes from several supply centres to several demand centres. The objective is to determine the number of units of an item (commodity or product) that should be shipped from an origin to a destination in order to satisfy the required quantity of goods or services at each destination centre.

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- recognize and formulate a transportation problem involving a large number of shipping routes.
- drive initial feasible solution using several methods.
- drive optimal solution by using Modified Distribution Method.
- handle the problem of degenerate and unbalanced transportation problem.
- examine multiple optimal solutions, and prohibited routes in the transportation problem.
- construct the initial transportation table for a trans-shipment problem.
- solve a profit maximization transportation problem using suitable changes in the transportation algorithm.

CHAPTER OUTLINE

9.1 Introduction
9.2 Mathematical Model of Transportation Problem
9.3 The Transportation Algorithm
9.4 Methods of Finding Initial Solution
- Conceptual Questions A
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9.1 INTRODUCTION

One important application of linear programming is in the area of physical distribution (transportation) of goods and services from several supply centres to several demand centres. A transportation problem when expressed in terms of an LP model can also be solved by the simplex method. However a transportation problem involves a large number of variables and constraints, solving it using simplex methods takes a long time. Two transportation algorithms, namely *Stepping Stone Method* and the *MODI* (modified distribution) *Method* have been developed for solving a transportation problem.

The structure of transportation problem involves a large number of shipping routes from several supply centres to several demand centres. Thus, objective is to determine shipping routes between supply centres and demand centres in order to satisfy the required quantity of goods or services at each destination centre, with available quantity of goods or services at each supply centre at the minimum transportation cost and/or time.

The transportation algorithms help to minimize the total cost of transporting a homogeneous commodity (product) from supply centres to demand centres. However, it can also be applied to the maximization of total value or utility.

There are various types of transportation models and the simplest of them was first presented by F L Hitchcock (1941). It was further developed by T C Koopmans (1949) and G B Dantzig (1951). Several extensions of transportation models and methods have been subsequently developed.

The study of **transportation problem** helps to identify optimal transportation routes along with units of commodity to be shipped in order to minimize total transportation cost.

9.2 MATHEMATICAL MODEL OF TRANSPORTATION PROBLEM

Let us consider Example 9.1 to illustrate the mathematical model formulation of transportation problem of transporting a single commodity from three sources of supply to four demand destinations. The *sources of supply* are production facilities, warehouses, or supply centres, each having certain amount of commodity to supply. The *destinations* are consumption facilities, warehouses or demand centres each having certain amount of requirement (or demand) of the commodity.

Example 9.1 A company has three production facilities S_1, S_2 and S_3 with production capacity of 7, 9 and 18 units (in 100s) per week of a product, respectively. These units are to be shipped to four warehouses D_1, D_2, D_3 and D_4 with requirement of 5, 6, 7 and 14 units (in 100s) per week, respectively. The transportation costs (in rupees) per unit between factories to warehouses are given in the table below:

	D_1	D_2	D_3	D_4	*Supply (Availability)*
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand (Requirement)	5	8	7	14	34

Formulate this transportation problem as an LP model to minimize the total transportation cost.

Model formulation Let x_{ij} = number of units of the product to be transported from a production facility i (i = 1, 2, 3) to a warehouse j (j = 1, 2, 3, 4)

The transportation problem is stated as an LP model as follows:

$$\text{Minimize (total transportation cost) } Z = 19x_{11} + 30x_{12} + 50x_{13} + 10x_{14} + 70x_{21} + 30x_{22} + 40x_{23} + 60x_{24} + 40x_{31} + 8x_{32} + 70x_{33} + 20x_{34}$$

subject to the constraints

$$\left.\begin{aligned} x_{11} + x_{12} + x_{13} + x_{14} &= 7 \\ x_{21} + x_{22} + x_{23} + x_{24} &= 9 \\ x_{31} + x_{32} + x_{33} + x_{34} &= 18 \end{aligned}\right\} \text{(Supply)}$$

$$\left.\begin{aligned} x_{11} + x_{21} + x_{31} &= 5 \\ x_{12} + x_{22} + x_{32} &= 8 \\ x_{13} + x_{23} + x_{33} &= 7 \\ x_{14} + x_{24} + x_{34} &= 14 \end{aligned}\right\} \text{(Demand)}$$

and $x_{ij} \geq 0$ for $i = 1, 2, 3$ and $j = 1, 2, 3$, and 4.

In the above LP model, there are $m \times n = 3 \times 4 = 12$ decision variables, x_{ij} and $m + n = 7$ constraints, where m are the number of rows and n are the number of columns in a general transportation table.

9.2.1 General Mathematical Model of Transportation Problem

Let there be m sources of supply, $S_1, S_2, \ldots, S_m$ having a_i ($i = 1, 2, \ldots, m$) units of supply (or capacity), respectively to be transported to n destinations, $D_1, D_2, \ldots, D_n$ with b_j ($j = 1, 2, \ldots, n$) units of demand (or requirement), respectively. Let c_{ij} be the cost of shipping one unit of the commodity from source i to destination j. If x_{ij} represents number of units shipped from source i to destination j, the problem is to determine the transportation schedule so as to minimize the total transportation cost while satisfying the supply and demand conditions. Mathematically, the transportation problem, in general, may be stated as follows:

$$\text{Minimize (total cost) } Z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij}\, x_{ij} \tag{1}$$

subject to the constraints

$$\sum_{j=1}^{n} x_{ij} = a_i, \; i = 1, 2, \ldots, m \text{ (supply constraints)} \tag{2}$$

$$\sum_{i=1}^{m} x_{ij} = b_j, \; j = 1, 2, \ldots, n \text{ (demand constraints)} \tag{3}$$

and $$x_{ij} \geq 0 \text{ for all } i \text{ and } j. \tag{4}$$

For easy presentation and solution, a transportation problem data is generally presented as shown in Table 9.1.

Transportation table is a convenient way to summarize data.

Existence of feasible solution A necessary and sufficient condition for a feasible solution to the transportation problems is:

$$\text{Total supply} = \text{Total demand}$$

$$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j \text{ (also called } \textit{rim conditions}\text{)}$$

For proof, see Appendix at the end of this chapter.

To / From	D_1	D_2	...	D_n	*Supply* a_i
S_1	c_{11} (x_{11})	c_{12} (x_{12})	...	c_{1n} (x_{1n})	a_1
S_2	c_{21} (x_{21})	c_{22} (x_{22})	...	c_{2n} (x_{2n})	a_2
⋮	⋮	⋮		⋮	⋮
S_m	c_{m1} (x_{m1})	c_{m2} (x_{m2})	...	c_{mn} (x_{mn})	a_m
Demand b_j	b_1	b_2	...	b_n	$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$

Table 9.1 General Transportation Table.

In this problem, there are $(m + n)$ constraints, one for each source of supply, and distinction and $m \times n$ variables. Since all $(m + n)$ constraints are equations, therefore, one of these equations is extra (redundant). The extra constraint (equation) can be derived from the other constraints (equations), without affecting the feasible solution. It follows that any feasible solution for a transportation problem must have exactly $(m + n - 1)$ non-negative basic variables (or allocations) x_{ij} satisfying the rim conditions.

Remarks 1. When the total supply is equal to the total demand, the problem is called a *balanced transportation problem,* otherwise it is called an *unbalanced transportation problem*. The unbalanced transportation problem can be made balanced by adding a dummy supply centre (row) or a dummy demand centre (column) as the need arises.

2. When the number of positive allocations (values of decision variables) at any stage of the feasible solution is less than the required number (rows + columns – 1), i.e. number of independent constraint equations, the solution is said to be *degenerate,* otherwise *non-degenerate*. For proof, see Appendix at the end of this chapter.

When total demand equals total supply, the transportation problem is said to be balanced.

3. Cells in the transportation table having positive allocation, i.e., $x_{ij} > 0$ are called *occupied cells,* otherwise are known as *non-occupied (or empty) cells.*

9.3 THE TRANSPORTATION ALGORITHM

The algorithm for solving a transportation problem may be summarized into the following steps:

Step 1: Formulate the problem and arrange the data in the matrix form The formulation of the transportation problem is similar to the LP problem formulation. In transportation problem, the objective function is the total transportation cost and the constraints are the amount of supply and demand available at each source and destination, respectively.

Step 2: Obtain an initial basic feasible solution In this chapter, following three different methods are discussed to obtain an initial solution:

- *North-West Corner Method,*
- *Least Cost Method,* and
- *Vogel's Approximation* (*or Penalty*) *Method.*

The initial solution obtained by any of the three methods must satisfy the following conditions:

(i) The solution must be feasible, i.e. it must satisfy all the supply and demand constraints (also called *rim conditions*).

(ii) The number of positive allocations must be equal to $m + n - 1$, where m is the number of rows and n is the number of columns.

Any solution that satisfies the above conditions is called *non-degenerate basic feasible solution,* otherwise, *degenerate solution.*

Step 3: Test the initial solution for optimality In this chapter, the Modified Distribution (MODI) method is discussed to test the optimality of the solution obtained in Step 2. If the current solution is optimal, then stop. Otherwise, determine a new improved solution.

Step 4: Updating the solution Repeat Step 3 until an optimal solution is reached.

9.4 METHODS OF FINDING INITIAL SOLUTION

There are several methods available to obtain an initial basic feasible solution. In this chapter, we shall discuss only following three methods:

9.4.1 North-West Corner Method (NWCM)

This method does not take into account the cost of transportation on any route of transportation. The method can be summarized as follows:

Step 1: Start with the cell at the upper left (north-west) corner of the transportation table (or matrix) and allocate commodity equal to the minimum of the rim values for the first row and first column, i.e. min (a_1, b_1).

Step 2: (a) If allocation made in Step 1 is equal to the supply available at first source (a_1, in first row), then move vertically down to the cell (2, 1), i.e., second row and first column. Apply Step 1 again, for next allocation.

(b) If allocation made in Step 1 is equal to the demand of the first destination (b_1 in first column), then move horizontally to the cell (1, 2), i.e., first row and second column. Apply Step 1 again for next allocation.

(c) If $a_1 = b_1$, allocate $x_{11} = a_1$ or b_1 and move diagonally to the cell (2, 2).

Step 3: Continue the procedure step by step till an allocation is made in the south-east corner cell of the transportation table.

Remark If during the process of making allocation at a particular cell, the supply equals demand, then the next allocation of magnitude zero can be made in a cell either in the next row or column. This condition is known as degeneracy.

Example 9.2 Use North-West Corner Method (NWCM) to find an initial basic feasible solution to the transportation problem using data of Example 9.1

Solution The cell (S_1, D_1) is the north-west corner cell in the given transportation table. The rim values for row S_1 and column D_1 are compared. The smaller of the two, i.e. 5, is assigned as the first allocation; otherwise it will violate the feasibility condition. This means that 5 units of a commodity are to be transported from source S_1 to destination D_1. However, this allocation leaves a supply of 7 – 5 = 2 units of commodity at S_1.

Move horizontally and allocate as much as possible to cell (S_1, D_2). The rim value for row S_1 is 2 and for column D_2 is 8. The smaller of the two, i.e. 2, is placed in the cell. Proceeding to row S_2, since the demand of D_1 is fullfilled. The unfulfilled demand of D_2 is now 8 – 2 = 6 units. This can be fulfilled by S_2 with capacity of 9 units. So 6 units are allocated to cell (S_2, D_2). The demand of D_2 is now satisfied and a balance of 9 – 6 = 3 units remains with S_2.

Table 9.2 Initial Solution using NWCM

	D_1	D_2	D_3	D_4	*Supply*
S_1	19 (5)	30 (2)	50	10	7
S_2	70	30 (6)	40 (3)	60	9
S_3	40	8	70 (4)	20 (14)	18
Demand	5	8	7	14	34

Continue to move horizontally and vertically in the same manner to make desired allocations. Once the procedure is over, count the number of positive allocations. These allocations (occupied cells) should be equal to $m + n - 1 = 3 + 4 - 1 = 6$. If yes, then solution is non-degenerate feasible solution. Otherwise degenerate solution.

The total transportation cost of the initial solution is obtained by multiplying the quantity x_{ij} in the occupied cells with the corresponding unit cost c_{ij} and adding all the values together. Thus, the total transportation cost of this solution is

Total cost = 5 × 19 + 2 × 30 + 6 × 30 + 3 × 40 + 4 × 70 + 14 × 20 = Rs 1,015

9.4.2 Least Cost Method (LCM)

Since the main objective is to minimize the total transportation cost, transport as much as possible through those routes (cells) where the unit transportation cost is lowest. This method takes into account the minimum unit cost of transportation for obtaining the initial solution and can be summarized as follows:

Step 1: Select the cell with the lowest unit cost in the entire transportation table and allocate as much as possible to this cell. Then eliminate (line out) that row or column in which either the supply or demand is fulfilled. If a row and a column are both satisfied simultaneously, then crossed off either a row or a column.

In case the smallest unit cost cell is not unique, then select the cell where the maximum allocation can be made.

Step 2: After adjusting the supply and demand for all uncrossed rows and columns repeat the procedure to select a cell with the next lowest unit cost among the remaining rows and columns of the transportation table and allocate as much as possible to this cell. Then crossed off that row and column in which either supply or demand is exhausted.

Step 3: Repeat the procedure until the available supply at various sources and demand at various destinations is satisfied. The solution so obtained need not be non-degenerate.

Example 9.3 Use Least Cost Method (LCM) to find initial basic feasible solution to the transportation problem using data of Example 9.1.

Solution The cell with lowest unit cost (i.e., 8) is (S_3, D_2). The maximum units which can be allocated to this cell is 8. This meets the complete demand of D_2 and leave l0 units with S_3, as shown in Table 9.3.

In the reduced table without column D_2, the next smallest unit transportation cost, is 10 in cell (S_1, D_4). The maximum which can be allocated to this cell is 7. This exhausts the capacity of S_1 and leaves 7 units with D_4 as unsatisfied demand. This is shown in Table 9.3.

	D_1	D_2	D_3	D_4	*Supply*
S_1	19	30	50	10 (7)	7
S_2	70	30	40	60	9
S_3	40	8 (8)	70	20	18
Demand	5	8	7	14	34

Table 9.3

In Table 9.3, the next smallest cost is 20 in cell (S_3, D_4). The maximum units that can be allocated to this cell is 7 units. This satisfies the entire demand of D_4 and leaves 3 units with S_3, as the remaining supply, shown in Table 9.4.

In Table 9.4, the next smallest unit cost cell is not unique. That is, there are two cells – (S_2, D_3) and (S_3, D_1) – that have the same unit transportation cost of 40. Allocate 7 units in cell (S_2, D_3) first because it can accommodate more units as compared to cell (S_3, D_1). Then allocate 3 units (only supply left with S_3) to cell (S_3, D_1). The remaining demand of 2 units of D_1 is fulfilled from S_2. Since supply and demand at each supply centre and demand centre is exhausted, the initial solution is arrived at, and is shown in Table 9.4.

	D_1	D_2	D_3	D_4	*Supply*
S_1	19	30	50	10 (7)	7
S_2	70 (2)	30	40 (7)	60	9
S_3	40 (3)	8 (8)	70	20 (7)	18
Demand	5	8	7	14	34

Table 9.4

The total transportation cost of the initial solution by LCM is calculated as given below:

Total cost = 7 × 10 + 2 × 70 + 7 × 40 + 3 × 40 + 8 × 8 + 7 × 20 = Rs 814

The total transportation cost obtained by LCM is less than the cost obtained by NWCM.

9.4.3 Vogel's Approximation Method (VAM)

Vogel's approximation (penalty or regret) is preferred over NWCR and LCM methods. In this method, an allocation is made on the basis of the opportunity (or penalty or extra) cost that would have been incurred if the allocation in certain cells with minimum unit transportation cost were missed. Hence, allocations are made in such a way that the penalty cost is minimized. An initial solution obtained by using this method is nearer to an optimal solution or is the optimal solution itself. The steps of VAM are as follows:

Step 1: Calculate the penalties for each row (column) by taking the difference between the smallest and next smallest unit transportation cost in the same row (column). This difference indicates the penalty or extra cost that has to be paid if decision-maker fails to allocate to the cell with the minimum unit transportation cost.

Step 2: Select the row or column with the largest penalty and allocate as much as possible in the cell that has the least cost in the selected row or column and satisfies the rim conditions. If there is a tie in the values of penalties, it can be broken by selecting the cell where the maximum allocation can be made.

Step 3: Adjust the supply and demand and cross out the satisfied row or column. If a row and a column are satisfied simultaneously, only one of them is crossed out and the remaining row (column) is assigned a zero supply (demand). Any row or column with zero supply or demand should not be used in computing future penalties.

Step 4: Repeat Steps 1 to 3 until the available supply at various sources and demand at various destinations is satisfied.

Example 9.4 Use Vagel's Approximation Method (VAM) to find the initial basic feasible solution to the transportation problem using the data of Example 9.1.

Solution The differences (penalty costs) for each row and column have been calculated as shown in Table 9.5. In the first round, the maximum penalty, 22 occurs in column D_2. Thus the cell (S_3, D_2) having the least transportation cost is chosen for allocation. The maximum possible allocation in this cell is 8 units and it satisfies demand in column D_2. Adjust the supply of S_3 from 18 to 10 (18 – 8 = 10).

	D_1	D_2	D_3	D_4	*Supply*	*Row differences*			
S_1	19 (5)	30	50	10 (2)	7	9	9	40	40
S_2	70	30	40 (7)	60 (2)	9	10	20	20	20
S_3	40	8 (8)	70	20 (10)	18	12	20	50	–
Demand	5	8	7	14	34				
Column differences	21	22	10	10					
	21	–	10	10					
	–	–	10	10					
	–	–	10	50					

Table 9.5 Initial Solution Using VAM

The new row and column penalties are calculated except column D_2 because D_2's demand has been satisfied. In the second round, the largest penalty, 21 appears at column D_1. Thus the cell (S_1, D_1) having the least transportation cost is chosen for allocating 5 units as shown in Table 9.5. After adjusting the supply and demand in the table, we move to the third round of penalty calculations.

In the third round, the maximum penalty 50 appears at row S_3. The maximum possible allocation of 10 units is made in cell (S_3, D_4) that has the least transportation cost of 20 as shown in Table 9.5.

The process is continued with new allocations till a complete solution is obtained. The initial solution using VAM is shown in Table 9.5. The total transportation cost associated with this method is:

$$\text{Total cost} = 5 \times 19 + 2 \times 10 + 7 \times 40 + 2 \times 60 + 8 \times 8 + 10 \times 20 = \text{Rs } 779$$

Example 9.5 A dairy firm has three plants located in a state. The daily milk production at each plant is as follows:

Plant 1 : 6 million litres, Plant 2 : 1 million litres, and Plant 3 : 10 million litres

Each day, the firm must fulfil the needs of its four distribution centres. The minimum requirement of each centre is as follows:

Distribution centre 1 : 7 million litres, Distribution centre 2 : 5 million litres,

Distribution centre 3 : 3 million litres, and Distribution centre 4 : 2 million litres

Cost (in hundreds of rupees) of shipping one million litre from each plant to each distribution centre is given in the following table:

		Distribution Centre			
		D_1	D_2	D_3	D_4
Plant	P_1	2	3	11	7
	P_2	1	0	6	1
	P_3	5	8	15	9

Find the initial basic feasible solution for given problem by using following methods:

(a) North-west corner rule

(b) Least cost method

(c) Vogel's approximation method

Solution (a) ***North-West Corner Rule***

		Distribution Centre				
		D_1	D_2	D_3	D_4	*Supply*
Plant	P_1	2 (6)	3	11	7	$6 = a_1$
	P_2	1 (1)	0	6	1	$1 = a_2$
	P_3	5	8 (5)	15 (3)	9 (2)	$10 = a_3$
Demand		$7 = b_1$	$5 = b_2$	$3 = b_3$	$2 = b_4$	

Table 9.6 Initial Solution by NWCR

(i) Comparing a_1 and b_1, since $a_1 < b_1$; allocate $x_{11} = 6$. This exhausts the supply at P_1 and leaves 1 unit as unsatisfied demand at D_1.

(ii) Move to cell (P_2, D_1). Compare a_2 and b_1 (i.e. 1 and 1). Since $a_2 = b_1$, allocate $x_{21} = 1$.

(iii) Move to cell (P_3, D_2). Since supply at P_3, is equal to the demand at D_2, D_3 and D_4, therefore, allocate $x_{32} = 5$, $x_{33} = 3$ and $x_{34} = 2$.

It may be noted that the number of allocated cells (also called *basic cells*) are 5 which is one less than the required number $m + n - 1$ $(3 + 4 - 1 = 6)$. Thus, this solution is the degenerate solution. The transportation cost associated with this solution is:

Total cost = Rs $(2 \times 6 + 1 \times 1 + 8 \times 5 + 15 \times 3 + 9 \times 2) \times 100$ = Rs 11,600

(b) ***Least Cost Method***

Table 9.7 Initial Solution by LCM

		Distribution Centre				
		D_1	D_2	D_3	D_4	*Supply*
Plant	P_1	2 (6)	3	11	7	6
	P_2	1	0 (1)	6	1	1
	P_3	5 (1)	8 (4)	15 (3)	9 (2)	10
Demand		7	5	3	2	

(i) The lowest unit cost in Table 9.7 is 0 in cell (P_2, D_2), therefore the maximum possible allocation that can be made is 1 unit. Since this allocation exhausts the supply at plant P_2, therefore row 2 is crossed off.

(ii) The next lowest unit cost is 2 in cell (P_1, D_1). The maximum possible allocation that can be made is 6 units. This exhausts the supply at plant P_1, therefore, row P_1 is crossed off.

(iii) Since the total supply at plant P_3 is now equal to the unsatisfied demand at all the four distribution centres, therefore, the maximum possible allocations satisfying the supply and demand conditions, are made in cells (P_3, D_1), (P_3, D_2), (P_3, D_3) and (P_3, D_4).

The number of allocated cells in this case are six, which is equal to the required number $m + n - 1$ ($3 + 4 - 1 = 6$). Thus, this solution is non-degenerate. The transportation cost associated with this solution is

Total cost = Rs ($2 \times 6 + 5 \times 1 + 8 \times 4 + 15 \times 3 + 9 \times 2$) $\times$ 100 = Rs 11,200

(c) ***Vogel's Approximation Method:*** First calculating penalties as per rules and then allocations are made in accordance of penalties as shown in Table 9.8.

Table 9.8 Initial Solution by VAM

		D_1	D_2	D_3	D_4	*Supply*	*Row penalty*		
Plant	P_1	2 (1)	3 (5)	11	7	6	1	1	5
	P_2	1	0	6	1 (1)	1	0	–	–
	P_3	5 (6)	8	15 (3)	9 (1)	10	3	3	4
Demand		7	5	3	2				
Column penalty		1	3	5	6				
		3	5	4	2				
		3	–	4	2				
		Distribution Centre							

The number of allocated cells in Table 9.8 are six, which is equal to the required number $m + n - 1$ ($3 + 4 - 1 = 6$), therefore, this solution is non-degenerate. The transportation cost associated with this solution is:

Total cost = Rs ($2 \times 1 + 3 \times 5 + 1 \times 1 + 5 \times 6 + 15 \times 3 + 9 \times 1$) $\times$ 100 = Rs 10,200

Remark: Total transportation cost found by VAM is lower than the costs of transportation determined by the NWCR and LCM methods. Therefore, it is of advantage to use this method in order to reduce computational time required to obtain optimum solution.

CONCEPTUAL QUESTIONS A

1. Show that all the bases for a transportation problem are triangular.
2. With reference to a transportation problem define the following terms:
   (i) Feasible solution (ii) Basic feasible solution
   (iii) Optimal solution
   (iv) Non-degenerate basic feasible solution
3. Given a mathematical formulation of the transportation problem and the simplex methods, what are the differences in the nature of problems that can be solved by using these methods?
4. Prove that there are only $m + n - 1$ independent equations in a transportation problem, m and n being the number of origins and destination, and only one equation can be dropped as being redundant. (For proof see Appendix).
5. Describe the transportation problem with its general mathematical formulation.
6. Show that a transportation problem is a special type of LP problem. In what areas of management can the transportation model be effectively used? Discuss.
7. What are the characteristics of transportation problem of linear programming?
8. What is meant by the triangular form of a system of linear equations? When does a system of linear equations have a triangular basis? (See Appendix for proof.)
9. What is meant by non-degenerate basic feasible solution of a transportation problem?
10. Explain in brief three, methods of initial feasible solution for transportation problem.
11. Explain the various steps involved in solving transportation problem using (i) Least cost method, and (ii) Vogel's approximation method.
12. Explain the (i) North-West Corner method, (ii) Least-Cost method, and (iii) Vogel's Approximation method, for obtaining an initial basic feasible solution of a transportation problem.
13. State the transportation problem. Describe clearly the steps involved in solving it.
14. Is the transportation model an example of decision-making under certainty or under uncertainty? Why?
15. Why does Vogel's approximation method provide a good initial feasible solution? Can the North-West Corner method ever be able to provide an initial solution with a cost as low as this?

SELF PRACTICE PROBLEMS A

1. Determine an initial basic feasible solution to the following transportation problem by using (a) NWCR, (b) LCM and (c) VAM.

Destination

		D_1	D_2	D_3	D_4	*Supply*
Source	S_1	21	16	15	3	11
	S_2	17	18	14	23	13
	S_3	32	27	18	41	19
	Demand	6	6	8	23	

2. Determine an initial basic feasible solution to the following transportation problem by using (a) the least cost method, and (b) Vogel's approximation method.

Destination

		D_1	D_2	D_3	D_4	*Supply*
Source	S_1	1	2	1	4	30
	S_2	3	3	2	1	30
	S_3	4	2	5	9	40
	Demand	20	40	30	10	

3. Determine an initial basic feasible solution to the following transportation problem by using (a) NWCM, (b) LCM, and (c) VAM.

Destination

		D_1	D_2	D_3	D_4	*Supply*
Source	*A*	11	13	17	14	250
	B	16	18	14	10	300
	C	21	24	13	10	400
	Demand	200	225	275	250	

4. Determine an initial basic feasible solution to the following transportation problem by using the North-West corner rule, where O_i and D_j represent ith origin and jth destination, respectively.

Destination

		D_1	D_2	D_3	D_4	*Supply*
Source	O_1	6	4	1	5	14
	O_2	8	9	2	7	16
	O_3	4	3	6	2	5
	Demand	6	10	15	4	

HINTS AND ANSWERS

1. $x_{14} = 11$, $x_{21} = 6$, $x_{22} = 3$, $x_{24} = 4$, $x_{32} = 3$, $x_{33} = 4$, $x_{34} = 12$; Total cost = 686.
2. (a) and (b): $x_{11} = 20$, $x_{13} = 10$, $x_{22} = 20$, $x_{33} = 20$, $x_{24} = 10$, $x_{32} = 20$; Total cost = 180.
3. (a) $x_{11} = 200$, $x_{12} = 50$, $x_{22} = 175$, $x_{23} = 125$, $x_{33} = 150$, $x_{34} = 250$; Total cost = 12,200.
   (b) $x_{11} = 200$, $x_{12} = 50$, $x_{22} = 175$, $x_{23} = 125$, $x_{33} = 150$, $x_{34} = 250$; Total cost = 12,200.
   (c) $x_{11} = 200$, $x_{12} = 50$, $x_{22} = 175$, $x_{24} = 125$, $x_{33} = 275$, $x_{34} = 125$; Total cost = 12,075.
4. $x_{11} = 6$; $x_{12} = 8$; $x_{22} = 2$; $x_{23} = 14$; $x_{33} = 1$; $x_{34} = 4$; Total cost = Rs 128.

9.5 TEST FOR OPTIMALITY

Once an initial solution is obtained, the next step is to check its optimality in terms of feasibility of the solution and total minimum transportation cost.

The test of optimality begins by calculating an opportunity cost associated with each unoccupied cell (represents unused route) in the transportation table. An unoccupied cell with the largest negative opportunity cost is selected to include in the new set of transportation routes (allocations). *This value indicates the per unit cost reduction that can be achieved by making appropriate allocation in the unoccupied cell.* This cell is also known as an *incoming cell* (*or variable*). The outgoing cell (or variable) from the current solution is the occupied cell (basic variable) where allocation will become zero as allocation is made in the unoccupied cell with the largest negative opportunity cost. Such an exchange reduces the total transportation cost. The process is continued until there is no negative opportunity cost. That is, the current solution is an optimal solution.

The negative opportunity cost indicates the per unit cost reduction that can be achieved by raising the shipment allocation in the unoccupied cell from its present level of zero.

The *Modified-distribution* (MODI) method (also called *u-v method* or *method of multipliers*) is used to calculate opportunity cost associated with each unoccupied cell and then improving the current solution leading to an optimal solution. The steps of MODI method based on the concept of duality are summarized in section 9.5.3.

9.5.1 Dual of Transportation Model

For a given basic feasible solution if we associate numbers (also called *dual variables* or *multipliers*) u_i and v_j with row i ($i = 1, 2, \ldots, m$) and column j ($j = 1, 2, \ldots, n$) of the transportation table, respectively, then u_i and v_j must satisfy the equation

$$u_i + v_j = c_{ij}, \text{ for each occupied cell } (i, j)$$

These equations yield $m + n - 1$ equations in $m + n$ unknown dual variables. The values of these variables can be determined by arbitrarily assigning a zero value to any one of these variables. The value of the remaining $m + n - 2$ variables can then be obtained algebraically by using the above equation for the occupied cells. The opportunity cost of each unoccupied cell (called *non-basic variable* or *unused route*) is calculated by using following equation that involves u_i and v_j values.:

$$d_{rs} = c_{rs} - (u_r + v_s), \text{ for each unoccupied cell } (r, s)$$

This equation also indicates the per unit reduction in the total transportation cost for the route (r, s). To prove these two results, consider the general transportation model:

$$\text{Minimize } Z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij}\, x_{ij}$$

subject to the constraints

$$\sum_{j=1}^{n} x_{ij} = a_i, \quad i = 1, 2, \ldots, m \quad \text{(Supply)}$$

$$\sum_{i=1}^{m} x_{ij} = b_j, \quad j = 1, 2, \ldots, n \quad \text{(Demand)}$$

and $$x_{ij} \geq 0 \quad \text{for all } i \text{ and } j$$

Modi method helps in comparing the relative advantage of alternative allocations for all the unoccupied cells simultaneously.

Since all of the constraints are equalities, write each equality constraint equivalent to two inequalities as follows:

$$\left.\begin{aligned} \sum_{j=1}^{n} x_{ij} &\geq a_i, \quad i = 1, 2, \ldots, m \\ \sum_{j=1}^{n} -(x_{ij}) &\geq -a_i \end{aligned}\right\} \text{(Supply constraints)}$$

$$\left.\begin{aligned} \sum_{i=1}^{m} x_{ij} &\geq b_j, \quad j = 1, 2, \ldots, m \\ \sum_{i=1}^{m} -(x_{ij}) &\geq -b_j \end{aligned}\right\} \text{(Demand constraints)}$$

Let u_i^+ and u_i^- be the dual variables, one for each supply constraint i. Similarly v_j^+, v_j^- be the dual variables one for each demand constraint j. Then, the dual of the transportation model can be written as:

Maximize $Z^* = \sum_{i=1}^{m} (u_i^+ - u_i^-)\, a_i + \sum_{j=1}^{n} (v_j^+ - v_j^-)\, b_j$

subject to the constraints

$$(u_i^+ - u_i^-) + (v_j^+ - v_j^-) \le c_{ij}$$

and $u_i^+, u_i^-, v_j^+, v_j^- \ge 0$, for all i and j.

The variables u_i^+ and u_i^- that appear in the objective function, may take positive, negative or zero values. Thus, either of these will appear in the optimal basic feasible solution because one is the negative of the other. The same argument may be given for v_j^+ and v_j^-. Thus, let

$$u_i = u_i^+ - u_i^-, \quad i = 1, 2, \ldots, m$$
$$v_j = v_j^+ - v_j^-, \quad j = 1, 2, \ldots, n`$$

The values of u_i and v_j will then be unrestricted in sign. Hence, the dual of the transportation model can now be written as

Maximize $Z^* = \sum_{i=1}^{m} u_i\, a_i + \sum_{j=1}^{n} v_j\, b_j$

subject to the constraints

$$u_i + v_j \le c_{ij}$$

and u_i, v_j unrestricted in sign for all i and j.

The relationship $(c_{ij} - u_i - v_j)\, x_{ij} = 0$ is known as *complementary slackness* for a transportation problem and indicates that

(a) if $x_{ij} > 0$ and solution is feasible, then $c_{ij} - u_i - v_j = 0$ or $c_{ij} = u_i + v_j$, for each occupied cell,
(b) if $x_{ij} = 0$ and $c_{ij} > u_i + v_j$, then it is not desirable to have $x_{ij} > 0$ in the solution mix because it would cost more to transport on a route (i, j),
(c) if $c_{ij} \le u_i + v_j$ for some $x_{ij} = 0$, then x_{ij} can be brought into the solution mix.

9.5.2 Economic Interpretation of u_i's and v_j's

The u_i values measures the comparative advantage of additional unit of supply or shadow price (or value) of available supply at centre i. This may also be termed as *location rent.* Similarly, the v_j values measures the comparative advantage of an additional unit of commodity demanded at demand centre j. This may also be termed as *market price.*

The **dual variables** u_is and v_js represent the shadow price (value of the commodity) for the supply centres and demand centres, respectively.

Illustration The concept of duality in transportation problem is applied on Example 9.1 in the following manner:

Reproducing transportation data of Example 9.1 for ready reference in Table 9.9. In Table 9.9, there are $m = 3$ rows and $n = 4$ columns. Let u_1, u_2 and u_3 be dual variables corresponding to each of the supply constraint in that order. Similarly, v_1, v_2, v_3 and v_4 be dual variables corresponding to each of demand constraint in that order. The dual problem then becomes

	D_1	D_2	D_3	D_4	*Supply*	u_i
S_1	19	30	50	10	7	u_1
S_2	70	30	40	60	9	u_2
S_3	40	8	70	20	18	u_3
Demand	5	8	7	14	34	
v_j	v_1	v_2	v_3	v_4		

Table 9.9

Maximize $Z = (7u_1 + 9u_2 + 18u_3) + (5v_1 + 8v_2 + 7v_3 + 14v_4)$

subject to the constraints

(i) $u_1 + v_1 \le 19$, (ii) $u_1 + v_2 \le 30$, (iii) $u_1 + v_3 \le 50$, (iv) $u_1 + v_4 \le 10$,
(v) $u_2 + v_1 \le 70$, (vi) $u_2 + v_2 \le 30$, (vii) $u_2 + v_3 \le 40$, (viii) $u_2 + v_4 \le 60$,
(ix) $u_3 + v_1 \le 40$, (x) $u_3 + v_2 \le 8$, (xi) $u_3 + v_3 \le 70$, (xii) $u_3 + v_4 \le 20$,

and u_i, v_j unrestricted in sign for all i and j.

Interpretation Consider the dual constraint $u_1 + v_1 \leq 19$ or $v_1 \leq 19 - u_1$. This represents the delivered market value of the commodity at destination D_1 which should be less than or equal to the unit cost of transportation from S_1 to D_1 minus the per unit value of commodity at D_1. A similar interpretation can also be given for other constraints.

The optimal value of dual variables can be obtained either by simplex method or by reading values of these variables from the optimal solution of transportation problem. It may be noted that the total transportation cost at optimal solution would be the same as obtained by putting values of u_i's and v_j's from optimal solution of transportation problem in the dual objective function:

$$\text{Maximize } Z = \sum_{i=1}^{3} a_i u_i + \sum_{j=1}^{4} b_j v_j$$

9.5.3 Steps of MODI Method (Transportation Algorithm)

The steps to evaluate unoccupied cells are as follows:

Step 1: For an initial basic feasible solution with $m + n - 1$ occupied cells, calculate u_i and v_j for rows and columns. The initial solution can be obtained by any of the three methods discussed earlier.

To start with, any one of u_is or v_js is assigned the value zero. It is better to assign zero to a particular u_i or v_j where there are maximum number of allocations in a row or column respectively, as this will reduce the considerably arithmetic work. The value of u_is and v_js for other rows and columns is calculated by using the relationship.

$$c_{ij} = u_i + v_j, \quad \text{for all occupied cells } (i, j).$$

Step 2: For unoccupied cells, calculate the opportunity cost by using the relationship

$$d_{ij} = c_{ij} - (u_i + v_j), \quad \text{for all } i \text{ and } j.$$

Step 3: Examine sign of each d_{ij}

(i) If $d_{ij} > 0$, then the current basic feasible solution is optimal.

(ii) If $d_{ij} = 0$, then the current basic feasible solution will remain unaffected but an alternative solution exists.

(iii) If one or more $d_{ij} < 0$, then an improved solution can be obtained by entering an unoccupied cell (i, j) into the solution mix (basis). An unoccupied cell having the largest negative value of d_{ij} is chosen for entering into the solution mix (new transportation schedule).

Changing the shipping route involves adding to cells on the closed path with plus signs and subtracting from cells with negative signs.

Step 4: Construct a closed-path (or loop) for the unoccupied cell with largest negative value of d_{ij}. Start the closed path with the selected unoccupied cell and mark a plus sign (+) in this cell. Trace a path along the rows (or columns) to an occupied cell, mark the corner with a minus sign (–) and continue down the column (or row) to an occupied cell. Then mark the corner with plus sign (+) and minus sign (–) alternatively. Close the path back to the selected unoccupied cell.

Step 5: Select the smallest quantity amongst the cells marked with minus sign on the corners of closed loop. Allocate this value to the selected unoccupied cell, add it to occupied cells marked with plus signs, and subtract it from the occupied cells marked with minus signs.

Step 6: Obtain a new improved solution by allocating units to the unoccupied cell according to Step 5 and calculate the new total transportation cost.

Step 7: Test optimality of the revised solution. The procedure terminates when all $d_{ij} \geq 0$ for unoccupied cells.

Remarks 1. The closed-loop (path) starts and ends at the selected unoccupied cell. It consists of successive horizontal and vertical (connected) lines whose end points must be occupied cells, except an end point associated with entering unoccupied cell. This means that every corner element of the loop must be an occupied cell.

It is immaterial whether the loop is traced in a clockwise or anti-clockwise direction and whether it starts up, down, right or left (but never diagonally). However, for a given solution only one loop can be constructed for each unoccupied cell.

2. There can only be one plus (+) sign and only one minus (–) sign in any given row or column.
3. The closed path indicates changes involved in reallocating the shipments.

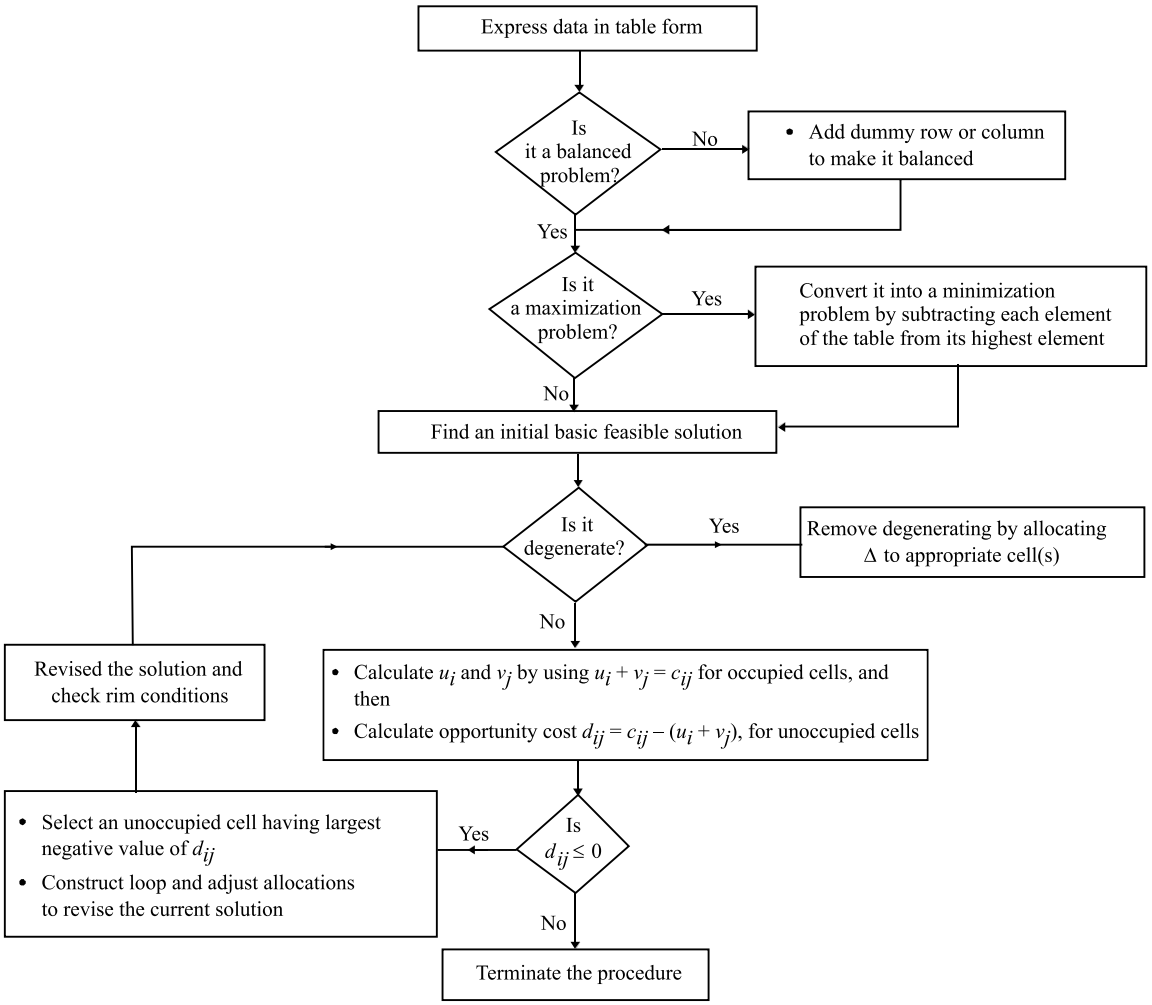


Fig. 9.1 Flow Chart of MODI Method

The steps of MODI method for solving a transportation problem are summarized in the flow chart shown in Fig. 9.1.

9.5.4 Close-Loop in Transportation Table and its Properties

Any basic feasible solution must contain $m + n - 1$ non-zero allocations provided.

(i) any two adjacent cells of the ordered set lie either in the same row or in the same column, and

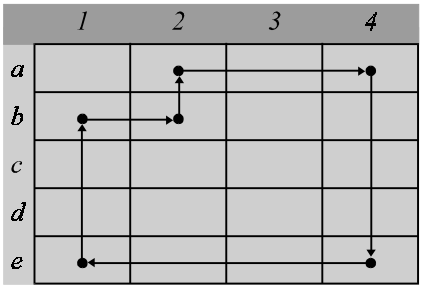
(ii) no three or more adjacent cells in the ordered set lie in the same row or column. The first cell of the set must follow the last in the set, i.e. each cell (except the last) must appear only once in the ordered set.

An ordered set of at least four cells in a transportation table forms a **loop.**

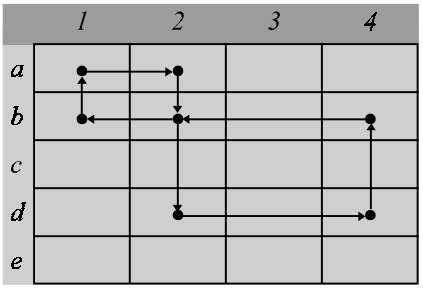
Consider the following two cases represented in Tables 9.10(a) and 9.10(b). In Table 9.10(a), if we join the positive allocations by horizontal and vertical lines, then a closed loop is obtained. The ordered set of cells forming a loop is:

$$L = \{(a, 2), (a, 4), (e, 4), (e, 1), (b, 1), (b, 2), (a, 2)\}$$

The loop in Table 9.10(b) is not allowed because it does not satisfy the conditions in the definition of a loop. That is, the cell $(b, 2)$ appears twice.



(a)



(b)

Table 9.10

Remarks 1. Every loop has an even number of cells and has at least four cells.
2. The allocations are said to be in independent position if it is not possible to increase or decrease any individual allocation without changing the positions of these allocations, or if a closed loop cannot be formed through these allocations without violating the rim conditions.
3. Each row and column in the transportation table should have only one plus and minus sign. All cells that have a plus or a minus sign, except the starting unoccupied cell, must be occupied cells.
4. Closed loops may or may not be in the shape of a square.

Example 9.6 Apply MODI method to obtain optimal solution of transportation problem using the data of Example 9.1.

	D_1	D_2	D_3	D_4	*Supply*
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	34

Solution Applying Vogel's approximation method to obtain an initial basic feasible solution. This solution is shown in Table 9.11 [for ready reference see Table 9.5].

1. In Table 9.11, since number of occupied cells are $m + n - 1 = 3 + 4 - 1 = 6$ (as required), therefore this initial solution is non-degenerate. Thus, an optimal solution can be obtained. The total transportation cost associated with this solution is Rs 779.

2. In order to calculate the values of u_is ($i = 1, 2, 3$) and v_js ($j = 1, 2, 3, 4$) for each occupied cell, assigning arbitrarily, $v_4 = 0$ in order to simplify calculations. Given $v_4 = 0$, u_1, u_2 and u_3 can be immediately computed by using the relation $c_{ij} = u_i + v_j$ for occupied cells, as shown in Table 9.11.

	D_1	D_2	D_3	D_4	*Supply*	u_i
S_1	19 (5)	30 +32	50 + 60	10 (2)	7	$u_1 = 10$
S_2	70 + 1	30 (+) – 18	40 (7)	60 (2) (–)	9	$u_2 = 60$
S_3	40 + 11	8 (–) (8)	70 + 70	20 (10) (+)	18	$u_3 = 20$
Demand	5	8	7	14	34	
v_j	$v_1 = 9$	$v_2 = -12$	$v_3 = -20$	$v_4 = 0$		

Table 9.11 Initial Solution, VAM

$$c_{34} = u_3 + v_4 \quad \text{or} \quad 20 = u_3 + 0 \quad \text{or} \quad u_3 = 20$$
$$c_{24} = u_2 + v_4 \quad \text{or} \quad 60 = u_2 + 0 \quad \text{or} \quad u_2 = 60$$
$$c_{14} = u_1 + v_4 \quad \text{or} \quad 10 = u_1 + 0 \quad \text{or} \quad u_1 = 10$$

Given u_1, u_2, and u_3, value of v_1, v_2 and v_3 can also be calculated as shown below:

$$c_{11} = u_1 + v_1 \quad \text{or } 19 = 10 + v_1 \text{ or } v_1 = 9$$
$$c_{23} = u_2 + u_3 \quad \text{or } 40 = 60 + v_3 \text{ or } v_3 = -20$$
$$c_{32} = u_3 + v_2 \quad \text{or } 8 = 20 + v_2 \text{ or } v_2 = -12$$

3. The opportunity cost for each of the occupied cell is determined by using the relation $d_{ij} = c_{ij} - (u_i + v_j)$ and is shown below.

$$\begin{aligned}
d_{12} &= c_{12} - (u_1 + v_2) = 30 - (10 - 12) = 32 \\
d_{13} &= c_{13} - (u_1 + v_3) = 50 - (10 - 20) = 60 \\
d_{21} &= c_{21} - (u_2 + v_1) = 70 - (60 + 9) = 1 \\
d_{22} &= c_{22} - (u_2 + v_2) = 30 - (60 - 12) = -18 \\
d_{31} &= c_{31} - (u_3 + v_1) = 40 - (20 + 9) = 11 \\
d_{33} &= c_{33} - (u_3 + v_3) = 70 - (20 - 20) = 70
\end{aligned}$$

4. According to the optimality criterion for cost minimizing transportation problem, the current solution is not optimal, since the opportunity costs of the unoccupied cells are not all zero or positive. The value of $d_{22} = -18$ in cell (S_2, D_2) is indicating that the total transportation cost can be reduced in the multiple of 18 by shifting an allocation to this cell.

5. A closed-loop (path) is traced along row S_2 to an occupied cell (S_3, D_2). A plus sign is placed in cell (S_2, D_2) and minus sign in cell (S_3, D_2). Now take a right-angle turn and locate an occupied cell in column D_4. An occupied cell (S_3, D_4) exists at row S_3, and a plus sign is placed in this cell.

   Continue this process and complete the closed path. The occupied cell (S_2, D_3) must be bypassed otherwise they will violate the rules of constructing closed path.

6. In order to maintain feasibility, examine the occupied cells with minus sign at the corners of closed loop, and select the one that has the smallest allocation. This determines the maximum number of units that can be shifted along the closed path. The minus signs are in cells (S_3, D_2) and (S_2, D_4). The cell (S_2, D_4) is selected because it has the smaller allocation, i.e. 2. The value of this allocation is then added to cell (S_2, D_2) and (S_3, D_4), which carry plus signs. The same value is subtracted from cells (S_2, D_4) and (S_3, D_2) because they carry minus signs.

7. The revised solution is shown in Table 9.12. The total transportation cost associated with this solution is:

   Total cost = 5 × 19 + 2 × 10 + 2 × 30 + 7 × 40 + 6 × 8 + 12 × 20 = Rs 743

	D_1	D_2	D_3	D_4	*Supply*	u_i
S_1	19 (5)	30 +32	50 +42	10 (2)	7	$u_1 = 0$
S_2	70 + 19	30 (2)	40 (7)	60 +14	9	$u_2 = 32$
S_3	40 + 11	8 (6)	70 + 52	20 (12)	18	$u_3 = 10$
Demand	5	8	7	14	34	
v_j	$v_1 = 19$	$v_2 = -2$	$v_3 = 8$	$v_4 = 10$		

Table 9.12
Optimal Solution

8. Test the optimality of the revised solution once again in the same way as discussed in earlier steps. The values of u_is, v_js and d_{ij}s are shown in Table 9.12. Since each of $d_{ij}s$ is positive, therefore, the current basic feasible solution is optimal with a mi]nimum total transportation cost of Rs 743.

Example 9.7 A company has factories at F_1, F_2, and F_3 that supply products to warehouses at W_1, W_2 and W_3. The weekly capacities of the factories are 200, 160 and 90 units, respectively. The weekly warehouse requirements are 180, 120 and 150 units, respectively. The unit shipping costs (in rupees) are as follows:

		Warehouse			
		W_1	W_2	W_3	*Supply*
Factory	F_1	16	20	12	200
	F_2	14	8	18	160
	F_3	26	24	16	90
	Demand	180	120	150	450

Determine the optimal distribution for this company in order to minimize its total shipping cost.

Solution Initial basic feasible solution obtained by North-West Corner Rule is given in Table 9.13. Since, as required, this initial solution has $m + n - 1 = 3 + 3 - 1 = 5$ allocations, it is a non-degenerate solution. The optimality test can, therefore, be performed. The total transportation cost associated with this solution is:

Table 9.13 Initial Solution

	W_1	W_2	W_3	*Supply*
F_1	16 (180)	20 (20)	12	200
F_2	14	8 (100)	18 (60)	160
F_3	26	24	16 (90)	90
Demand	180	120	150	450

Total cost = $16 \times 180 + 20 \times 20 + 8 \times 100 + 18 \times 60 + 16 \times 90$ = Rs 6,800

Determine the values of u_is and v_js as usual, by arbitrarily assigning $u_1 = 0$. Given $u_1 = 0$, the values of others variables obtained by using the equation $c_{ij} = u_i + v_j$ for occupied cells, are shown in Table 9.14.

At each step a non-occupied cell with largest negative opportunity cost is selected to get maximum reduction in total transportation cost.

Table 9.14

	W_1	W_2	W_3	*Supply*	u_i
F_1	16 (180)	20 (–) (20)	12 (+) – 18	200	$u_1 = 0$
F_2	14 +10	8 (+) (100)	18 (60) (–)	160	$u_2 = -12$
F_3	26 +24	24 +18	16 (90)	90	$u_3 = -14$
Demand	180	120	150	450	
v_j	$v_1 = 16$	$v_2 = 20$	$v_3 = 30$		

$c_{11} = u_1 + v_1$ or $16 = 0 + v_1$ or $v_1 = 16$
$c_{12} = u_1 + v_2$ or $20 = 0 + v_2$ or $v_2 = 20$
$c_{22} = u_2 + v_2$ or $8 = u_2 + 20$ or $u_2 = -12$
$c_{23} = u_2 + v_3$ or $18 = -12 + v_3$ or $v_3 = 30$
$c_{33} = u_3 + v_3$ or $16 = u_3 + 30$ or $u_3 = -14$

The opportunity cost for each of the unoccupied cells is determined by using the equation, $d_{ij} = c_{ij} - (u_i + v_j)$ as follows:

$$d_{13} = c_{13} - (u_1 + v_3) = 12 - (0 + 30) = -18$$
$$d_{21} = c_{21} - (u_2 + v_1) = 14 - (-12 + 16) = 10$$
$$d_{31} = c_{31} - (u_3 + v_1) = 26 - (-14 + 16) = 24$$
$$d_{32} = c_{32} - (u_3 + v_2) = 24 - (-14 + 20) = 18$$

The value of $d_{13} = -18$ in the cell (F_1, W_3) indicates that the total transportation cost can be reduced in a multiple of 18 by introducing this cell in the new transportation schedule. To see how many units of the commodity could be allocated to this cell (route), form a closed path as shown in Table 9.14.

The largest number of units of the commodity that should be allocated to the cell (F_1, W_3) is 20 units because it does not violate the supply and demand restrictions (minimum allocation among the occupied cells bearing negative sign at the corners of the loop). The new transportation schedule (solution) so obtained is shown in Table 9.15.

	W_1	W_2	W_3	*Supply*
F_1	16 (180)	20	12 (20)	200
F_2	14	8 (120)	18 (40)	160
F_3	26	24	16 (90)	90
Demand	180	120	150	450

Table 9.15 Revised Schedule

The total transportation cost associated with this solution is

Total cost = 16 × 180 + 12 × 20 + 8 × 120 + 18 × 40 + 16 × 90 = Rs 6,240

To test the optimality of the new solution shown in Table 9.15, again calculate the opportunity cost of each unoccupied cell in the same manner as discussed earlier. The calculations for u_is, v_js and d_{ij}s are shown in Table 9.16.

$$c_{13} = u_1 + v_3 \quad \text{or} \quad 12 = u_1 + 0 \quad \text{or} \quad u_1 = 12$$
$$c_{23} = u_2 + v_3 \quad \text{or} \quad 18 = u_2 + 0 \quad \text{or} \quad u_2 = 18$$
$$c_{33} = u_3 + v_3 \quad \text{or} \quad 16 = u_3 + 0 \quad \text{or} \quad u_3 = 16$$
$$c_{11} = u_1 + v_1 \quad \text{or} \quad 16 = 12 + v_1 \quad \text{or} \quad v_1 = 4$$
$$c_{22} = u_2 + v_2 \quad \text{or} \quad 8 = 18 + v_2 \quad \text{or} \quad v_2 = -10$$

$$d_{12} = c_{12} - (u_1 + v_2) \quad \text{or} \quad 20 - (12 - 10) = 18$$
$$d_{21} = c_{21} - (u_2 + v_1) \quad \text{or} \quad 14 - (18 + 4) = -8$$
$$d_{31} = c_{31} - (u_3 + v_1) \quad \text{or} \quad 26 - (16 + 4) = 6$$
$$d_{32} = c_{32} - (u_3 + v_2) \quad \text{or} \quad 24 - (16 - 10) = 18$$

The value of $d_{21} = -8$ in the cell (F_2, W_1) indicates that the total cost of transportation can further be reduced in a multiple of 8 by introducing this cell in the new transportation schedule. The new solution is obtained by introducing 40 units of the commodity in the cell (F_2, W_1), as indicated in Table 9.16. The new solution is shown in Table 9.17.

	W_1	W_2	W_3	*Supply*	u_i
F_1	16 (–) (180)	20 18	12 (20) (+)	200	$u_1 = 12$
F_2	14 (+) – 8	8 (120)	18 (40) (–)	160	$u_2 = 18$
F_3	26 6	24	16 (90)	90	$u_3 = 16$
Demand	180	120	150		
v_j	$v_1 = 4$	$v_2 = -10$	$v_3 = 0$		

Table 9.16

	W_1	W_2	W_3	*Supply*
F_1	16 (140)	20	12 (60)	200
F_2	14 (40)	8 (120)	18	160
F_3	26	24	16 (90)	90
Demand	180	120	150	

Table 9.17
Revised Schedule

The total transportation cost associated with this solution is

Total cost = 16 × 140 + 12 × 60 + 14 × 40 + 8 × 120 + 16 × 90 = Rs 5,920

To test the optimality of the new solution shown in Table 9.17, calculate again the opportunity cost of each unoccupied cell in the same manner as discussed earlier. The calculations are shown in Table 9.18.

	W_1	W_2	W_3	*Supply*	u_i
F_1	16 (140)	20 +10	12 (60)	200	$u_1 = 16$
F_2	14 (40)	8 (120)	18 +8	160	$u_2 = 14$
F_3	26 +6	24 +10	16 (90)	90	$u_3 = 20$
Demand	180	120	150		
v_j	$v_1 = 0$	$v_2 = -6$	$v_3 = -4$		

Table 9.18

$d_{12} = c_{12} - (u_1 + v_2)$ or $20 - (16 - 6) = 10$
$d_{23} = c_{23} - (u_2 + v_3)$ or $18 - (14 - 4) = 8$
$d_{31} = c_{31} - (u_3 + v_1)$ or $26 - (20 + 0) = 6$
$d_{32} = c_{32} - (u_3 + v_2)$ or $24 - (20 - 6) = 10$

Since none of the unoccupied cells in Table 9.18 has a negative opportunity cost value, therefore, the total transportation cost cannot be reduced further. Thus, the solution shown in Table 9.18 is the optimal solution, giving the optimal transportation schedule with a total cost of Rs 5,920.

Example 9.8 The following table provides all the necessary information on the availability of supply to each warehouse, the requirement of each market, and the unit transportation cost (in Rs) from each warehouse to each market.

Market

Warehouse	*P*	*Q*	*R*	*S*	*Supply*
A	6	3	5	4	22
B	5	9	2	7	15
C	5	7	8	6	8
Demand	7	12	17	9	45

The shipping clerk of the shipping agency has worked out the following schedule, based on his own experience: 12 units from A to Q, 1 unit from A to R, 9 units from A to S, 15 units from B to R, 7 units from C to P and 1 unit from C to R.

(a) Check and see if the clerk has the optimal schedule.

(b) Find the optimal schedule and minimum total transport cost.

(c) If the clerk is approached by a carrier of route C to Q, who offers to reduce his rate in the hope of getting some business, by how much should the rate be reduced before the clerk would offer him the business.

Solution (a) The shipping schedule determined by the clerk based on his experience is shown in Table 9.19. The total transportation cost associated with this solution is

$$\text{Total cost} = 3 \times 12 + 5 \times 1 + 4 \times 9 + 2 \times 15 + 5 \times 7 + 8 \times 1 = \text{Rs } 150$$

Since the number of occupied cells (i.e. 6) is equal to the required number of occupied cells (i.e. $m + n - 1$) in a feasible solution, therefore the solution is non-generate feasible solution. Now, to test the optimality of the solution given in Table 9.19, evaluate each unoccupied cell in terms of the opportunity cost associated with it. This is done in the usual manner and is shown in Table 9.20.

	P	*Q*	*R*	*S*	*Supply*
A	6	3 (12)	5 (1)	4 (9)	22
B	5	9	2 (15)	7	15
C	5 (7)	7	8 (1)	6	8
Demand	7	12	17	9	45

Table 9.19
Initial Solution

In Table 9.20, cell (C, S) has a negative opportunity cost (i.e. – 1). Thus, this solution is not the optimal solution and, therefore, the schedule prepared by the shipping clerk is not optimal.

(b) By forming a closed-loop to introduce the cell (C, S) into the new transportation schedule as shown in Table 9.20, we get a new solution that is shown in Table 9.21.

	P	*Q*	*R*	*S*	*Supply*	u_i
A	6 +4	3 (12)	5 (+) (1)	4 (9) (–)	22	$u_1 = 0$
B	5	9 +9	2 (15)	7 +6	15	$u_2 = -3$
C	5 (7)	7 +1	8 (–) (1)	6 (+) –1	8	$u_3 = 3$
Demand	7	12	17	9	45	
v_j	$v_1 = 2$	$v_2 = 3$	$v_3 = 5$	$v_4 = 4$		

Table 9.20

While testing the optimality of the improved solution shown in Table 9.21, we found that the opportunity costs in all the unoccupied cells are positive. Thus the current solution is optimal and the optimal schedule is to transport 12 units from A to Q; 2 units from A to R; 8 units from A to S; 15 units from B to R; 7 units from C to P and 1 unit from *C* to *S*. The total minimum transportation cost associated with this solution is

Total cost = $3 \times 12 + 5 \times 2 + 4 \times 8 + 2 \times 15 + 5 \times 7 + 6 \times 1$ = Rs 149

	P	*Q*	*R*	*S*	*Supply*	u_i
A	6 +3	3 (12)	5 (2)	4 (8)	22	$u_1 = 0$
B	5 +5	9 +9	2 (15)	7 +6	15	$u_2 = -3$
C	5 (7)	7 +2	8 +1	6 (1)	8	$u_3 = 2$
Demand	7	12	17	9	45	
v_j	$v_1 = 3$	$v_2 = 3$	$v_3 = 5$	$v_4 = 4$		

Table 9.21
Optimal Solution

(c) The total transportation cost will increase by Rs 2 (opportunity cost) if one unit of commodity is transported from C to Q. This means that the rate of the carrier on the route C to Q should be reduced by Rs 2, i.e. from Rs 7 to Rs 5 so as to get some business of one unit of commodity only.

In case all the 8 units available at C are shipped through the route (C, Q), then the solution presented in Table 9.21 may be read as shown in Table 9.22.

	P	*Q*	*R*	*S*	*Supply*
A	6 (7)	3 (4)	5 (2)	4 (9)	22
B	5	9	2 (15)	7	15
C	5	7 (8)	8	6	8
Demand	7	12	17	9	45

Table 9.22

The total cost of transportation associated with this solution is

Total cost = 6 × 7 + 3 × 4 + 5 × 2 + 4 × 9 + 2 × 15 + 7 × 8 = Rs 186.

Thus, the additional cost of Rs 37 (= 186 – 149) should be reduced from the transportation cost of 8 units from C to Q. Hence transportation cost per unit from C to Q should be at the most 7 – (37/8) = Rs 2.38.

Example 9.10 ABC Limited has three production shops that supply a product to five warehouses. The cost of production varies from shop to shop and cost of transportation from one shop to a warehouse also varies. Each shop has a specific production capacity and each warehouse has certain amount of requirement. The costs of transportation are given below:

		Warehouse					
		I	*II*	*III*	*IV*	*V*	*Supply*
Shop	*A*	6	4	4	7	5	100
	B	5	6	7	4	8	125
	C	3	4	6	3	4	175
	Demand	60	80	85	105	70	400

The cost of manufacturing the product at different production shops is

Shop	*Variable Cost*	*Fixed Cost*
A	14	7,000
B	16	4,000
C	15	5,000

Find the optimum quantity to be supplied from each shop to different warehouses at the minimum total cost. [*Delhi Univ., MBA, 2007*]

Solution In this case, the fixed cost data is of no use. The transportation cost matrix will include the given transportation cost plus the variable cost, as shown in Table 9.23.

	I	*II*	*III*	*IV*	*V*	*Supply*
A	6 + 14 = 20	4 + 14 = 18	4 + 14 = 18	7 + 14 = 21	5 + 14 = 19	100
B	5 + 16 = 21	6 + 16 = 22	7 + 16 = 23	4 + 16 = 20	8 + 16 = 24	125
C	3 + 15 = 18	4 + 15 = 19	6 + 15 = 21	3 + 15 = 18	4 + 15 = 19	175
Demand	60	80	85	105	70	400

Table 9.23

The optimal solution obtained by applying MODI method is shown in Table 9.24.

	I	*II*	*III*	*IV*	*V*	*Supply*	u_i
A	20 +3	18 (15)	18 (85)	21 +5	19 +1	100	$u_1 = 18$
B	21 0	22 (20)	23 +1	20 (105)	24 +2	125	$u_2 = 22$
C	18 (60)	19 (45)	21 +2	18 +1	19 (70)	175	$u_3 = 19$
Demand	60	80	85	105	70		
v_j	$v_1 = -1$	$v_2 = 0$	$v_3 = 0$	$v_4 = -2$	$v_5 = 0$		

Table 9.24
Optimal Solution

The transportation cost associated with the solution is

Total cost = 18 × 15 + 18 × 85 + 22 × 20 + 20 × 105 + 18 × 60 + 19 × 45 + 19 × 70 = Rs 7,605

CONCEPTUAL QUESTIONS B

1. Describe the computational procedure of the optimality test in a transportation problem.
2. Indicate how you will test for optimality of initial feasible solution of a transportation problem.
3. Let S_i and D_j (i = 1, 2, . . ., m; j = 1, 2, . . ., n) be the supply and demand available, respectively, for a commodity at m godowns and n markets. Let c_{ij} be the cost of transporting one unit of the commodity from godown i to market j. Assuming that

$$\sum_{i=1}^{m} S_i = \sum_{j=1}^{n} D_j$$

Symbolically state the transportation problem. Establish that the optimal solution is not altered when the c_{ij}'s are replaced by c_{ij}^*s, where $c_{ij}^* = c_{ij} + u_i + v_j$, u_i (i = 1, 2, . . ., m) and v_j (j = 1, 2, . . ., n) are arbitrary real numbers.

SELF PRACTICE PROBLEMS B

1. Consider four bases of operation B_i and three targets T_j. The tons of bombs per aircraft from any base that can be delivered to any target are given in the following table:

Target (T_j)

Base (B_i)	T_1	T_2	T_3
B_1	8	6	5
B_2	6	6	6
B_3	10	8	4
B_4	8	6	4

The daily sortie capability of each of the four bases is 150 sorties per day. The daily requirement of sorties spread over each individual target is 200. Find the allocation of sorties from each base to each target which maximizes the total tonnage over all three targets. Explain each step in the process.

2. A company has four warehouses, a, b, c and d. It is required to deliver a product from these warehouses to three customers A, B and C. The warehouses have the following amounts in stock:

Warehouse : a b c d
No. of units : 15 16 12 13

and the customers' requirements are

Customer : A B C
No. of units : 18 20 18

The table below shows the costs of transporting one unit from warehouse to the customer.

Warehouse

Customer	*a*	*b*	*c*	*d*
A	8	9	6	3
B	6	11	5	10
C	3	8	7	9

Find the optimal transportation routes.

3. A firm manufacturing a single product has three plants I, II and III. They have produced 60, 35 and 40 units, respectively during this month. The firm had made a commitment to sell 22 units to customer A, 45 units to customer B, 20 units to customer C, 18 units to customer D and 30 units to customer E. Find the minimum possible transportation cost of shifting the manufactured product to the five customers. The net unit cost of transporting from the three plants to the five customers is given below:

Customers

Plants	*A*	*B*	*C*	*D*	*E*
I	4	1	3	4	4
II	2	3	2	2	3
III	3	5	2	4	4

4. The following table gives the cost of transporting material from supply points A, B, C and D to demand points E, F, G, H and I.

To

From	*E*	*F*	*G*	*H*	*I*
A	8	10	12	17	15
B	15	13	18	11	9
C	14	20	6	10	13
D	13	19	7	5	12

The present allocation is as follows:
A to E 90; A to F 10; B to F 150; C to F 10; C to G 50; C to I 120; D to H 210; D to I 70.

(a) Check if this allocation is optimum. If not, find an optimum schedule.
(b) If in the above problem, the transportation cost from A to G is reduced to 10, what will be the new optimum schedule?

5. A whole selling company has three warehouses from which the supplies are drawn for four retail customers. The company deals in a single product, the supplies of which at each warehouse are:

Warehouse Number	*Supply* (*units*)	*Customer Number*	*Demand* (*units*)
1	20	1	15
2	28	2	19
3	17	3	13
		4	18

Total supply at the warehouses is equal to total demand from the customers. The table below gives the transportation costs, per unit, shipped from each warehouse to each customer.

Customer

Warehouse	C_1	C_2	C_3	C_4
W_1	3	6	8	5
W_2	6	1	2	5
W_3	7	8	3	9

Determine what supplies should be dispatched from each of the warehouses to each customer so as to minimize the overall transportation cost.

6. A manufacturer has distribution centres at Agra, Allahabad and Kolkata. These centres have availability of 40, 20 and 40 units of his product, respectively. His retail outlets at A, B, C, D and E require 25, 10, 20, 30 and 15 units of the products, respectively. The transport cost (in rupees) per unit between each centre outlet is given below:

Distribution Centre	*Retail Outlets*				
	A	*B*	*C*	*D*	*E*
Agra	55	30	40	50	40
Allahabad	35	30	100	45	60
Kolkata	40	60	95	35	30

Determine the optimal distribution so as to minimize the cost of transportation. [*Delhi Univ., MBA, 2002*]

7. A manufacturer has distribution centres located at Agra, Allahabad and Kolkata. These centres have available 40, 20 and 40 units of his product. His retail outlets at A, B, C, D and E requires 25, 10, 20, 30 and 15 units of the product, respectively. The shipping cost per unit (in rupees) between each centre and outlet is given in the following table.

Distribution Centre	*Retail Outlets*				
	A	*B*	*C*	*D*	*E*
Agra	55	30	40	50	40
Allahabad	35	30	100	45	60
Kolkata	40	60	95	35	30

Determine the optimal shipping cost.
[*Delhi Univ. MBA, 2003*]

8. A steel company is concerned with the problem of distributing imported ore from three ports to four steel mills. The supplies of ore arriving at the ports are:

Port	*Tonnes per week*
a	20,000
b	38,000
c	16,000

The demand at the steel mills is as follows:

Steel mills	:	A	B	C	D
Tonnes per week	:	10,000	18,000	22,000	24,000

The transportation cost is Re 0.05 per tonne per km. The distance between the ports and the steel mills is as given below:

	A	*B*	*C*	*D*
a	50	60	100	50
b	80	40	70	50
c	90	70	30	50

Calculate a transportation plan that will minimize the distribution cost for the steel company. State the cost of this distribution plan.

9. A company has three factories at Amethi, Baghpat and Gwalior that have a production capacity of 5,000, 6,000, and 2,500 tonnes, respectively. Four distribution centres at Allahabad, Bombay, Kolkata and Delhi, require 6,000 tonnes, 4,000 tonnes, 2,000 tonnes and 1,500 tonnes, respectively, of the product. The transportation costs per tonne from different factories to different centres are given below:

Factories	*Distribution Centres*			
	Allahabad	*Bombay*	*Kolkata*	*Delhi*
Amethi	3	2	7	6
Baghpat	7	5	2	3
Gwalior	2	5	4	5

Suggest an optimum transportation schedule and find the minimum cost of transportation.

10. A company has three plants and four warehouses. The supply and demand in units and the corresponding transportation costs are given. The table below has been taken from the solution procedure of a transportation problem:

Warehouses

Plants	*I*	*II*	*III*	*IV*	*Supply*
A	5	10	4 (10)	5	10
B	6 (20)	8	7	2 (5)	25
C	4 (5)	2 (10)	5 (5)	7	20
Demand	25	10	15	5	55

Answer the following questions, giving brief reasons for the same:

(a) Is this solution feasible?
(b) Is this solution degenerate?
(c) Is this solution optimum?
(d) Does this problem have more than one optimum solution? If so, show all of them.
(e) If the cost for the route B-III is reduced from Rs 7 to Rs 6 per unit, what will be the optimum solution?

11. A baking firm can produce a special bread in either of its two plants, the details of which are as follows:

Plant	*Production Capacity Loaves*	*Production Cost Rs/Loaf*
A	2,500	2.30
B	2,100	2.50

Four restaurant chains are willing to purchase this bread; their demand and the prices they are willing to pay are as follows:

Chain	*Maximum Demand Loaves*	*Price Offered Rs/Loaf*
1	1,800	3.90
2	2,300	3.70
3	550	4.00
4	1,750	3.60

The cost (in paise) of shipping a loaf from a plant to a restaurant chain is:

	Chain 1	*Chain 2*	*Chain 3*	*Chain 4*
Plant A	6	8	11	9
Plant B	12	6	8	5

Determine a delivery schedule for the baking firm that will maximize its profit from this bread.

Write the dual of this transportation problem and use it for checking the optimal solution to the given problem.

[*Delhi Univ., MBA, 2005*]

HINTS AND ANSWERS

1. The initial solution obtained by VAM is also the optimal solution: $x_{11} = 50$, $x_{12} = 100$, $x_{21} = 150$, $x_{33} = 150$, $x_{42} = 100$, $x_{43} = 50$. Maximum total tonnage = 3,300.
2. (A, b) = 5, (A, d) = 13, (B, b) = 8, (B, c) = 12, (C, a) = 15 and (C, b) = 3, Total cost = Rs 301.
3. (I, B) = 45, (I, F) = 15, (II, A) = 17, (II, D) = 18, (III, A) = 5, (III, C) = 20 and (III, E) = 15, Total cost = Rs 290.
4. (a) (A, F) = 100, (B, F) = 70, (B, I) = 80, (C, E) = 90, (C, G) = 50, (C, I) = 40, (D, H) = 210, (D, I) = 70, Total cost = Rs 6,600.
   (b) When transportation cost from A to G is reduced to 10, the optimal schedule given in (a) remains unchanged.
5. $x_{11} = 15$, $x_{14} = 5$, $x_{22} = 19$, $x_{24} = 9$, $x_{33} = 13$, $x_{34} = 4$, Total cost = Rs 209.
6. $x_{11} = 5$, $x_{12} = 10$, $x_{13} = 20$, $x_{14} = 5$, $x_{21} = 20$, $x_{34} = 25$, $x_{35} = 15$, Total cost = Rs 3,650.
7. $x_{12} = 2$, $x_{13} = 4$, $x_{15} = 2$, $x_{21} = 4$, $x_{31} = 1$, $x_{34} = 6$, $x_{35} = 1$, Total cost = Rs 720.
8. $x_{11} = 10{,}000$, $x_{14} = 10{,}000$, $x_{22} = 18{,}000$, $x_{23} = 6{,}000$, $x_{24} = 14{,}000$, $x_{33} = 16{,}000$, Total cost = Rs 1,66,000.
9. $x_{11} = 3{,}500$, $x_{12} = 1{,}500$, $x_{22} = 2{,}500$, $x_{23} = 2{,}000$, $x_{24} = 1{,}500$, $x_{31} = 2{,}500$, Total cost = Rs 39,500.
10. (a) The solution is feasible because it satisfies supply and demand constraints.
    (b) The solution is non-degenerate because the number of occupied cells are equal to the required number of $(m + n - 1)$ of occupied cells in the solution.
    (c) Solution is optimal.
    (d) The problem has alternative optimal solution because opportunity cost for cell (B, III) is zero; $x_{11} = 10$, $x_{21} = 15$, $x_{23} = 5$, $x_{24} = 5$, $x_{31} = 10$ and $x_{32} = 10$, Total cost = Rs 235.
    (e) If cell (B, III) has a unit cost of 6, the opportunity cost in this cell will be – 1 and, therefore, the given solution will not be optimal. The new solution obtained will be: $x_{13} = 10$, $x_{21} = 15$, $x_{23} = 5$, $x_{24} = 5$, $x_{31} = 10$, $x_{32} = 10$, Total cost = Rs 230.

9.6 VARIATIONS IN TRANSPORTATION PROBLEM

Some of the variations that often arise while solving a transportation problem are as follows.

9.6.1 Unbalanced Supply and Demand

For a feasible solution to exist, it is necessary that the total supply must equal the total demand. That is,

$$\text{Total supply} = \text{Total demand}$$

$$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$$

A **dummy source** or destination is added to balance transportation problem where demand is not equal to supply.

But a situation may arise when the total available supply is not equal to the total demand [For proof see appendix]. The following two cases may arise:

(a) If the total supply exceeds the total demand, then an additional column (called a *dummy demand centre*) can be added to the transportation table in order to absorb the excess supply. The unit transportation cost for the cells in this column is set equal to zero because these represent product items that are neither made nor sent.

(b) If the total demand exceeds the total supply, a dummy row (called a *dummy supply centre*) can be added to the transportation table to account for the excess demand quality. The unit transportation cost in such a case also, for the cells in the dummy row is set equal to zero.

Example 9.10 A company has received a contract to supply gravel to three new construction projects located in towns A, B and C. The construction engineers have estimated that the required amounts of gravel which will be needed at these construction projects are:

Project Location	*Weekly Requirement (Truckloads)*
A	72
B	102
C	41

The company has 3 gravel pits located in towns X, Y and Z. The gravel required by the construction projects can be supplied by three pits. The amount of gravel that can be supplied by each pit is as follows:

Plant	:	X	Y	Z
Amount available (truckloads)	:	76	82	77

The company has computed the delivery cost from each pit to each project site. These costs (in Rs) are shown in the following table:

		Project Location		
		A	*B*	*C*
Pit	*X*	4	8	8
	Y	16	24	16
	Z	8	16	24

Schedule the shipment from each pit to each project in such a manner that it minimizes the total transportation cost within the constraints imposed by pit capacities and project requirements. Also find the minimum cost.

Solution The total plant availability of 235 truckloads exceeds the total requirement of 215 truckloads by 20 truckloads. The excess truckload capacity of 20 is handled by adding a dummy project location (column), D_{excess} with a requirement of 20 truck loads, Assigning unit transportation costs to the dummy project location, modified transportation table is shown in Table 9.25.

	A	*B*	*C*	D_{excess}	*Supply*
W	4	8 (35)	8 (41)	0	76
X	16	24 (62)	16	0 (20)	82
Y	8 (72)	16 (5)	24	0	77
Demand	72	102	41	20	235

Table 9.25 Initial Solution

The initial solution is obtained by using Vogel's approximation method as shown in Table 9.25. It may be noted that 20 units are allocated from pit X to dummy project location D. This means pit X is short by 20 units.

To test the optimality of the initial solution shown in Table 9.25, calculate u_is and v_js corresponding to rows and columns respectively. These values are shown in Table 9.26.

	A	*B*	*C*	D_{excess}	*Supply*	u_i
W	4 +4	8 (+) (35)	8 (41) (–)	0 +16	76	$u_1 = 8$
X	16	24 (–) (62)	16 (+) –8	0 (20)	82	$u_2 = 24$
Y	8 (72)	16 (5)	24 +8	0 +8	77	$u_3 = 16$
Demand	72	102	41	20	235	
v_j	$v_1 = -8$	$v_2 = 0$	$v_3 = 0$	$v_4 = -24$		

Table 9.26

In Table 9.26, opportunity cost shown in the cell (X, C) is negative, the current solution is not optimal. Thus, the unoccupied cell (X, C), where $d_{23} = -8$ must enter into the basis and cell (W, C) must leave the basis, as shown by the closed path in Table 9.26. The new solution is shown in Table 9.27.

	A	B	C	D_{excess}	*Supply*	u_i
W	4	8	8	0		$u_1 = -16$
	+4	(76)	+8	+16	76	
X	16	24	16	0		$u_2 = 0$
	0	(21)	(41)	(20)	82	
Y	8	16	24	0		$u_3 = -8$
	(72)	(5)	+16	+8	77	
Demand	72	102	41	20	235	
v_j	$v_1 = 16$	$v_2 = 24$	$v_3 = 16$	$v_4 = 0$		

Table 9.27
Optimal Solution

Since all opportunity costs, d_{ij} are non-negative in Table 9.27, the current solution is optimal. The total minimum transportation cost associated with this solution is:

Total cost = 8 × 76 + 24 × 21 + 16 × 41 + 0 × 20 + 8 × 72 + 16 × 5 = Rs 2,424.

Example 9.11 A product is manufactured at four factories A, B, C and D. Their unit production costs are Rs 2, Rs 3, Re 1 and Rs 5, respectively. Their production capacities are 50, 70, 30 and 50 units, respectively. These factories supply the product to four stores, demands of which are 25, 35, 105 and 20 units respectively. Unit transportation cost in rupees from each factory to each store is given in the table below.

		Stores			
		I	*II*	*III*	*IV*
Factories	*A*	2	4	6	11
	B	10	8	7	5
	C	13	3	9	12
	D	4	6	8	3

Determine the extent of deliveries from each of the factories to each of the stores, so that the total production and transportation cost is the minimum.

Solution The new transportation costs that include both the production and the transportation costs is given in Table 9.28.

	I	*II*	*III*	*IV*	*Supply*
A	2 + 2 = 4	4 + 2 = 6	6 + 2 = 8	11 + 2 = 13	50
B	10 + 3 = 13	8 + 3 = 11	7 + 3 = 10	5 + 3 = 8	70
C	13 + 1 = 14	3 + 1 = 4	9 + 1 = 10	12 + 1 = 13	30
D	4 + 5 = 9	6 + 5 = 11	8 + 5 = 13	3 + 5 = 8	50
Demand	25	35	105	20	200 / 185

Table 9.28

Since the total supply of 200 units exceeds the total demand of 185 units by 15 units, a dummy destination (store) is added (or created) to absorb the excess capacity. The associated cost coefficients in dummy store are taken as zero. This may be due to the reason that the surplus quantity remains lying in the respective factories and is not shipped at all. The modified table is shown in Table 9.29.

	I	*II*	*III*	*IV*	*Dummy*	*Supply*
A	4 (25)	6 (5)	8 (20)	13	0	50
B	13	11	10 (70)	8	0	70
C	14	4 (30)	10	13	0	30
D	9	11	13 (15)	8 (20)	0 (15)	50
Demand	25	35	105	20	15	200

Table 9.29 Initial Solution

Using the VAM method the initial solution is shown in Table 9.29. It can be seen that 15 units are allocated to dummy store from factory D. This means that the company may cut down the production by 15 units at the factory that is proving to be uneconomical. Now to test the optimality of the solution shown in Table 9.29, evaluate each unoccupied cell in terms of opportunity cost associated with it as shown in Table 9.30.

	I	*II*	*III*	*IV*	*Dummy*	*Supply*	u_i
A	4 (25)	6 (5)	8 (20)	13 +10	0 +5	50	$u_1 = -5$
B	13 +7	11 +3	10 (70)	8 +3	0 +3	70	$u_2 = -3$
C	14 +12	4 (30)	10 +4	13 +12	0 +7	30	$u_3 = -7$
D	9 0	11 0	13 (15)	8 (20)	0 (15)	50	$u_4 = 0$
Demand	25	35	105	20	15	200	
v_j	$v_1 = 9$	$v_2 = 11$	$v_3 = 13$	$v_4 = 8$	$v_5 = 0$		

Table 9.30

Since the opportunity cost in all the unoccupied cells is positive, the initial solution is an optimal solution. The total cost of transportation associated with this solution is

Total cost = 4 × 25 + 6 × 5 + 8 × 20 + 10 × 70 + 4 × 30 + 13 × l5 + 8 × 20 + 0 × l5 = Rs l,465.

9.6.2 Degeneracy and its Resolution

A basic feasible solution for the general transportation problem must have exactly $m + n - 1$ (number of rows + number of columns – 1) positive allocations in the transportation table. If the number of occupied cells is less than the required number, $m + n - 1$, then such a solution is called degenerate solution. In such cases, the current solution cannot be improved further because it is not possible to draw a closed path for every occupied cell. Also, the values of dual variables u_i and v_j that are used to test the optimality cannot be computed. Thus, degeneracy needs to be removed in order to improve the given solution. The degeneracy in the transportation problems may occur at two stages:

Degeneracy arises when the number of occupied cells are less than the number of rows + columns – 1.

(a) At initial basic feasible solution the number of occupied cells may be less than $m + n - 1$ allocations.
(b) At any stage while moving towards optimal solution two or more occupied cells may become simultaneously unoccupied.

Case 1: Degeneracy at the initial solution To resolve degeneracy at the initial solution, allocate a very small quantity close to zero to one or more (if needed) unoccupied cells so as to get $m + n - 1$ number of occupied cells. This amount is denoted by a Greek letter ε (epsilon) or Δ (delta). This quantity

would neither affect the total cost nor the supply and demand values. In a minimization transportation problem it is better to allocate Δ to unoccupied cells that have lowest transportation costs, whereas in maximization problems it should be allocated to a cell that has a high payoff value. In some cases, Δ must be added in one of those unoccupied cells that may help in the determination of u_i and v_j values.

The quantity Δ is considered to be so small that if it is transferred to an occupied cell it does not change the quantity of allocation. That is,

$$x_{ij} + \Delta = x_{ij} \qquad - \Delta = x_{ij}$$
$$\Delta - \Delta = 0; \qquad \Delta + \Delta = \Delta$$
$$0 + \Delta = \Delta; \qquad k \times \Delta = \Delta$$

The Δ value also does not affect the total transportation cost. Hence, the quantity Δ is used to evaluate unoccupied cells and to reduce the number of improvement cycles necessary to reach an optimal solution. Once the purpose is over, Δ can be removed from the transportation table.

Example 9.12 A manufacturer wants to ship 22 loads of his product as shown below. The matrix gives the kilometres from sources of supply to the destinations.

Destination

		D_1	D_2	D_3	D_4	D_5	*Supply*
Source	S_1	5	8	6	6	3	8
	S_2	4	7	7	6	5	5
	S_3	8	4	6	6	4	9
	Demand	4	4	5	4	8	25 / 22

The shipping cost is Rs 10 per load per km. What shipping schedule should be used in order to minimize the total transportation cost? [*Delhi Univ., MBA, 2001*]

Solution Since the total destination requirement of 25 units exceeds the total resource capacity of 22 by 3 units, the problem is unbalanced. The excess requirement is handled by adding a dummy plant, S_{excess} with a capacity equal to 3 units. Assign zero unit transportation cost to the dummy plant, the modified transportation table is shown in Table 9.31.

	D_1	D_2	D_3	D_4	D_5	*Supply*
S_1	5	8	6 (5)	6	3 (3)	8
S_2	4 (4)	7	7	6 (1)	5	5
S_3	8	4 (4)	6	6	4 (5)	9
S_{excess}	0	0	0	0 (3)	0	3
Demand	4	4	5	4	8	25

Table 9.31 Initial Solution, but Degenerate

The initial solution is obtained by using Vogel's approximation method as shown in Table 9.31. Since solution does not have required number of $m + n - 1 = 4 + 5 - 1 = 8$ occupied cells, therefore, the initial solution is degenerate. In order to remove degeneracy, assign Δ to unoccupied cell (S_2, D_5), which has the minimum cost amongst the unoccupied cells, as shown in Table 9.32.

	D_1	D_2	D_3	D_4	D_5	*Supply*	u_i
S_1	5 +3	8 +5	6 (–) (5)	6 +2	3 (3) (+)	8	$u_1 = 0$
S_2	4 (4)	7 +2	7 – 1	6 (1) (+)	5 (Δ) (–)	5	$u_2 = 2$
S_3	8 +5	4 (4)	6 – 1	6 +1	4 (5)	9	$u_3 = 1$
S_{excess}	0 +2	0 +1	0 (+) – 2	0 (3) (–)	0 +1	3	$u_4 = -4$ 1
Demand	4	4	5	4	8	25	
v_j	$v_1 = 2$	$v_2 = 3$	$v_3 = 6$	$v_4 = 4$	$v_5 = 3$		

Table 9.32

Determine u_i and v_j for occupied cells as shown in Table 9.32. Since opportunity cost in the cell (S_{excess}, D_3) is largest negative, it is entered into the new solution mix and the cell (S_2, D_5) leaves the current solution mix. The new solution is shown in Table 9.33.

	D_1	D_2	D_3	D_4	D_5	*Supply*	u_i
S_1	5 +1	8 +5	6 (–) (5)	6 0	3 (3) (+)	8	$u_1 = 0$
S_2	4 (4)	7 +4	7 +1	6 (1)	5 +2	5	$u_2 = 0$
S_3	8 +3	4 (4)	6 –1	6 (+) – 1	4 (5) (–)	9	$u_3 = 1$
S_{excess}	0 +2	0 +3	0 (+) (Δ)	0 (3) (–)	0 +3	3	$u_4 = -6$
Demand	4	4	5	4	8	25	
v_j	$v_1 = 4$	$v_2 = 3$	$v_3 = 6$	$v_4 = 6$	$v_5 = 3$		

Table 9.33

Repeat the procedure of testing optimality of the solution given in Table 9.33. The optimal solution is shown in Table 9.34.

	D_1	D_2	D_3	D_4	D_5	*Supply*
S_1	5	8	6	6	3 (8)	8
S_2	4 (4)	7	7	6 (1)	5	5
S_3	8	4 (4)	6 (2)	6 (3)	4	9
S_{excess}	0	0	0 (3)	0	0	3
Demand	4	4	5	4	8	25

Table 9.34
Optimal Solution

The minimum total transportation cost associated with this solution is

$$\text{Total cost} = (3 \times 8 + 4 \times 4 + 6 \times 1 + 4 \times 4 + 6 \times 2 + 6 \times 3) \times 10 = \text{Rs } 920$$

Case 2: Degeneracy at subsequent iterations To resolve degeneracy, which occurs during optimality test, the quantity may be allocated to one or more cells that have recently been unoccupied, to have $m + n - 1$ number of occupied cells in the new solution.

Example 9.13 Goods have to be transported from sources S_1, S_2 and S_3 to destinations D_1, D_2 and D_3. The transportation cost per unit, capacities of the sources, and the requirements of the destinations are given in the following table.

	D_1	D_2	D_3	*Supply*
S_1	8	5	6	120
S_2	15	10	12	80
S_3	3	9	10	80
Demand	150	80	50	

Determine a transportation schedule so that cost is minimized.

Solution Using North-West Corner Method, the non-degenerate initial basic feasible solution so obtained is given in Table 9.35.

	D_1	D_2	D_3	*Supply*
S_1	8 (120)	5	6	120
S_2	15 (30)	10 (50)	12	80
S_3	3	9 (30)	10 (50)	80
Demand	150	80	50	280

Table 9.35
Initial Solution

To test the optimality of the solution given in Table 9.35, calculate u_i, v_j and d_{ij} as usual as shown in Table 9.36.

	D_1	D_2	D_3	*Supply*	u_i
S_1	8 (120)	5 +2	6 +2	120	$u_1 = -7$
S_2	15 (–) (30)	10 (50) (+)	12 +1	80	$u_1 = 0$
S_3	3 (+) –11	9 (30) (–)	10 (50)	80	$u_1 = -1$
Demand	150	80	50	280	
v_j	$v_1 = 15$	$v_2 = 10$	$v_3 = 11$		

Table 9.36

Since the unoccupied cell (S_3, D_1) has the largest negative opportunity cost of – 11, therefore, cell (S_3, D_1) is entered into the new solution mix. The closed path for (S_3, D_1) is shown in Table 9.36. The maximum allocation to (S_3, D_1) is 30. However, when this amount is allocated to (S_3, D_1) both cells (S_2, D_1) and (S_3, D_2) become unoccupied because these two have same allocations. Thus, the number of positive allocations become less than the required number, $m + n - 1 = 3 + 3 - 1 = 5$. Hence, this is a degenerate solution as shown in Table 9.37.

	D_1	D_2	D_3	*Supply*
S_1	8 (120)	5	6	120
S_2	15	10 (80)	12	80
S_3	3 (30)	9 (Δ)	10 (50)	80
Demand	150	80	50	280

Table 9.37 Degenerate Solution

To remove degeneracy a quantity Δ is assigned to one of the cells that has become unoccupied so that there are $m + n - 1$ occupied cells. Assign Δ to (S_3, D_2) and proceed with the usual solution procedure. The optimal solution is given in Table 9.38.

	D_1	D_2	D_3	*Supply*
S_1	8 (70)	5 (Δ)	6 (50)	120
S_2	15	10 (80)	12	80
S_3	3 (80)	9	10	80
Demand	150	80	50	280

Table 9.38 Optimal Solution

The total minimum transportation cost associated with this solution is:

Total cost = 8 × 70 + 6 × 50 + 10 × 80 + 3 × 80 = Rs 1,900

9.6.3 Alternative Optimal Solutions

The existence of alternative optimal solutions can be determined by an inspection of the opportunity costs, d_{ij} for the unoccupied cells. If $d_{ij} = 0$, for an unoccupied cell in an optimal solution, then an alternative optimal solution exists and can be obtained by bringing such an unoccupied cell in the solution mix without increasing the total transportation cost.

Illustration Consider the optimal solution of Example 9.10 given in Table 9.27. For ready reference Table 9.27 is reproduced as Table 9.39.

	A	*B*	*C*	D_{excess}	*Supply*	u_i
W	4 +4	8 (76)	8 + 8	0 + 16	76	$u_1 = -16$
X	16 0	24 (21)	16 (41)	0 (20)	82	$u_2 = 0$
Y	8 (72)	16 (5)	24 + 16	0 +8	77	$u_3 = -8$
Demand	72	102	41	20	235	
v_j	$v_1 = 16$	$v_2 = 24$	$v_3 = 16$	$v_4 = 0$		

Table 9.39

The opportunity costs in all unoccupied cells are positive except for the cell (X, A) where $d_{21} = 0$. This means, if (X, A) is entered into the new solution mix, no change in the transportation cost would occur. To determine this alternative solution, form a closed path for cell (X, A) as shown in Table 9.40.

Table 9.40 Optimal Solution

	A	*B*	*C*	D_{excess}	*Supply*	u_i
W	4 +4	8 (76)	8 + 8	0 +16	76	$u_1 = -16$
X	16 (+) 0	24 (21) (–)	16 (41)	0 (20)	82	$u_2 = 0$
Y	8 (–) (72)	16 (5) (+)	24 + 16	0 + 8	77	$u_3 = -8$
Demand	72	102	41	20	235	
v_j	$v_1 = 16$	$v_2 = 24$	$v_3 = 16$	$v_4 = 0$		

The maximum quantity that can be allocated to cell (*X*, *A*) is 21. After this change, the new solution is shown in Table 9.41.

Since all d_{ij} values are positive or zero, the solution given in Table 9.41 is optimal with a minimum total transportation cost of Rs 2,424, which is same as in the previous solution.

Table 9.41 Alternative Optimal Solution

	A	*B*	*C*	D_{excess}	*Supply*	u_i
W	4 + 4	8 (76)	8 + 8	0 + 16	76	$u_1 = -16$
X	16 (21)	24 0	16 (41)	0 (20)	82	$u_2 = -0$
Y	8 (51)	16 (26)	24 + 16	0 + 8	77	$u_3 = -8$
Demand	72	102	41	20	235	
v_j	$v_1 = 16$	$v_2 = 24$	$v_3 = 16$	$v_4 = 0$		

Example 9.14 XYZ tobacco company purchases tobacco and stores in warehouses located in the following four cities:

Warehouse Location (City) :	A	B	C	D
Capacity (Tonnes) :	90	50	80	60

The warehouses supply tobacco to cigarette companies in three cities that have the following demand:

Cigarette Company	*Demand (Tonnes)*
Bharat	120
Janata	100
Red Lamp	110

The following railroad shipping costs per tonne (in hundred rupees) have been determined:

Warehouse Location	*Bharat*	*Janata*	*Red Lamp*
A	7	10	5
B	12	9	4
C	7	3	11
D	9	5	7

Because of railroad construction, shipments are temporarily prohibited from warehouse at city A to Bharat Cigarette company.

(a) Find the optimum distribution for XYZ tobacco company.
(b) Are there multiple optimum solutions? If yes, identify them.
(c) Write the dual of the given transportation problem and use it for checking the optimum solution.

[*Delhi Univ., MBA, 2003, 2005, 2008*]

Solution Since the total demand of 330 units exceeds the total capacity of 280 units by 50 units of the product, a dummy company is created to handle the excess demand. The associated cost coefficients for the dummy warehouse location are taken as zero. Further, the cost element (i.e. 7) on the route city A–Bharat company is replaced by M, since the route is prohibited. The modified table is shown in Table 9.42.

	Bharat	*Janata*	*Red Lamp*	*Supply*
A	M	10	5	90
B	12	9	4	50
C	7	3	11	80
D	9	5	7	60
Dummy	0	0	0	50
Demand	120	100	110	330

Table 9.42

Using the VAM method, the initial solution is shown in Table 9.43. To test the optimality of this solution evaluate each unoccupied cell in terms of opportunity cost associated with it as shown in Table 9.43.

	Bharat	*Janata*	*Red Lamp*	*Supply*	u_i
A	M M – 13	10 +1	5 (90)	90	$u_1 = 13$
B	12 (30)	9	4 (20)	50	$u_2 = 12$
C	7 (+) 0	3 (80) (–)	11 +12	11 80	$u_3 = 7$
D	9 (–) (40)	5 (20) (+)	7 + 6	60	$u_4 = 9$
Dummy	0 (50)	0 + 4	0 + 8	50	$u_5 = 0$
Demand	120	100	110	330	
v_j	$v_1 = 0$	$v_2 = -4$	$v_3 = -8$		

Table 9.43
Optimal Solution

Since the opportunity cost in all the unoccupied cells is positive, the initial solution shown in Table 9.43 is also an optimal solution. The total transport cost associated with this solution is

Total cost = $5 \times 90 + 12 \times 30 + 4 \times 20 + 3 \times 80 + 9 \times 40 + 5 \times 20$ = Rs 1,59,000

(b) Since opportunity cost in cell (C, Bharat), $d_{31} = 0$, there exists an alternative optimal solution:

$$x_{13} = 90,\ x_{21} = 30,\ x_{23} = 20,\ x_{31} = 40,\ x_{32} = 40,\ x_{42} = 60 \text{ and } x_{51} = 50$$

and total cost = Rs 1,59,000

(c) The dual of the given problem is

$$\text{Maximize } Z = (90\ u_1 + 50\ u_2 + 80\ u_3 + 60\ u_4 + 50\ u_5) + (120\ v_1 + 100\ v_2 + 110\ v_3)$$

subject to the constraints

$$\begin{array}{llll} u_1 + v_1 \le \text{M} & u_2 + v_1 \le 12 & u_3 + v_1 \le 7 & u_4 + v_1 \le 9 \\ u_1 + v_2 \le 10 & u_2 + v_2 \le 9 & u_3 + v_2 \le 3 & u_4 + v_2 \le 5 \\ u_1 + v_3 \le 5 & u_2 + v_3 \le 4 & u_3 + v_3 \le 11 & u_4 + v_3 \le 7 \end{array}$$

and u_i, v_j unrestricted in sign, for all i and j.

Substituting the values of u_is and v_js from the optimal solution of transportation problem shown in Table 9.43, we get

Maximize $Z = 90 \times 13 + 50 \times 12 + 80 \times 7 + 60 \times 9 + 50 \times 0 + 120 \times 0 + 100 \times -4 + 110 \times -8$ = Rs 1,59,000

which is the same value as obtained earlier.

9.6.4 Prohibited Transportation Routes

If situations like road hazards (snow, flood, etc.), traffic regulations, etc., arise, then it may not be possible to transport goods from certain sources to certain destinations. Such situations can be handled by assigning a very large cost, say M (or ∞) to such a route(s) (or cell).

Example 9.15 Consider the problem of scheduling the weekly production of certain items for the next four weeks. The production cost of the item is Rs 10 for the first two weeks and Rs 15 for the last two weeks. The weekly demands are 300, 700, 900 and 800, which must be met. The plant can produce a maximum of 700 units per week. In addition, the company can use overtime during the second and third week. This increases the weekly production by an additional 200 units, but the production cost also increases by Rs 5. Excess production can be stored at a unit cost of Rs 3 per week. How should the production be scheduled so as to minimize the total cost?

Solution The given information is presented as a transportation problem in Table 9.44. The cost elements in each cell are determined by adding the production cost, the overtime cost of Rs 5, and the storage cost of Rs 3. Thus, in the first row, the cost of Rs 3 is added during second week onward. Since the output of any period cannot be used in a period preceding it, the cost element is written in the appropriate cells. A dummy column has been added because the supply exceeds demand.

Week (*Origin*)	*Production Cost per Week* (*Destination*)					*Supply*
	I	*II*	*III*	*IV*	*Dummy*	
R_1	10	13	16	19	0	700
R_2	–	10	13	16	0	700
O_2	–	15	18	21	0	200
R_3	–	–	15	18	0	700
O_3	–	–	20	23	0	200
R_4	–	–	–	15	0	700
Demand	300	700	900	800	500	3,200

Table 9.44

Apply Vogel's method to get initial solution of the given transportation problem and then apply MODI method to get optimal transportation schedule. Degeneracy occurs at the initial basic feasible solution stage. Degeneracy may be removed by adding Δ in the cell (R_2, Dummy). The optimal solution is shown in Table 9.45.

	I	*II*	*III*	*IV*	*Dummy*	*Supply*
R_1	10 (300)	13	16 (200)	19 (100)	0 (100)	700
R_2	–	10 (700)	13 (Δ)	16	0	700
O_2	–	15	18	21	0 (200)	200
R_3	–	–	15 (700)	18	0	700
O_3	–	–	20	23	0 (200)	200
R_4	–	–	–	15 (700)	0	700
Demand	300	700	900	800	500	3,200

Table 9.45 Optimal Solution

The production schedule is given in Table 9.46.

Production in Week	*Units*	*For Use in Week*			
		I	*II*	*III*	*IV*
I	700	300	–	200	100
II	R_2 700 O_2 Nil	–	700	–	–
III	R_3 700 O_3 Nil	–	–	700	–
IV	700	–	–	–	700
Demand		300	700	900	800

Table 9.46 Production Schedule

The total minimum cost for the optimal production schedule given in Table 9.47 is

Total cost = 10 × 300 + 16 × 200 + 19 × 100 + 10 × 700 + 15 × 700 + 15 × 700 = Rs 36,100

Example 9.16 ABC company wishes to develop a monthly production schedule for the next three months. Depending upon the sales commitments, the company can either keep the production constant, allowing fluctuations in inventory, or inventories can be maintained at a constant level, with fluctuating production. Fluctuating production necessitates working overtime, the cost of which is estimated to be double the normal production cost of Rs 12 per unit. Fluctuating inventories result in inventory carrying cost of Rs 2 per unit. If the company fails to fulfill its sales commitment, it incurs a shortage cost of Rs 4 per unit per month. The production capacities for the next three months are shown in the following table:

	Production Capacity		
Month	*Regular*	*Overtime*	*Sales*
M_1	50	30	60
M_2	50	0	120
M_3	60	50	40

Determine the optimal production schedule. [*Delhi Univ, MBA, 2002, 2005, AMIE 2009*]

Solution The given information is presented as a transportation problem in Table 9.47. The cost elements in each cell are determined as follows:

(i) If items are produced in a month for sales during the same month, there will be no inventory carrying cost. Thus, the total cost will either be the normal production cost or the overtime production cost. Thus the cost elements for cells (R_1, 1), (R_2, 2) and (R_3, 3) are Rs 12 each and for cells (O_1, 1), (O_2, 3) are Rs 24 each.

(ii) If items are produced in a particular month for sales during the subsequent month, in addition to the production costs (normal or overtime) inventory carrying cost at the rate of Rs 2 per month will be incurred.

Cell	*Production Cost* (*Rs*)	*Inventory Carrying Cost* (*Rs*)	*Total Cost* (*Rs*)
(R_1, 2)	12	2	14
(R_1, 3)	12	4	16
(O_1, 2)	24	2	26
(O_1, 3)	24	4	28
(R_2, 3)	12	2	14

(iii) If the company fails to fulfil its sales commitment, it incurs a shortage cost of Rs 4 per unit per month, in addition to the production costs (normal or overtime), carrying (storage) cost at the rate of Rs 2 per month will be incurred.

Cell	*Production Cost* (*Rs*)	*Shortage Cost* (*Rs*)	*Total Cost* (*Rs*)
(R_2, 1)	12	4	16
(R_3, 2)	12	4	16
(R_3, 1)	12	8	20
(O_3, 2)	24	4	28
(O_3, 1)	24	8	32

The solution is left as an exercise for the reader. The initial basic feasible solution obtained by Vogel's method (shown in Table 9.47) is also the optimal solution.

Table 9.47

	M_1	M_2	M_3	*Dummy*	*Product Supply*
R_1	12 (50)	14	16	0	50
O_1	24 (10)	26 (20)	28	0	30
R_2	16	12 (50)	14	0	50
R_3	20	16 (20)	12 (40)	0	60
O_3	32	28 (30)	24	0 (20)	50
Sales Demand	60	120	40	20	240

The production schedule is given in Table 9.48.

Production in Month		Units	For Use in Month M_1	M_2	M_3
M_1	R_1	50	50	–	–
	O_1	30	10	20	–
M_2	R_2	50	–	50	–
M_3	R_3	60	–	20	40
	O_3	30	–	30	–
	Demand		60	120	40

Table 9.48 Production Schedule

The total minimum cost for the optimal production schedule given in Table 9.49 is

Total cost = 12 × 50 + 24 × 10 + 16 × 20 + 12 × 50 + 16 × 20 + 12 × 40 + 28 × 30

= Rs 3,400.

Example 9.17 The following is the information that concerns the operations of the XYZ manufacturing company. The production cost of the company is estimated to be Rs 5 per unit.

	Month 1	Month 2
Units on order	800	1,400
Production Capacity		
Regular time	920	920
Overtime	250	250
Excess cost/unit (overtime)	1.25	1.25
Storage cost/unit	0.50	0.50

Formulate and solve the above problem as transportation problem. [*Delhi Univ., MBA, 2004; AMIE 2006*]

Solution The storage cost of Re 0.50 per unit per month is charged only if production during the month 1 is used for supplies during month 2.

Costs for supplies against first month's order from the previous month have been assumed at infinity as this is treated not only as prohibitive but also as undesirable. This is because the order quantity of first month is even less than a regular time production capacity for the same month.

This problem is unbalanced as net supply is of 2,340 units while the demand is only of 2,200 units. A dummy demand centre of 140 units with supply cost zero is added to solve the problem. The data are summarized in Table 9.49 along with initial solution.

The initial basic solution obtained by Vogel's method (shown in Table 9.49) is updated in order to obtain optimal solution, which is shown in Table 9.50.

The least cost production schedule to meet the sale demand is shown below:

	M_1	M_2	Dummy	Supply
M_1	5.0 (800)	5.5 (120)	0	920
$M_1(OT)$	6.25	6.75 (250)	0	250
M_2	∞	5.0 (920)	0	920
$M_2(OT)$	∞	6.25 (110)	0 140	250
Demand	800	1,400	140	

Table 9.49 Initial Solution

	M_1	M_2	Dummy	Supply
M_1	5.0 (800)	5.5 (120)	0	920
$M_1(OT)$	6.25	6.75 (110)	0 140	250
M_2	∞	5.0 (920)	0	920
$M_2(OT)$	∞	6.25 (250)	0	250
Demand	800	1,400	140	

Table 9.50 Optimal Solution

	Production		Supply	
	M_1	M_2	M_1	M_2
Regular time	920	110	800	230
Overtime	920	250	–	1,170
			800	1,400

9.7 MAXIMIZATION TRANSPORTATION PROBLEM

In general, the transportation model is used for cost minimization problems. However, it may also be used to solve problems in which the objective is to maximize total profit. That is, instead of unit cost c_{ij}, the unit profit or payoff p_{ij} associated with each route, (i, j) is given. The objective function in terms of total profit (or payoff) is then stated as follows:

$$\text{Maximize } Z = \sum_{i=1}^{m} \sum_{j=1}^{n} p_{ij}\, x_{ij}$$

The procedure for solving such problems is same as that for the minimization problem. However, a few adjustments in Vogel's approximation method (VAM) for finding initial solution and in the MODI optimality test are required.

For finding the initial solution by VAM, the penalties are computed as difference between the largest and next largest payoff in each row or column. In this case, row and column differences represent payoffs. Allocations are made in those cells where the payoff is largest, corresponding to the highest row or column difference.

Since it is a maximization problem, the criterion of optimality is the converse of the rule for minimization. The rule is: *A solution is optimal if all opportunity costs d_{ij} for the unoccupied cells are zero or negative.*

Example 9.18 A company has four manufacturing plants and five warehouses. Each plant manufactures the same product, which is sold at different prices in each warehouse area. The cost of manufacturing and cost of raw materials are different in each plant due to various factors. The capacities of the plants are also different. The relevant data is given in the following table:

	Plant			
Item	*1*	*2*	*3*	*4*
Manufacturing cost (Rs) per unit	12	10	8	8
Raw material cost (Rs) per unit	8	7	7	5
Capacity per unit time	100	200	120	80

The company has five warehouses. The sale prices, transportation costs and demands are given in the following table:

Warehouse	*Transportation Cost (Rs) per Unit*				*Sale Price*	*Demand per Unit (Rs)*
	1	*2*	*3*	*4*		
A	4	7	4	3	30	80
B	8	9	7	8	32	120
C	2	7	6	10	28	150
D	10	7	5	8	34	70
E	2	5	8	9	30	90

(a) Formulate this problem as a transportation problem in order to maximize profit.
(b) Find the solution using VAM method.
(c) Test for optimality and find the optimal solution.

Solution Based on the given data, the profit matrix can be derived by using following equation.

Profit = Sales price – Production cost – Raw material cost – Transportation cost

The matrix, so obtained, is shown in Table 9.51.

Converting data on profit into cost by subtracting all the profit values in the table from the highest profit value. Since highest profit value is 15, therefore, subtracting all cell values including itself from it. The new values so obtained are shown in Table 9.52. The problem now becomes a usual cost minimizing transportation problem.

	1	*2*	*3*	*4*	*Dummy*	*Demand*
A	6	6	11	15	0	80
B	4	6	10	12	0	120
C	6	4	7	6	0	150
D	4	10	14	14	0	70
E	8	8	7	9	0	90
Supply	100	200	120	80	10	510

Table 9.51
Profit Matrix

Apply Vogel's method to find the initial basic feasible solution, as shown in Table 9.52.

	1	*2*	*3*	*4*	*Dummy*	*Demand*	u_i
A	9 + 7	9 + 5	4 + 4	0 (80)	15 + 12	80	$u_1 = 4$
B	11 + 4	9 (70)	5 (–) (50)	3 (+) – 2	15 + 7	120	$u_2 = 9$
C	9 (100)	11 (40)	8 +1	11 + 4	15 (10)	150	$u_3 = 11$
D	11 + 8	5 0	1 (+) (70)	1 Δ (–)	15 +11	70	$u_4 = 5$
E	7 + 2	7 (90)	8 + 5	6 + 3	15 + 9	90	$u_5 = 7$
Supply	100	200	120	80	10	510	
v_j	$v_1 = -2$	$v_2 = 0$	$v_3 = -4$	$v_4 = -4$	$v_5 = -1$		

Table 9.52
Initial Feasible Solution

Since at initial solution stage, the number of occupied cells are only 8, which is one less than the required number, $m + n - 1 = 9$, therefore, the solution is degenerate. However, after making an allocation of Δ to the cell (D, 4), the initial solution has now become eligible for optimality test.

Apply MODI method to evaluate each unoccupied cell in terms of opportunity cost associated with it in the usual manner. This is shown in Table 9.52.

The cell (B, 4) has a negative opportunity cost (i.e. – 2) as shown in Table 9.52. Introduce it into the new solution by constructing a loop shown in Table 9.52. The new solution is given in Table 9.53, where Δ has shifted from cell (D, 4) to cell (B, 4).

	1	2	3	4	Dummy	Demand	u_i
A	9 +5	9 +3	4 +2	0 (80)	15 +5	80	$u_1 = -5$
B	11 +4	9 (70)	5 (50)	3 (Δ)	15 +2	120	$u_2 = -2$
C	9 (100)	11 (40)	8 +1	11 +6	15 (10)	150	$u_3 = 0$
D	11 +8	5 0	1 (70)	1 +2	15 +6	70	$u_4 = -6$
E	7 +2	7 (90)	8 +5	6 +5	15 +4	90	$u_5 = -4$
Supply	100	200	120	80	10	510	
v_j	$v_1 = 9$	$v_2 = 11$	$v_3 = 7$	$v_4 = 5$	$v_5 = 15$		

Table 9.53
Optimal Solution

Since there is no negative opportunity cost in any of the unoccupied cells in Table 9.53, therefore, this solution is the optimal solution. However, the zero opportunity cost in cell (D, 2) indicates the existence of an alternative solution. The total maximization profit associated with the solution is

Total profit = 9 × 70 + 5 × 50 + 9 × 100 + 11 × 40 + 15 × 10 + 1 × 70 + 7 × 90 = Rs 4,580.

9.8 TRANS-SHIPMENT PROBLEM

In a transportation problem, the shipment of a commodity takes place among sources and destinations. But instead of direct shipments to destinations, the commodity can be transported to a particular destination through one or more intermediate or trans-shipment points. Each of these points, in turn supply to other points. Thus, when the shipments are done from destination to destination and from source to source, such type of transportation problems are referred as trans-shipment problems. A trans-shipment problem involving four sources and three destinations is shown diagrammatically in Figs. 9.2(a) and (b).

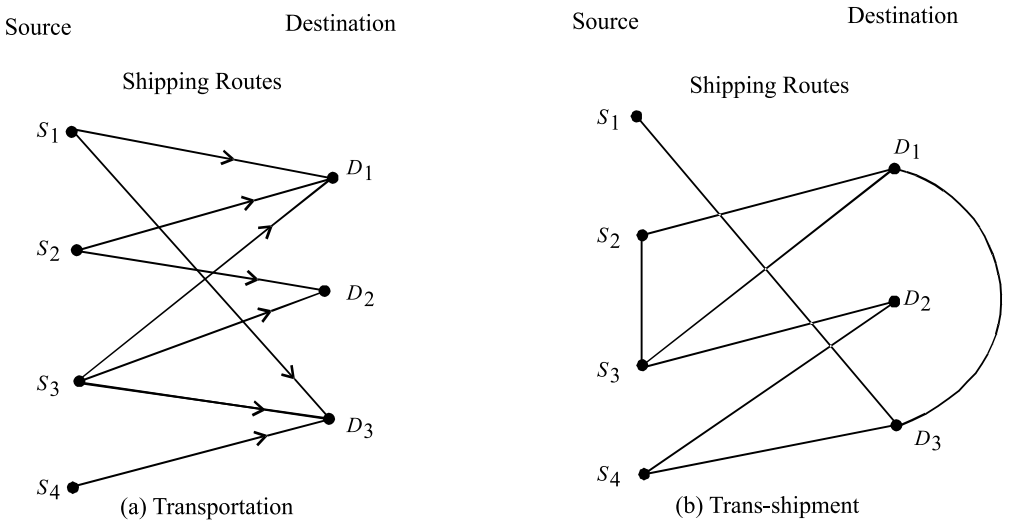


Fig. 9.2

Since the flow of commodity can be in both directions, arrows are not shown in Fig. 9.2(b). The solution to this problem can be obtained by using the transportation model. The solution procedure is as follows: If there are m sources and n destinations, a transportation table shall be of size $(m + n) \times (m + n)$ instead

of $m \times n$. If the total number of units transported from all sources to all destinations is N, then the given supply at each source and demand at each destination are added to N. The demand at source and the supply at each destination are set to be equal to N. The problem can then be solved by the usual MODI method for transportation problems. In the final solution, ignore the units transported from a point to itself, i.e. diagonal cells, because they do not have any physical meaning (no transportation).

Example 9.19 Consider a firm having two factories. The firm is to ship its products from the factories to three retail stores. The number of units available at factories X and Y are 200 and 300, respectively, while those demanded at retail stores A, B and C are 100, 150 and 250, respectively. Rather than shipping the products directly from factories to retail stores, it is asked to investigate the possibility of trans-shipment. The transportation cost (in rupees) per unit is given in the table:

		Factory		*Retail Store*		
		X	*Y*	*A*	*B*	*C*
Factory	*X*	0	8	7	8	9
	Y	6	0	5	4	3
Retail Store	*A*	7	2	0	5	1
	B	1	5	1	0	4
	C	8	9	7	8	0

Find the optimal shipping schedule.

Solution The number of units available at X and Y are 200 and 300, respectively and the demand at A, B and C is 100, 150 and 250, respectively. The maximum amount which can be transported through a factory or retail store is the total supply and demand, i.e. N = 500 units. If all these 500 units are not transported through a factory or retail store, then the remaining units will play the role of dummy. The trans-shipment table is shown in Table 9.54 where 500 units have been added to the supply and demand at factory and to a retail store.

	X	*Y*	*A*	*B*	*C*	*Supply*
X	0	8	7	8	9	200 + 500
Y	6	0	5	4	3	300 + 500
A	7	2	0	5	1	500
B	1	5	1	0	4	500
C	8	9	7	8	0	500
Demand	500	500	100 + 500	150 + 500	250 + 500	

Table 9.54

The initial solution to trans-shipment problem given in Table 9.55 can be obtained by putting 500 units to each route on the diagonal and making allocation in the matrix, by using Vogel's approximation method.

	A	*B*	*C*	*Supply*
X	7	8	9	700
Y	5	4	3	800
Demand	600	650	750	

Applying the MODI method to test the optimality of the solution given in Table 9.55. Since the opportunity cost corresponding to each unoccupied cell is positive, the solution given in Table 9.55 is also optimal. In order to interpret the optimal solution, allocations in the diagonal cells are ignored as these values show the extra dummies that have been added in order to allow as much as flow possible.

	X	*Y*	*A*	*B*	*C*	*Supply*	u_i
X	0 (500)	8 +4	7 (100)	8 (100)	9 +2	700	$u_1 = 4$
Y	6 +10	0 (500)	5 +2	4 (50)	3 (250)	800	$u_2 = 0$
A	7 +14	2 +5	0 (500)	5 +4	1 +1	500	$u_3 = -3$
B	1 +9	5 +9	1 +2	0 (500)	4 +5	500	$u_4 = -4$
C	8 +15	9 +12	7 +7	8 +7	0 (500)	500	$u_5 = -3$
Demand	500	500	600	650	750	3,000	
v_j	$v_1 = -4$	$v_2 = 0$	$v_3 = 3$	$v_4 = 4$	$v_5 = 3$		

Table 9.55
Optimal Solution

CONCEPTUAL QUESTIONS C

1. What is meant by unbalanced transportation problem? Explain the method for solving such a problem.
2. Explain how a profit maximization transportation problem can be converted to an equivalent cost minimization transportation problem.
3. What is degeneracy in transportation problems? How is a transportation problem solved when the demand and supply are not equal?
4. (a) Explain how to resolve degeneracy in a transportation problem.
   (b) How does the problem of degeneracy arise in a transportation problem? Explain how one can overcome it.
5. State a transportation problem in general terms and explain the problem of degeneracy. How does one overcome it?
6. Explain a trans-shipment problem.
7. What are the main characteristics of a transshipment problem?
8. Explain how a trans-shipment problem can be solved as a transportation problem.
9. What is a trans-shipment problem? Explain how it can be formulated and solved as a transportation problem.
   [*Delhi Univ., MBA, 2006*]
10. Explain the method for solving trans-shipment problem.

SELF PRACTICE PROBLEMS C

1. A steel company has three open hearth furnaces and five rolling mills. The transportation costs (rupees per quintal) for shipping steel from furnaces to rolling mills are given in the following table:

	M_1	M_2	M_3	M_4	M_5	*Supply*
F_1	4	2	3	2	6	8
F_2	5	4	5	2	1	12
F_3	6	5	4	7	7	14
Demand	4	4	6	8	8	

   What is the optimal shipping schedule?
2. Consider the following unbalanced transportation problem.

		To *I*	*II*	*III*	*Supply*
From	*A*	5	1	7	10
	B	6	4	6	80
	C	3	2	5	15
Demand		75	20	50	

   Since there is not enough supply, some of the demands at these destinations may not be satisfied. Suppose there are penalty costs for every unsatisfied demand unit which are given by 5, 3 and 2 for destinations I, II and III, respectively. Find the optimal solution.

3. A company produces a small component for an industrial product and distributes it to five wholesalers at a fixed delivered price of Rs 2.50 per unit. The sales forecasts indicate that the monthly deliveries will be 3,000, 3,000, 10,000, 5,000 and 4,000 units to wholesalers I, II, III, IV and V, respectively. The company's monthly production capacities are 5,000, 10,000 and 12,500 at plants 1, 2 and 3, respectively. Respective direct costs of production of each unit are Re 1.00, Re 0.90, and Re 0.80 at plants W, X and Y. Transportation costs of shipping a unit from a plant to a wholesaler are as follows.

Wholesaler

	I	*II*	*III*	*IV*	*V*
W	0.05	0.07	0.10	0.25	0.15
Plant *X*	0.08	0.06	0.09	0.12	0.14
Y	0.10	0.09	0.08	0.10	0.15

Find how many components each plant should supply to each wholesaler in order to maximize its profit.

4. ABC Tool Company has a sales force of 25 men, who operate from three regional offices. The company produces four basic product lines of hand tools. Mr Jain, the sales manager, feels that 6 salesmen are needed to distribute product line *I*, 10 to distribute product line *II*, 4 for product line III and 5 salesmen for product line IV. The cost (in Rs) per day of assigning salesmen from each of the offices for selling each of the product lines are as follows:

Product Lines

		I	*II*	*III*	*IV*
	A	20	21	16	18
Regional Office	*B*	17	28	14	16
	C	29	23	19	20

At the present time, 10 salesmen are allocated to office *A*, 9 to office *B* and 7 salesmen to office *C*. How many salesmen should be assigned from each office to sell each product line in order to minimize costs? Identify alternate optimum solutions, if any. [*Delhi Univ., MBA, 2001*]

5. The Purchase Manager, Mr Shah, of the State Road Transport Corporation must decide on the amount of fuel that should be bought from three possible vendors. The corporation refuels its buses regularly at the four depots within the area of its operations.

The oil companies have said that they can furnish up to the following amounts of fuel during the coming month: 2,75,000 litres by oil company 1; 5,50,000 litres by oil company 2; and 6,60,000 litres by oil company 3. The required amount of the fuel is 1,10,000 litres by depot 1; 2,20,000 litres at depot 2; 3,30,000 litres at depot 3; and 4,40,000 litres at depot 4.

When the transportation costs are added to the bid price per litre supplied, the combined cost per litre for fuel from each vendor, servicing a specific depot, is as under:

	Company 1	*Company 2*	*Company 3*
Depot 1	25.00	24.75	24.25
Depot 2	25.00	25.50	26.75
Depot 3	24.50	26.00	25.00
Depot 4	25.50	26.00	24.50

Determine the optimal schedule. [*Delhi Univ., MBA, 2002*]

6. A departmental store wishes to purchase the following quantities of sarees:

Types of sarees	:	A	B	C	D	E
Quantity	:	150	100	75	250	200

Tenders are submitted by four different manufacturers who undertake to supply not more than the quantities mentioned below (all types of sarees combined);

Manufacturer	:	W	X	Y	Z
Total quantity	:	300	250	150	200

Sarees

		A	*B*	*C*	*D*	*E*
	W	275	350	425	225	150
	X	300	325	450	175	100
Manufacture	*Y*	250	350	475	200	125
	Z	325	275	400	250	175

How should the orders be placed?

7. A company has four factories F_1, F_2, F_3 and F_4 that manufacture the same product. Production and raw material costs differ from factory to factory and are given in the following table in the first two rows. The transportation costs from the factories to the sales depots, S_1, S_2 and S_3 are also given. The last two columns in the table give the sales price and the total requirement at each depot. The production capacity of each factory is given in the last row.

		F_1	F_2	F_3	F_4	*Sales Price per Unit*	*Requirement*
Production cost/unit		15	18	14	13		
Raw material cost/unit		10	9	12	9		
Transportation cost/unit	S_1	3	9	5	4	34	80
	S_2	1	7	4	5	32	120
	S_3	5	8	3	6	31	150
Supply		50	150	50	100		

Determine the most profitable production and distribution schedule and the corresponding profit. The deficit production should be taken to yield zero profit. Write the dual of this transportation problem and use it for checking the optimal solution. [*Delhi Univ., MBA, 2000*]

8. Link Manufacturing Company has several plants, three of which manufacture two principal products – a standard card table and deluxe card table. A new deluxe card table will be introduced, which must be considered in terms of selling price and costs. The selling prices are: Standard, Rs 14.95, deluxe Rs 18.85 and New Deluxe, Rs 21.95.

		Variable Costs (in Rs)				
Model	*Qty.*	*Plant A*	*Plant B*	*Plant C*	*Plant*	*Capacity*
Standard	450	8.00	7.95	8.10	A	800
Deluxe	1,050	8.50	8.60	8.45	B	600
New Deluxe	600	9.25	9.20	9.30	C	700

Solve this problem by the transportation technique for the maximum contribution.

9. City Super Market keeps five different patterns of a particular size of readymade garment for sale. There are four manufacturers

available to the manager of the market to whom be can order the required quantity. The demand for the coming season and the maximum quantity that can be produced by a manufacturer is as follows:

Pattern	*1 U. Cut*	*2 P. Form*	*3 Maxi*	*4 Mini*	*5 Midi*
Quantity required	200	150	200	150	300
Manufacturer	*A*	*B*	*C*	*D*	
Capacity	200	300	250	250	

The following quotations have been submitted by the manufacturers for different patterns:

	1 U. Cut	*2 P. Form*	*3 Maxi*	*4 Mini*	*5 Midi*
A	Rs 100	Rs 130	Rs 160	Rs 70	Rs 140
B	100	120	150	75	130
C	115	140	155	80	120
D	125	145	140	60	125

The super market is selling different patterns at the following prices:

Pattern	:	1	2	3	4	5
Selling Price (Rs)	:	120	150	170	80	140

How should the super market manager place the order? Write the dual of this transportation problem and give its economic interpretation.

10. A company wishes to determine an investment strategy for each of the next four years. Five investment types have been selected. The investment capital has been allocated for each of the coming four years, and maximum investment levels have been established for each investment type. There is an assumption that amounts invested in any year will remain invested until the end of the planning horizon of four years. The following table summarizes the data of this problem. The values in the body of the table represent the net return on investment of one rupee up to the end of the planning horizon. For example, a rupee invested in investment type B at the beginning of year will grow to Rs 1.90 by the end of the fourth year, yielding a net return of Re 0.90.

Investment Made at the Beginning of Year	*Investment Type*					*Rupees Available (in '000s)*
	A	*B*	*C*	*D*	*E*	
1	0.80	0.90	0.60	0.75	1.00	500
2	0.55	0.65	0.40	0.60	0.50	600
3	0.30	0.25	0.30	0.50	0.20	750
4	0.15	0.12	0.25	0.35	0.10	800
Maximum rupees investment (in '000s)	750	600	500	800	1,000	

The objective in this problem is to determine the amount to be invested at the beginning of each year in an investment type so as to maximize the net rupee return for the four year period.

Solve the above transportation problem and get an optimal solution. Also calculate the net return on investment for the planning horizon of four-year period.

11. XYZ Company has provided the following data and seek your advice on the optimum investment strategy:

Investment Made at the Beginning of Year	*Net Return Data (in paise) of Selected Investment*				*Amount Available (lacs)*
	P	*Q*	*R*	*S*	
1	95	80	70	60	70
2	75	65	60	50	40
3	70	45	50	40	90
4	60	40	40	30	30
Max. Investment (Rs lakh)	40	50	60	60	–

The following additional information is also provided:

(a) P, Q, R and S represent the selected investments.
(b) The company has decided to have a four years investment plan.
(c) The policy of the company is that the amount invested in any year will remain so until the end of fourth year.
(d) The values (paise) in the table represent net return on investment of one rupee till the end of the planning horizon (for example, a rupee in investment P at the beginning of year will grow to Rs 1.95 by the end of the fourth year, yielding a return of 95 paise).

Using the above, determine the optimum investment strategy.

12. A manufacturer must produce a certain product in sufficient quantity in order to meet contracted sales of the next four months. The production facilities available for this product are limited, and vary in different months. The unit cost of production also changes accordingly to the facilities and personnel available. The product may be produced in one month and then held for sale in a later month, at an estimated storage cost of Re 1 per unit per month. No storage cost is incurred for goods that are sold in the same month in which they are produced. Presently there is no inventory of this product and none is desired at the end of four months. Given the following table, show how much to produce in each of four months in order to minimize total cost.

Month	*Contracted Sales (in units)*	*Maximum Production (in units)*	*Unit Cost of Production (Rs)*	*Unit Storage Cost per Month (Rs)*
1	20	40	14	1
2	30	50	16	1
3	50	30	15	1
4	40	50	17	1

Formulate the problem as a transportation problem and solve it.

13. A company has factories at A, B and C, which supply its products to warehouses at D, E, F and G. The factory capacities are 230, 280 and 180, respectively for regular production. If overtime production is utilized, the capacities can be increased to 300, 360 and 190, respectively. Increment unit overtime costs are Rs 5, Rs 4 and Rs 6, respectively. The current warehouse requirements are 165, 175, 205 and 165, respectively. Unit shipping costs in rupees between the factories and the warehouses are:

		To *D*	*E*	*F*	*G*
	A	6	7	8	10
From	*B*	4	10	7	6
	C	3	22	2	11

Determine the optimum distribution for the company to minimize costs.

14. The products of three plants X, Y and Z are to be transported to four warehouses I, II, III and IV. The cost of transportation of each unit from plants to the warehouses along with the normal capacities of plants and warehouses are indicated below:

		Warehouse				
		I	*II*	*III*	*IV*	*Available*
Plant	*X*	25	17	25	14	300
	Y	15	10	18	24	500
	Z	16	20	8	13	600
Required		300	300	500	500	

(a) Solve the problem for minimum cost of transportation. Are there any alternative solutions? If any, explain the methodology.
(b) Overtime can be used in each plant to raise the capacity by 50 per cent of the normal but the corresponding cost of trans-shipment will also increase by 10, 15 and 20 to the unit costs of production at each plant.

15. A firm manufacturing industrial chemicals has got 3 plants P_1, P_2 and P_3 each having capacities to produce 300 kg, 200 kg, and 500 kg, respectively of a particular chemical per day. The production costs per kg in plants P_1, P_2 and P_3, respectively are Re 0.70, Re 0.60 and Re 0.66. Four bulk consumers have placed orders for the products on the following basis:

Consumer	*kg Required per Day*	*Price Offered Rs/kg*
I	400	1.00
II	250	1.00
III	350	1.02
IV	150	1.03

Shipping costs (paise per kg) from plants to consumers are given in the table below:

		Consumer			
		C_1	C_2	C_3	C_4
Plant	P_1	3	5	4	6
	P_2	8	11	9	12
	P_3	4	6	2	8

Work out an optimal schedule for the above situation. Under what conditions would you change schedule?

16. The personnel manager of a manufacturing company is in the process of filling 175 jobs in six different entry level skills due to the establishment of a third shift by the company. Union wage scale and requirement for the skills are shown in the following table:

	Pay Scales and Skills Requirements					
Entry Level Skills	*A*	*B*	*C*	*D*	*E*	*F*
Wage scale Rs/month	1,000	1,100	1,200	1,300	1,400	1,500
No. required	25	29	31	40	33	17

230 applicants for the jobs have been tested and their aptitudes and skills for the jobs in question have been matched against company standards and evaluated. The applicants have been grouped into four categories by their abilities; the grouping and values of each category to the company are shown in the table below:

Applicant	*Category Value (Rs per month)*						*Number of*
Category	*A*	*B*	*C*	*D*	*E*	*F*	*Applicants*
I	1,000	1,100	1,500	1,400	1,400	1,450	54
II	1,200	1,250	1,200	1,350	1,400	1,400	54
III	1,000	1,100	1,200	1,400	1,500	1,600	45
IV	1,500	1,500	1,600	1,400	1,400	1,500	74

How many applicants of each category should the personnel manager hire and for which jobs?

17. Debonair Private Ltd. is in the business of manufacturing and selling office shirts for men. It has four factories located in different parts of the country and the monthly capacities of the factories in thousand are as given below. The shirts are made in a few standard designs and colours, and each factory can make all types of shirts in any size subject to the overall capacity of the factory.

Factories	:	I	II	III	IV
Monthly capacity	:	3	4.5	2.5	5

From the factories, the shirts are transported to five warehouses located in five different regions in India. The warehouses in turn supply to the distributors and the retailers. The monthly demand of shirts (in thousand) from the warehouses is as follows:

Warehouses	:	A	B	C	D	E
Monthly capacity	:	3	5	1.5	2	2.5

The cost of transporting a shirt from a factory to a warehouse depends on the distance between them and the cost of transporting a shirt from each factory to each warehouse is given in the table below:

		Warehouse				
		A	*B*	*C*	*D*	*E*
Factory	*I*	6	3	4	2	5
	II	11	7	5	10	9
	III	10	7	1	2	8
	IV	12	10	5	3	5

How many shirts are to be produced, in which factory, and how are these to be dispatched to the warehouse so that the total cost involved in transportation is minimized.
(a) Use the North-West Corner Method to get an initial feasible solution.
(b) Check if the solution obtained in (a) above is an optimal allocation and if not, then find the optimal solution.

18. A manufacturer of jeans is interested in developing an advertising campaign that will reach four different age groups. Advertising campaigns can be conducted through TV, radio and magazines. The following table gives the estimated cost in paise per exposure for each age group according to the medium employed. In addition, maximum exposure levels possible in each of the media, namely TV, radio and magazines are 40, 30 and 20 millions, respectively. Also the minimum desired exposure within each age group, namely 13–18, 19–25, 26–35, 36 and older are 30, 25, 15 and 10 millions. The objective is to minimize the cost of attaining the minimum exposure level in each age group.

Media	Age Groups			
	13–18	l9–25	26–35	36 and older
TV	12	7	10	10
Radio	10	9	12	10
Magazine	14	12	9	12

(a) Formulate the above as a transportation problem, and find the optimal solution.
(b) Solve this problem if the policy is to provide at least 4 million exposures through TV in the 13–18 age group and at least 8 million exposures through TV in the age group 19–25.

19. Two drug companies have inventories of 1.1 and 0.9 million doses of a particular flu vaccine, and an epidemic of the flu seems imminent in three states. Since the flu could be fatal to senior citizens, it is imperative that they be vaccinated first; others will be vaccinated on a first-come-first-served basis while the vaccine supply lasts. The amounts of vaccine (in millions of doses) each state estimates it could administer are as follows:

	State 1	State 2	State 3
Elders	0.325	0.260	0.195
Others	0.750	0.800	0.650

The shipping costs (paise per dose) between drug companies and states are as follows:

	State 1	State 2	State 3
Company 1	30	30	60
Company 2	10	40	70

Determine a minimum-cost shipping schedule which will provide each state with at least enough vaccine to care for its senior citizens. Write the dual of this transportation problem and use it for checking the optimal solution.

[*Delhi Univ., MBA (HCA), 2000, 2005*]

20. A leading firm has three auditors. Each auditor can work up to 160 hours during the next month, during which time three projects must be completed. Project 1 will take 130 hours, project 2 will take 140 hours, and project 3 will take 160 hours. The amount per hour that can be billed for assigning each auditor to each project is given in the table.

	Project		
Auditor	1(Rs)	2(Rs)	3(Rs)
1	1,200	1,500	1,900
2	1,400	1,300	1,200
3	1,600	1,400	1,500

Formulate this as a transportation problem and find the optimal solution. Also find out the maximum total billing during the next month.

Degeneracy

21. Obtain an optimum basic feasible solution to the following degenerate transportation problem.

		To			
		A	B	C	Supply
From	X	7	3	4	2
	Y	2	1	3	3
	Z	3	4	6	5
Demand		4	1	5	

22. A manufacturer wants to ship 8 loads of his product as shown in the table. The matrix gives the mileage from origin to destination. Shipping costs are Rs 10 per load per mile. What shipping schedule should be used?

	D_1	D_2	D_3	Supply
O_1	50	30	220	1
O_2	90	45	170	3
O_3	250	200	50	4
Demand	4	2	2	

Trans-shipment

23. Consider the following trans-shipment problem with two sources and three destinations, the cost for shipments (in rupees) is given below.

		Source		Destination			Supply
	S_1	S_2	D_1	D_2	D_3		
Source	S_1	0	80	10	20	30	100 + 300
	S_2	10	0	20	50	40	200 + 300
Destination	D_1	20	30	0	4	10	300
	D_2	40	20	10	0	20	300
	D_3	60	70	80	20	0	300
Demand			300	300	100 + 300	100 + 300	100 + 300

Determine the optimal shipping schedule.

24. A firm having two sources, S_1 and S_2 wishes to ship its product to two destinations, D_1 and D_2. The number of units available at S_1 and S_2 are 10 and 30 and the product demanded at D_1 and D_2 are 25 and 15 units, respectively. The firm instead of shipping from sources to destinations, decides to investigate the possibility of transshipment. The unit transportation cost (in Rs) is given in the following table:

		Source		Destination		Supply
		S_1	S_2	D_1	D_2	
Source	S_1	0	3	4	5	10 + 40
	S_2	3	0	3	5	30 + 40
Destination	D_1	4	3	0	2	40
	D_2	5	5	2	0	40
Demand		40	40	25 + 40	15 + 40	

Determine the optimal shipping schedule.

HINTS AND ANSWERS

1. Total requirement (30) < Total capacity (34), add a dummy mill with requirement (34 – 30) = 4. Degeneracy occur at the initial solution (VAM).

Ans. $x_{12} = 4$, $x_{14} = 4$, $x_{24} = 2$, $x_{25} = 8$, $x_{31} = 4$, $x_{33} = 6$ and $x_{36} = 4$. Total cost = Rs 80.

2. Demand (145) > Supply (105), add dummy source with supply (145 – 105) = 40 and transportation costs 5, 3 and 2 for destination 1, 2 and 3, respectively.
   Ans. $x_{12} = 0$, $x_{21} = 60$, $x_{22} = 10$, $x_{23} = 10$, $x_{31} = 5$ and $x_{43} = 40$. Transportation cost = Rs 515. Penalty for transportation of 40 units to destination 3 at the cost of Rs 2 per unit = Rs 80. Thus, total cost = 515 + 80 = Rs 595.
3. Maximization as well as unbalanced problem. Total capacity (27,500) > demand (25,000). Add a dummy wholesaler with demand 2,500 units. Since direct costs of production of each unit are Re 1, Re 0.90 and Re 0.80 at plants 1, 2 and 3, add this cost to each figure row wise to get exact data of the problem.
   Ans. $x_{11} = 2{,}500$, $x_{21} = 500$, $x_{22} = 3{,}000$, $x_{23} = 2{,}500$, $x_{25} = 4{,}000$, $x_{33} = 7{,}500$ and $x_{34} = 5{,}000$. Total cost Rs 23,730.
   Since in total 25,000 units are supplied to the wholesalers at a fixed price of Rs 2.50 per unit, therefore, the total cost is Rs 62,500.
   The net maximum profit to the manufacturer = Rs 62,500 – Rs 23,730 = Rs 38,770
4. Supply (26) > Demand (25), add dummy column with demand 1.
   Ans. $x_{12} = 4$, $x_{13} = 1$, $x_{14} = 5$; $x_{21} = 6$, $x_{23} = 3$, $x_{32} = 2$, $x_{35} = 1$ Total cost = Rs 472
   Alternative solution exists because opportunity cost is zero in cell (B, 4) and (C, 4).
5. Supply (14,85,000) > Demand (11,00,000), add dummy column with demand of 3,85,000 litres of oil; $x_{13} = 110$, $x_{21} = 55$, $x_{22} = 165$, $x_{31} = 220$, $x_{33} = 110$, $x_{43} = 440$, $x_{52} = 385$. Total cost = Rs 51,700.
6. Profit maximization problem. First add a dummy column with demand of 125 sarees and then subtract all elements of the profit matrix from the highest element 475.

Manufacturer	*Variety*	*Quantity*
W	*B, D, E*	25, 50, 200
X	*A*	150
Y	*B, C*	75, 75
Z	*D*	200

7. Profit maximization problem. First add dummy row with supply of 40 units and then subtract all elements of the profit matrix from highest element 8.
   Profit = Sales price – (Production cost + Raw material cost + Transportation cost)

Factory	*Sales Depot*	*Unit Profit*
F_1	S_2	6
F_2	S_2, S_3	– 2, – 4
F_3	S_3	2
F_4	S_1, S_2	8, 5

   Total profit = Rs 2,000
12. Transportation cost table and optimal solution

From/To	*1*	*2*	*3*	*4*	*Dummy*	*Supply*
1	14 (20)	15 (20)	16	17	0	40
2	∞	16 (10)	17 (20)	18	0 (20)	50
3	∞	∞	15 (30)	16	0	30
4	∞	∞	∞	17 (40)	0 (10)	50
Demand	20	30	50	40	30	170

13. Transportation cost table and optimal solution

	D	*E*	*F*	*G*	*Dummy*	*Supply*
A	6 (40)	7 (175)	8 (15)	10	0	230
B	4 (115)	10	7	6 (165)	0	280
C	3	22	2 (180)	11	0	180
A_1	11	12	13	15	0 (70)	70
B_1	8 (10)	14	11	10	0 (70)	80
C_1	9	28	8 (10)	17	0	10
Demand	165	175	205	165	140	850

22. Using NWCM, initial solution is degenerate.
Ans. $x_{13} = 2$, $x_{22} = 1$, $x_{23} = 2$, $x_{31} = 4$ and $x_{33} = 1$, Total cost = Rs 33.

23. ***Ans.*** $x_{12} = 10$, $x_{13} = 20$, $x_{15} = 10$, $x_{21} = 20$, $x_{31} = 5$, $x_{34} = 30$, $x_{35} = 5$, Total cost = Rs 3,600.

CHAPTER SUMMARY

The transportation problem (a special case of LP problem) is concerned with distributing a commodity from given sources to their respective destinations. Each source has a fixed supply of the commodity and each destination has a fixed demand. A basic assumption is that the cost of distribution from each source to each destination is directly proportional to the amount distributed. Formulating a transportation problem requires constructing a special purpose table that gives the unit costs of distribution, the supplies and the demands.

The objective of this chapter is to enable readers to recognize a problem that can be formulated and analyzed as a transportation problem or as a variant of one of these problem types.

CHAPTER CONCEPTS QUIZ

True or False

1. The test for degeneracy of a solution is to check if there are unused cells with an improvement index equal to zero. If so, degeneracy exists.
2. The advantage of the most method over the stepping stone method of computing improvement indices for unused cells lies in the greater computational efficiency.
3. If you have to solve a transportation problem with *m* rows and n columns using the simplex method, you would have to formulate the problem with *m* variables and *n* constraints.
4. A solution to the cost minimization transportation problem is optimal when all unused cells have an improvement index which is non-negative.
5. A dummy row or column is introduced in the transportation method is order to handle an unbalanced problem. The dummy serve, the same purpose as a slack variable in the simplex method.
6. Each iteration of the transportation method involves the elimination of one occupied cell and the introduction of one unoccupied cell which is similar to a pivot in the simplex method.
7. In the transportation problem, extra constraint equation can not be derived from any other constraint equations as it affects the feasible solution of the problem.
8. In the north-west corner method, the cost of transportation on any route of transportation is taken into account.
9. The unbalanced transportation problem can be balanced by adding a dummy supply row or a demand column as per the need.
10. In the transportation problem, the rim requirement for a row is the capacity of a supplier, white a rim requirement for a column is the demand of a user.

Fill in the Blanks

11. The northwest corner rule provides a _____ for obtaining an _____ solution to the transportation problem.
12. The degeneracy may occurs when there are two or more cells with the same smallest _____ value in a closed _____ for an incoming cell.
13. We have alternative optimal solutions to a minimization transportation problem whenever we find a solution where the improvement indices are all _____ with at least _____ to zero.
14. An improvement index in the transportation method is analogous to a value in the quantity column in the _____ method.
15. The _____ serves the same purpose for the transportation method as all slack variables in the simplex method.
16. An unused cell in the transportation table is analogous to a variable not in the solution column in the simplex method.
17. The end result of an iteration in the transportation method is the same as the end result in the simplex method in that value of the solution is _____.
18. In Vogel's approximation method, each allocation is made on the basis of the _____ cost that would have been incurred if allocation in certain cells with _____ unit transportation cost were missed.
19. _____ method is based on the concept of duality.
20. If the total supply _____ total demand, then an additional column known as _____ added to the transportation table to absorb the same.

Multiple Choice

21. The initial solution of a transportation problem can be obtained by applying any known method. However, the only condition is that
    (a) the solution be optimal
    (b) the rim conditions are satisfied
    (c) the solution not be degenerate
    (d) all of the above
22. The dummy source or destination in a transportation problem is added to
    (a) satisfy rim conditions
    (b) prevent solution from becoming degenerate
    (c) ensure that total cost does not exceed a limit
    (d) none of the above
23. The occurrence of degeneracy while solving a transportation problem means that
    (a) total supply equals total demand
    (b) the solution so obtained is not feasible
    (c) the few allocations become negative
    (d) none of the above
24. An alternative optimal solution to a minimization transportation problem exists whenever opportunity cost corresponding to unused route of transportation is:

(a) positive and greater than zero
(b) positive with at least one equal to zero
(c) negative with at least one equal to zero
(d) none of the above

25. One disadvantage of using North-West Corner Rule to find initial solution to the transportation problem is that
(a) it is complicated to use
(b) it does not take into account cost of trans-portation
(c) it leads to a degenerate initial solution
(d) all of the above

26. The solution to a transportation problem with m-rows (supplies) and n-columns (destination) is feasible if number of positive allocations are
(a) $m + n$
(b) $m \times n$
(c) $m + n - 1$
(d) $m + n + 1$

27. The calculation of opportunity cost in the MODI method is analogous to a
(a) $c_j - z_j$ value for non-basic variable columns in the simplex method
(b) value of a variable in $\mathbf{x_B}$-column of the simplex method
(c) variable in the **B**-column in the simplex method
(d) none of the above

28. An unoccupied cell in the transportation method is analogous to a
(a) $c_j - z_j$ value in the simplex table
(b) variable in the **B**-column in the simplex table
(c) variable not in the **B**-column in the simplex table
(d) value in the $\mathbf{x_B}$-column in the simplex table

29. If an opportunity cost value is used for an unused cell to test optimality, it should be
(a) equal to zero
(b) most negative number
(c) most positive number
(d) any value

30. During an iteration while moving from one solution to the next, degeneracy may occur when
(a) the closed path indicates a diagonal move
(b) two or more occupied cells are on the closed path but neither of them represents a corner of the path.
(c) two or more occupied cells on the closed path with minus sign are tied for lowest circled value
(d) either of the above

31. The large negative opportunity cost value in an unused cell in a transportation table is chosen to improve the current solution because
(a) it represents per unit cost reduction
(b) it represents per unit cost improvement
(c) it ensure no rim requirement violation
(d) none of the above

32. The smallest quantity is chosen at the corners of the closed path with negative sign to be assigned at unused cell because
(a) it improve the total cost
(b) it does not disturb rim conditions
(c) it ensure feasible solution
(d) all of the above

33. When total supply is equal to total demand in a transportation problem, the problem is said to be
(a) balanced
(b) unbalanced
(c) degenerate
(d) none of the above

34. Which of the following methods is used to verify the optimality of the current solution of the transportation problem,
(a) Least cost method
(b) Vagel's approximation method
(c) Modified distribution method
(d) all of the above

35. The degeneracy in the transportation problem indicates that
(a) dummy allocation (s) needs to be added
(b) the problem has no feasible solution
(c) the multiple optimal solution exist
(d) (a) and (b) but not (c)

Answers to Quiz

1. F 2. T 3. F 4. T 5. T 6. T 7. F 8. F 9. T 10. T
11. mechanism; initial 12. negative; path 13. non-negative; equals 14. simplex
15. north west corner rule 16. product mix 17. improved 18. opportunity; minimum
19. modified, distribution 20. exceed; dummy demand centre
21. (b) 22. (a) 23. (b) 24. (b) 25. (b) 26. (c) 27. (a) 28. (c) 29. (b) 30. (c)
31. (a) 32. (c) 33. (a) 34. (c) 35. (d)

CASE STUDY

Case 9.1: Asian Games*

For maintaining law and order during the Asian Games, 1982 police force has been requisitioned from the various central police organizations like CRPF, BSF, ITBP and other states such as Haryana, Himachal Pradesh, Uttar Pradesh, Orissa and Gujarat. Besides this there are 10 batalions of Delhi Police available for deployment. The outside and the local force have been stationed at various places in the Union Territory of Delhi. The figures are in the number of sections (10 Men) available for deployment.

Place	*Abbreviation*	*Force (Section)*
New Police Lines	NPL	400
Jharoda Kalan	JK	160
Pitam Pura	PP	160
Model Town	MT	90
Police Training School	PTS	90
Old Police Lines	OPL	120
New Kotwali	NK	20
Kamala Market	KMT	120
Parliament Street	PST	240
Rajpura Lines	RPL	120
Shakarpur	SP	120
Moti Nagar	MN	120
Mehram Nagar	MRN	240
	Total	2,000

The force is to be deployed at various stadia on the day of events. The requirement of the force has been calculated keeping in view the factors like capacity of the stadium, traffic problems, security of VIPs and participants etc. The requirement of force is also mentioned in number of sections.

Stadium	*Abbreviation*	*Force (Section)*
Chhatrasal	CS	60
Delhi University	DU	40
Indraprastha	IP	400
Yamuna Velodrome	YV	40
Talkatora Indoor	TKI	80
Talkatora Swimming	TKS	80
National	NS	240
Hall of sports	HS	60
Harbaksh	HB	60
Shivaji	SV	60
Jawaharlal Nehru	JLN	500
Hauz Khas	HK	20
Games Village	GV	60
Ambedkar	AK	200
Tughlakabad Range	TR	40
Golf Club	GC	20
Karnail Singh	KS	40
	Total	2,000

The cost of transportation of one section, i.e. 10 Men from every location to each stadium has been calculated as shown in the matrix below.

It is assumed that the availability and requirement of force is equal, i.e. 2,000 sections. It will not be possible on any day during the games as all stadia are not having games on all the days. Hence, there will always be a surplus force available. Consequently dummy stadium will have to be planned everyday so as to absorb the unutilized force. The cost (in Rs 100s for 10 persons – one section) matrix for the movement of the police force is given below:

* Based on the class assignment prepared by the MBA(FMS) student Mr M.S Upadhya, IPS.

	CS	*DU*	*IP*	*YV*	*TKI*	*TKS*	*NS*	*HS*	*HB*	*JLN*	*SV*	*HK*	*GV*	*AK*	*TR*	*GC*	*KS*	*Force Availability*
NPL	2	1	13	12	15	15	16	14	25	20	14	22	24	11	35	16	10	400
JK	28	31	36	37	130	30	35	36	16	42	30	35	36	35	50	35	36	160
PP	15	14	26	25	28	28	28	17	38	33	17	35	37	24	48	29	23	160
MT	4	3	15	14	17	17	18	16	27	22	16	24	26	13	37	18	12	90
PTS	26	24	15	16	13	13	12	15	15	12	15	33	3	16	10	17	16	90
OPL	7	2	8	7	10	10	11	13	20	15	9	17	13	6	30	11	6	120
NK	11	6	4	3	6	6	7	9	16	11	5	13	15	2	26	7	4	20
KMT	13	8	3	4	4	4	5	5	21	16	2	25	25	1	24	10	2	120
PST	15	12	5	6	2	2	4	5	16	10	12	12	12	4	22	4	3	240
RPL	4	1	11	10	13	13	14	14	23	18	12	20	22	9	33	14	8	120
SP	19	16	3	4	7	7	6	5	21	12	8	18	18	5	30	8	8	120
MN	10	12	16	17	13	13	15	16	20	20	15	25	25	14	35	18	12	120
MRN	30	28	18	19	15	15	15	16	6	22	15	12	12	20	20	20	20	240
Force Requirement	60	40	400	40	80	80	180	240	60	60	500	60	20	200	40	20	40	2,000

Suggest an optimal transportation schedule of the police force so as to minimize the total transportation cost.

APPENDIX: THEOREMS AND RESULTS

Theorem 9.1 (*Existence of Feasible Solution*) A necessary and sufficient condition for the existence of a feasible solution to the transportation problem is

$$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j \quad \text{(Rim condition)}$$

That is, the total capacity (or supply) must equal total requirement (or demand).

Proof (a) *Necessary Condition* Let there exist a feasible solution to the transportation problem. Then, we have

$$\sum_{i=1}^{m} \sum_{j=1}^{n} x_{ij} = \sum_{i=1}^{m} a_i \tag{1}$$

and

$$\sum_{j=1}^{n} \sum_{i=1}^{m} x_{ij} = \sum_{j=1}^{n} b_j \tag{2}$$

Since the left-hand side of (1) and (2) are the same, therefore $\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$

(b) *Sufficient Condition* Suppose

$$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j = k \text{ (say)}$$

If there exists a real number $\lambda_i \neq 0$ such $x_{ij} = \lambda_i b_j$ for all i and j, then value of λ_i is given by

$$\sum_{j=1}^{n} x_{ij} = \sum_{j=1}^{n} \lambda_i b_j = \lambda_i \sum_{j=1}^{n} b_j = k\lambda_i \quad \text{or} \quad \lambda_i = \frac{1}{k} \sum_{j=1}^{n} x_{ij} = \frac{a_i}{k}$$

Thus

$$x_{ij} = \lambda_i b_j = \frac{a_i b_j}{k} \text{ for all } i \text{ and } j.$$

Since $a_i > 0$ and $b_j > 0$ for all i and j, therefore $a_i b_j / k \geq 0$ and hence a feasible solution exists, i.e. $x_{ij} \geq 0$.

Theorem 9.2 (*Basic Feasible Solution*) The number of basic variables (positive allocations) in any basic feasible solution are $m + n - 1$ (the number of independent constraint equations) satisfying all the rim conditions.

Proof In the mathematical model of a transportation problem it can be seen that there are m rows (capacity or supply constraint equations) and n columns (requirement or demand constraint equations). Thus there are in total $m + n$ constraint equations. But because of Theorem 9.1 (total capacity be equal to the total requirement) out of $m + n$ constraint equations one of the equations is redundant and can also be eliminated. Thus there are $m + n - 1$ linearly independent equations. It can be verified by adding all the m rows equations and subtracting from the sum the first $n - 1$ column equations, thereby getting the last column equation. That is,

$$\sum_{i=1}^{m} \sum_{j=1}^{n} x_{ij} - \sum_{j=1}^{n-1} \sum_{i=1}^{m} x_{ij} = \sum_{i=1}^{m} a_i - \sum_{j=1}^{n-1} b_j$$

$$\sum_{i=1}^{m}\sum_{j=1}^{n} x_{ij} - \left[\sum_{j=1}^{n}\sum_{i=1}^{m} x_{ij} - \sum_{i=1}^{m} x_{in}\right] = \sum_{i=1}^{m} a_i - \left[\sum_{j=1}^{n} b_j - b_n\right]$$

$$\sum_{i=1}^{m} x_{in} = b_n; \qquad \text{since } \sum_{i=1}^{m} a_i = \sum_{j=1}^{m} b_j$$

Triangular Basis We know that the number of basic variables are equal to the number of constraints in linear programming. In the same way when capacity and requirement constraint equations are expressed in terms of basic variables and all non-basic variables are given zero value, the matrix of coefficients of variables in the system of equations is triangular, that is, there is an equation in which a single basic variable occurs; in the second equation one more basic variable is added but the total number of variables does not exceed two; similarly, in the third equation one more basic variable occurs, but total does not exceed three, and so on.

Theorem 9.3 The transportation problem has a triangular basis.

Proof To prove this theorem, consider capacity and requirement constraint equations represented in the tabular form (Table 9.1).

In the system of capacity and requirement constraint equations, every equation has a basic variable otherwise the equation cannot be satisfied for $a_i \neq 0$ or $b_j \neq 0$. Suppose, every row and column equation has at least two basic variables. Since there are m rows and n columns, therefore, the total number of basic variables in row equations and column equations will be at least $2m$ and $2n$, respectively. If N be the total number of basic variables, then obviously $N \geq 2m$, $N \geq 2n$. Now three cases may arise:

Case I : If $m > n$, then $m + m > m + n$, or $2m > m + n$. Thus $N \geq 2m \geq m + n$.
Case II : If $m < n$, then $m + n < n + n$ or $m + n < 2n$. Thus $N \geq 2n > m + n$.
Case III : If $m = n$, then $m + m = m + n$ or $2m = m + n$. Thus $N \geq 2m = m + n$.

In each of these cases, we observed that $N > m + n$. But the number of basic variables in the transportation problem are $N = m + n - 1$. This is a contradiction. Thus our assumption that every equation has at least two basic variables is wrong. Therefore, there is at least one equation, either row or column, having only one basic variable.

Let the rth equation have only one basic variable, and let x_{rt} be the only basic variable in row r and column t. Then $x_{rt} = a_r$. Eliminate rth row from the system of equations and substituting $x_{rt} = a_r$ in t-th column equation and replace b_t by $b_t' = b_t - a_r$.

After eliminating the r-th row, the system has $m - 1$ row equations and n column equations of which $m + n - 2$ are linearly independent. This implies that the number of basic variables are $m + n - 2$. Repeating the argument given earlier and conclude that in the reduced system of equation, there is an equation which has only one basic variable. But if this equation happens to be the t-th column equation in the original system, then it will have two basic variables. This indicates that in our original system of equations, there is an equation which has at least two basic variables.

Continue, repeating the argument to prove that the system has an equation which has at least three basic variables and so on.

Unbalanced Transportation Problem For a feasible solution to exist it is necessary that the total supply must equal total demand. That is,

$$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$$

But a situation may arise when the total available supply is not equal to the total requirement.

Case 1: When the supply exceeds demand, the constraints of the transportation problem will appear as,

$$\sum_{j=1}^{n} x_{ij} \leq a_i; \quad i = 1, 2, \ldots, m$$

$$\sum_{i=1}^{m} x_{ij} = b_j; \quad j = 1, 2, \ldots, n$$

$$x_{ij} \geq 0 \text{ for all } i, j$$

Adding slack variables $s_{i,\,n+1}$, $(i = 1, 2, \ldots, m)$ in the first m constraints, we get

$$\sum_{j=1}^{m} x_{ij} + s_{i,\,n+1} = a_i$$

$$\sum_{i=1}^{m}\left\{\sum_{j=1}^{n} x_{ij} + s_{i,n+1}\right\} = \sum_{i=1}^{m} a_i$$

$$\sum_{i=1}^{m} s_{i,n+1} = \sum_{i=1}^{m} a_i - \sum_{j=1}^{n} b_j = \text{excess supply available.}$$

If b_{n+1} denotes the excess supply available, then the modified transportation model can be stated as

$$\text{Minimize } Z = \sum_{i=1}^{m} \sum_{j=1}^{n} \left(c_{ij}x_{ij} + c_{i,n+1}\, s_{i,n+1}\right)$$

subject to

$$\sum_{j=1}^{n} x_{ij} + s_{i,n+1} = a_i; \quad i = 1, 2, \ldots, m$$

$$\sum_{j=1}^{n} x_{ij} = b_j \quad ; \quad j = 1, 2, \ldots, n+1$$

and $x_{ij} \geq 0$ for all i, j

where $c_{i,n+1} = 0$ $(i = 1, 2, \ldots, m)$ and

$$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j + b_{n+1} \quad \text{or} \quad b_{n+1} = \sum_{i=1}^{m} a_i - \sum_{j=1}^{n} b_j$$

It follows that, if $\sum_{i=1}^{m} a_i > \sum_{j=1}^{n} b_j$, then a dummy column (demand centre) can be added to the transportation table to absorb the excess supply. The unit transportation cost for the cells in this column is set equal to zero because these represent product items that are not being made and not being sent.

Case 2: When demand exceeds supply, the constraints of the transportation table will appear as

$$\sum_{j=1}^{n} x_{ij} = a_i; \quad i = 1, 2, \ldots, m$$

$$\sum_{i=1}^{m} x_{ij} \leq b_j; \quad j = 1, 2, \ldots, n$$

and $x_{ij} \geq 0$ for all i, j

Adding slack variables $s_{m+1,\,j}$ $(j = 1, 2, \ldots, n)$ in the last n constraints, we get

$$\sum_{j=1}^{n} x_{ij} = a_i \quad ; \quad i = 1, 2, \ldots, m$$

$$\sum_{i=1}^{m} x_{ij} + s_{m+1,j} = b_j; \quad j = 1, 2, \ldots, n$$

$$\sum_{j=1}^{n} s_{m+1,j} = \sum_{j=1}^{n} b_j - \sum_{i=1}^{m} a_i = \text{excess demand.}$$

If a_{m+1} denotes the excess demand, then the modified transportation model can be stated as:

$$\text{Minimize } Z = \sum_{i=1}^{m} \sum_{j=1}^{m} \left(c_{ij}x_{ij} + c_{m+1,j}\, s_{m+1,j}\right)$$

subject to

$$\sum_{j=1}^{n} x_{ij} = a_i \quad ; \quad i = 1, 2, \ldots, m+1$$

$$\sum_{i=1}^{m} x_{ij} + s_{m+1,j} = b_j; \quad j = 1, 2, \ldots, n$$

and $x_{ij} \geq 0$ for all i, j

where $c_{m+1,j} = 0$, for all j and

$$\sum_{i=1}^{m} a_i + a_{m+1} = \sum_{j=1}^{n} b_j \quad \text{or} \quad a_{m+1} = \sum_{j=1}^{n} b_j - \sum_{i=1}^{m} a_i$$

It follows that if $\sum_{j=1}^{n} b_j > \sum_{i=1}^{m} a_i$, then a dummy row (supply centre) can be added to the transportation table to account for excess demand quantity. The unit transportation cost here also for the cells in the dummy row is set equal to zero.

Chapter 10

Assignment Problem

"We don't have as many managers as we should, but we would rather have too few than too many."

– Larry Page

PREVIEW

An assignment problem is a particular case of a transportation problem where the given resources are allocated to an equal number of activities with an aim of either minimizing total cost, distance, time or maximizing profit. Travelling salesman is a specific application of assignment model.

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- understand the features of assignment problems and transportation problems.
- formulate an assignment problem as a square matrix.
- apply the Hungarian method to solve an assignment problem.
- make appropriate changes in the Hungarian method to solve an unbalanced assignment problem, profit maximization assignment problem, etc.
- solve a travelling salesman problem.

CHAPTER OUTLINE

10.1 Introduction
10.2 Mathematical Model of Assignment Problem
10.3 Solution Methods of Assignment Problem
- Conceptual Questions A
- Self Practice Problems A
- Hints and Answers

10.4 Variations of the Assignment Problem
- Conceptual Questions B
- Self Practice Problems B
- Hints and Answers

10.5 A Typical Assignment Problem
10.6 Travelling Salesman Problem
- Self Practice Problems C
- Hints and Answers

❑ Chapter Summary
❑ Chapter Concepts Quiz
❑ Case Study
❑ Appendix: Important Results and Theorems

10.1 INTRODUCTION

An assignment problem is a particular case of a transportation problem where the resources (say facilities) are assignees and the destinations are activities (say jobs). Given n resources (or facilities) and n activities (or jobs), with effectiveness (in terms of cost, profit, time, etc.) of each resource for each activity. Then problem becomes to assign (or allocate) each resource to only one activity (job) and vice-versa so that the given measure of effectiveness is optimized.

The problem of assignment arises because the resources that are available such as men, machines, etc., have varying degree of efficiency for performing different activities. Therefore, the cost, profit or time of performing different activities is also different. Thus, the problem becomes: *How should the assignments be made in order to optimize the given objective.*

Some of the problems where the assignment technique may be useful are assignment of (i) workers to machines, (ii) salesmen to different sales areas, (iii) clerks to various checkout counters, (iv) classes to rooms, (v) vehicles to routes, (vi) contracts to bidders, etc.

Assignment table is a convenient way to summarize available data.

10.2 MATHEMATICAL MODEL OF ASSIGNMENT PROBLEM

The general data matrix for assignment problem is shown in Table 10.1. It may be noted that this data matrix is the same as the transportation cost matrix except that the supply (or availability) of each of the resources and the demand at each of the destinations is taken to be one. It is due to this fact that assignments are made on a one-to-one basis.

Resources (*workers*)	*Activities* (*jobs*) J_1	J_2	. . .	J_n	*Supply*
W_1	c_{11}	c_{12}	. . .	c_{1n}	1
W_2	c_{21}	c_{22}	. . .	c_{2n}	1
.	.	.		.	.
.	.	.		.	.
.	.	.		.	.
W_n	c_{n1}	c_{n2}	. . .	c_{nn}	1
Demand	1	1	. . .	1	n

Table 10.1 Assignment Data Matrix

Suppose, x_{ij} represents the assignment of resource (facility) i to activity (job) j such that

$$x_{ij} = \begin{cases} 1 & \text{if resource } i \text{ is assigned to activity } j \\ 0 & \text{otherwise} \end{cases}$$

Then mathematical model of the assignment problem can be stated as:

$$\text{Minimize } Z = \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij} x_{ij}$$

subject to the constraints

$$\sum_{j=1}^{n} x_{ij} = 1, \text{ for all } i \text{ (resource availability)}$$

$$\sum_{i=1}^{n} x_{ij} = 1, \text{ for all } j \text{ (activity requirement)}$$

and $$x_{ij} = 0 \text{ or } 1, \text{ for all } i \text{ and } j$$

where c_{ij} represents the cost of assignment of resource i to activity j.

This mathematical model of assignment problem is a particular case of the transportation problem for two reasons: (i) the cost matrix is a square matrix, and (ii) the optimal solution table (matrix) for the problem would have only one assignment in a given row or a column.

Remark If a constant is added to or subtracted from every element of any row or column of the given cost matrix, then it would not change optimal assignments and value of objective function (See Appendix 10.A for proof.)

10.3 SOLUTION METHODS OF ASSIGNMENT PROBLEM

An assignment problem can be solved by any of the following methods:

- Enumeration method
- Transportation method
- Simplex method
- Hungarian method

1. **Enumeration Method** In this method, a list of all possible assignments among the given resources (men, machines, etc.) and activities (jobs, sales areas, etc.) is prepared. Then an assignment that involves the minimum cost (or maximum profit), time or distance is selected. If two or more assignments have the same minimum cost (or maximum profit), time or distance, the problem has multiple optimal solutions.

   In general, if an assignment problem involves n workers/jobs, then in total there are $n!$ possible assignments. For example, for an $n = 5$ workers/jobs problem, we have to evaluate a total of 5! or 120 assignments. However, when n is large, the method is unsuitable for manual calculations. Hence, this method is suitable only when the value of n is small.

2. **Simplex Method** Since each assignment problem can be formulated as a 0 or 1 integer linear programming problem, such a problem can also be solved by the simplex method. The general mathematical model of the assignment problem involves $n \times n$ decision variables and $n + n$ or $2n$ equalities. Thus, for any assignment problem that involves 5 workers/jobs, there will be 25 decision variables and 10 equalities. Solving such an assignment problem manually is difficult.

3. **Transportation Method** Since an assignment problem is a special case of the transportation problem, it can also be solved by using MODI method. However, every basic feasible solution of a general assignment problem that has a square matrix of order n should have $m + n - 1 = n + n - 1 = 2n - 1$ assignments. But due to the special structure of this problem, none of the solutions can have more than n assignments. Consequently, solutions will be degenerate. To remove degeneracy, $(n - 1)$ number of dummy allocations (deltas or epsilons) will be required in order to proceed with MODI method. Thus, the problem of degeneracy at each solution makes the procedure computationally inefficient for solving an assignment problem.

Hungarian method to solve an assignment problem is more efficient than using simplex method.

4. **Hungarian Method** The Hungarian method (developed by Hungarian mathematician D. Konig) is an efficient method of finding the optimal solution of an assignment problem without making a direct comparison of every solution. The method works on the principle of reducing the given cost matrix to a matrix of opportunity costs. Opportunity costs show the relative penalties associated with assigning a resource to an activity. Hungarian method reduces the cost matrix to the extent of having at least one zero in each row and column so as to make optimal assignments.

10.3.1 Hungarian Method for Solving Assignment Problem

The Hungarian method (minimization case) can be summarized in the following steps:

Step 1: Develop the cost matrix from the given problem If the number of rows are not equal to the number of columns, then add required number of dummy rows or columns. The cost element in dummy rows/columns are always zero.

Step 2: Find the opportunity cost matrix

(a) Identify the smallest element in each row of cost matrix and then subtract it from each element of that row, and

(b) In the reduced matrix obtained from 2(a), identify the smallest element in each column and then subtract it from each element of that column. Each row and column now have at least one zero element.

Step 3: Make assignments in the opportunity cost matrix The procedure of making assignments is as follows:

(a) First round for making assignments
- Identify rows successively from top to bottom until a row with exactly one zero element is found. Make an assignment to this single zero by making a square (□) around it. Then cross off (×) all other zeros in the corresponding column.
- Identify columns successively from left to right hand with exactly one zero element that has not been assigned. Make assignment to this single zero by making a square (□) around it and then cross off (×) all other zero elements in the corresponding row.

(b) Second round for making assignments
- If a row and/or column has two or more unmarked zeros and one cannot be chosen by inspection, then choose zero element arbitrarily for assignment.
- Repeat steps (a) and (b) successively until one of the following situations arise.

Step 4: Optimality criterion

(a) If all zero elements in the cost matrix are either marked with square (□) or are crossed off (×) and there is exactly one assignment in each row and column, then it is an optimal solution. The total cost associated with this solution is obtained by adding the original cost elements in the occupied cells.

(b) If a zero element in a row or column was chosen arbitrarily for assignment in Step 4(a), there exists an alternative optimal solution.

(c) If there is no assignment in a row (or column), then this implies that the total number of assignments are less than the number of rows/columns in the square matrix. In such a situation proceed to Step 5.

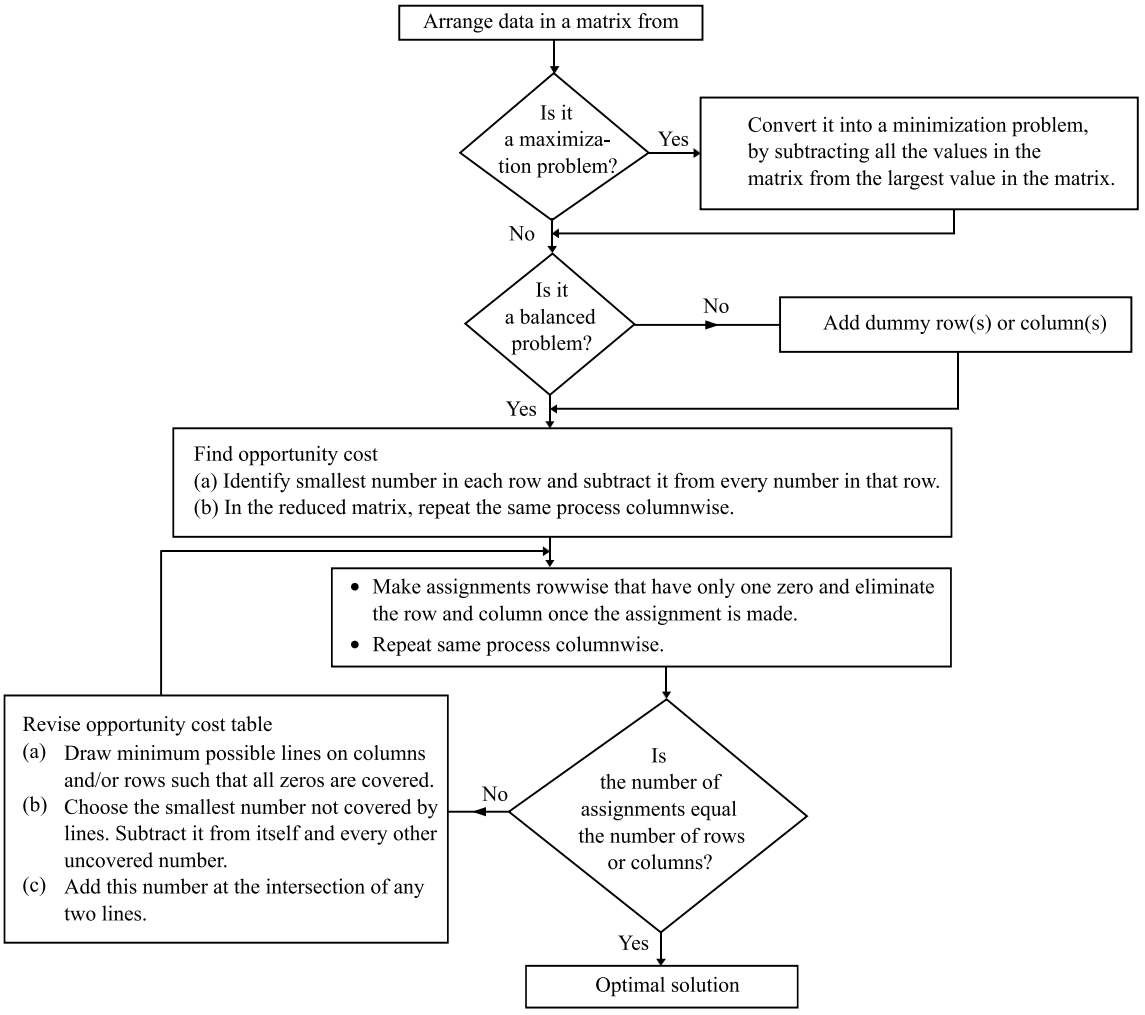


Fig. 10.1 Flow Chart of Hungarian Method

Step 5: Revise the opportunity cost matrix Draw a set of horizontal and vertical lines to cover all the zeros in the revised cost matrix obtained from Step 3, by using the following procedure:

(a) For each row in which no assignment was made, mark a tick (✓)
(b) Examine the marked rows. If any zero element is present in these rows, mark a tick (✓) to the respective columns containing zeros.
(c) Examine marked columns. If any assigned zero element is present in these columns, tick (✓) the respective rows containing assigned zeros.
(d) Repeat this process until no more rows or columns can be marked.
(e) Draw a straight line through each marked column and each unmarked row.

If the number of lines drawn (or total assignments) is equal to the number of rows (or columns), the current solution is the optimal solution, otherwise go to Step 6.

Zeros in the assignment table indicate opportunity cost and show the penalty of not making the least cost (or best) assignment.

Step 6: Develop the new revised opportunity cost matrix

(a) Among the elements in the matrix not covered by any line, choose the smallest element. Call this value k.
(b) Subtract k from every element in the matrix that is not covered by a line.
(c) Add k to every element in the matrix covered by the two lines, i.e. intersection of two lines.
(d) Elements in the matrix covered by one line remain unchanged.

Step 7: Repeat steps Repeat Steps 3 to 6 until an optimal solution is obtained.

The flow chart of steps in the Hungarian method for solving an assignment problem is shown in Fig. 10.1.

Example 10.1 A computer centre has three expert programmers. The centre wants three application programmes to be developed. The head of the computer centre, after carefully studying the programmes to be developed, estimates the computer time in minutes required by the experts for the application programmes as follows:

		Programmers		
		A	*B*	*C*
Programmes	*1*	120	100	80
	2	80	90	110
	3	110	140	120

Assign the programmers to the programmes in such a way that the total computer time is minimum.

Solution: Steps 1 and 2 The minimum time element in rows 1, 2 and 3 is 80, 80 and 110, respectively. Subtract these elements from all elements in their respective row. The reduced time matrix is shown in Table 10.2(a).

(a)

	A	*B*	*C*
1	40	20	0
2	0	10	30
3	0	30	10

(b)

	A	*B*	*C*
1	40	10	0
2	0	0	30
3	0	20	10

Table 10.2

In reduced Table 10.2(a) the minimum time element in columns A, B and C is 0, 10 and 0, respectively. Subtract these elements from all elements in their respective column in order to get the reduced time matrix. This is shown in Table 10.2(b).

Step 3 (a) Examine all the rows starting from the first, one-by-one, until a row containing single zero element is found. In Table 10.2(b) rows 1 and 3 have only one zero in the cells (1, C) and (3, A), respectively. Make an assignment in these cells and cross off all zero elements in the assigned column as shown in Table 10.3(a).

(a)

	A	*B*	*C*
1	40	20	[0]
2	~~0~~	0	30
3	[0]	30	10

(b)

	A	*B*	*C*
1	40	10	[0]
2	~~0~~	[0]	30
3	[0]	20	10

Table 10.3

(b) Now examine each column starting from *A* in Table 10.3(a). There is one zero in column *B* in the cell (2, *B*). Make an assignment in this cell as shown in Table 10.3(b).

(c) Since the number of assignments (= 3) equals the number of rows (= 3), the optimal solution is obtained.

The pattern of assignments among programmers and programmes with their respective time (in minutes) is given below:

Programmer	*Programme*	*Time* (*in minutes*)
1	C	80
2	B	90
3	A	110
	Total	280

Example 10.2 A department of a company has five employees with five jobs to be performed. The time (in hours) that each man takes to perform each job is given in the effectiveness matrix.

		Employees				
		I	*II*	*III*	*IV*	*V*
Jobs	*A*	10	5	13	15	16
	B	3	9	18	13	6
	C	10	7	2	2	2
	D	7	11	9	7	12
	E	7	9	10	4	12

How should the jobs be allocated, one per employee, so as to minimize the total man-hours?

Solution Applying Step 2 of Hungarian algorithm, the reduced opportunity time matrix is shown in Table 10.4(a).

(a)

	I	*II*	*III*	*IV*	*V*
A	5	0	8	10	11
B	0	6	15	10	3
C	8	5	0	0	0
D	0	4	2	0	5
E	3	5	6	0	8

(b)

	I	*II*	*III*	*IV*	*V*
A	5	[0]	8	10	11
B	[0]	6	15	10	3
C	8	5	[0]	~~0~~	~~0~~
D	~~0~~	4	2	~~0~~	5
E	3	5	6	[0]	8

Table 10.4

Steps 3 and 4: (a) Examine all the rows starting from A, one-by-one, until a row containing only single zero element is found. Rows *A*, *B* and *E* have only one zero element in the cells (*A*, *II*), (*B*, *I*) and (*E*, *IV*). Make an assignment in these cells, and cross off all zeros in the assigned columns as shown in Table 10.4(b).

(b) Now examine each column starting from column *I*. There is one zero in column III, cell (*C*, *III*). Assignment is made in this cell. Thus cell (*C*, *V*) is crossed off. All zeros in the table are now either assigned or crossed off as shown in Table 10.4(b). The solution is not optimal because only four assignments are made.

(a)

	I	*II*	*III*	*IV*	*V*	
A	~~5~~	[0]	8	10	11	
B	[0]	6	15	10	3	✓
C	~~8~~	~~5~~	[0]	~~0~~	~~0~~	
D	~~0~~	4	2	~~0~~	5	✓
E	~~3~~	5	6	[0]	8	✓
	✓			✓		

(b)

	I	*II*	*III*	*IV*	*V*
A	7	0	8	12	11
B	0	4	13	10	1
C	10	5	0	2	0
D	0	2	0	0	3
E	3	3	4	0	6

Table 10.5

Step 5: Cover the zeros with minimum number of lines (= 4) as explained below:
(a) Mark (✓) row *D* where there is no assignment.
(b) Mark (✓) columns *I* and *IV* since row *D* has zero element in these columns.
(c) Mark (✓) rows *B* and *E* since columns *I* and *IV* have an assignment in rows *B* and *E*, respectively.
(d) Since no other rows or columns can be marked, draw straight lines through the unmarked rows *A* and *C* and the marked columns *I* and *IV*, as shown in Table 10.5(a).

Step 6: Develop the revised matrix by selecting the smallest element among all uncovered elements by the lines in Table 10.5(a); viz., 2. Subtract $k = 2$ from uncovered elements including itself and add it to elements 5, 10, 8 and 0 in cells (*A*, *I*), (*A*, *IV*), (*C*, *I*) and (*C*, *IV*), respectively, which lie at the intersection of two lines. The revised matrix, so obtained is shown in Table 10.5(b).

Step 7: Repeat Steps 3 to 6 to find a new solution. The new assignments are shown in Table 10.6.

Table 10.6

	I	*II*	*III*	*IV*	*V*
A	7	[0]	8	12	11
B	[0]	4	13	10	1
C	10	5	~~0~~	2	[0]
D	~~0~~	2	[0]	~~0~~	3
E	3	3	4	[0]	6

Since the number of assignments (= 5) equals the number of rows (or columns), the solution is optimal. The pattern of assignments among jobs and employees with their respective time (in hours) is given below:

Job	*Employee*	*Time (in hours)*
A	*II*	5
B	*I*	3
C	*V*	2
D	*III*	9
E	*IV*	4
	Total	23

Example 10.3 A solicitors' firm employs typists on hourly piece-rate basis for their daily work. There are five typists and their charges and speed are different. According to an earlier understanding only one job was given to one typist and the typist was paid for a full hour, even if he worked for a fraction of an hour. Find the least cost allocation for the following data:

Typist	*Rate per hour (Rs)*	*No. of Pages Typed/Hour*
A	5	12
B	6	14
C	3	8
D	4	10
E	4	11

Job	*No. of Pages*
P	199
Q	175
R	145
S	298
T	178

[*Delhi Univ., MBA, 2000, 2002*]

Solution Develop a cost matrix based on the data of the problem as shown in Table 10.7(a), where elements represent the cost to be incurred due to assignment of jobs to various typists on a one-to-one basis.

Table 10.7

(a)

	P	*Q*	*R*	*S*	*T*
A	85	75	65	125	75
B	90	78	66	132	78
C	75	66	57	114	69
D	80	72	60	120	72
E	76	64	56	112	68

(b)

	P	*Q*	*R*	*S*	*T*
A	2	2	~~0~~	4	0
B	6	4	[0]	10	2
C	0	1	~~0~~	1	2
D	2	4	~~0~~	4	2
E	2	0	~~0~~	0	2

Applying Step 2 of Hungarian method, the reduced opportunity cost matrix is shown in Table 10.7(b).

To make assignments in Table 10.7(b), examine all the rows starting from *A* until a row containing only one zero element is found. Rows *B* and *D* have only one zero element in the cells (*B, R*) and (*D, R*), respectively. First make assignment in cells (*B*, *R*), and cross off all zeros in the assigned columns as shown in Table 10.7(b).

Now examine each column starting from column *P.* There is one zero in columns *P, Q, S* and *T* in the cells (*C, P*), (*E, Q*), (*E, S*) and (*A*, *T*). Assignment is made in these cells. All zeros in the matrix are either assigned or crossed off, as shown in Table 10.8(a).

(a)

	P	*Q*	*R*	*S*	*T*	
A	~~2~~	~~2~~	~~0~~	~~4~~	0	
B	6	4	0	10	2	✓
C	0	~~1~~	~~0~~	~~1~~	~~2~~	
D	2	4	~~0~~	4	2	✓
E	~~2~~	0	~~0~~	~~0~~	~~2~~	
			✓			

(b)

	P	*Q*	*R*	*S*	*T*
A	2	2	2	4	0
B	4	2	0	8	0
C	0	1	2	1	2
D	0	2	0	2	0
E	2	0	2	0	2

Table 10.8

The solution shown in Table 10.8(a) is not optimal since only four assignments are made. Thus, in order to get the next best solution, apply following steps.

(a) Mark (✓) row *D* since it has no assignment.
(b) Mark (✓) column *R* since row *D* has zero in this column.
(c) Mark (✓) row *B* since column *R* has an assignment in row *B*.
(d) Since no other rows or columns can be marked, therefore draw straight lines through the unmarked rows *A, C* and *E* and marked column *R* as shown in Table 10.8(a).

Develop the revised cost matrix by selecting the smallest element among all uncovered elements by the lines (i.e., $k = 2$) in Table 10.8(a). Subtract this element ($k = 2$) from all uncovered elements including itself and add it to elements in the cells (*A*, *R*), (*C*, *R*) and (*E, R*), respectively which lie at the intersection of two lines. Another revised cost matrix so obtained is shown in Table 10.8(b).

Again repeat the procedure to find a new solution. The new assignments are shown in Table 10.9(a).

(a)

	P	*Q*	*R*	*S*	*T*	
A	~~2~~	~~2~~	2	~~4~~	0	✓
B	~~4~~	~~2~~	0	~~8~~	~~0~~	✓
C	0	~~1~~	~~2~~	~~1~~	~~2~~	✓
D	~~0~~	~~2~~	~~0~~	~~2~~	~~0~~	✓
E	~~2~~	0	~~2~~	~~0~~	~~2~~	
	✓		✓		✓	

(b)

	P	*Q*	*R*	*S*	*T*
A	2	1	2	3	0
B	4	1	0	7	~~0~~
C	~~0~~	~~0~~	2	0	2
D	0	1	~~0~~	1	~~0~~
E	3	0	3	~~0~~	3

Table 10.9

The solution shown in Table 10.9(a) is also not optimal since only four assignments are made. Thus, to get the next best solution, follow Steps 6(a) to (d) of the algorithm to draw a minimum number of horizontal and vertical lines to cover all zero elements in Table 10.9(a). The new opportunity cost matrix obtained from Table 10.9(a) by subtracting the smallest element (= l) among all uncovered elements, including itself, by the lines and adding it to elements at the intersection of two lines is shown in Table 10.9(b).

The new solution obtained by repeating the procedure as explained earlier is shown in Table 10.9(b). Since both columns *Q* and *S* have two zeros, the arbitrary selection of a cell in any of these columns will give us an alternative solution with same total cost of assignment.

The pattern of assignments among typists and jobs, along with cost is as follows:

Typist	*Job*	*Cost (in Rs)*
A	*T*	75
B	*R*	66
C	*S*	114
D	*P*	80
E	*Q*	64
	Total	399

CONCEPTUAL QUESTIONS A

1. What is an assignment problem? Give two applications.
2. Give the mathematical formulation of an assignment problem. How does it differ from a transportation problem?
3. Explain the conceptual justification that an assignment problem can be viewed as a linear programming problem.
4. Explain the difference between a transportation problem and an assignment problem.
5. Specify the dual of an assignment problem. What are the techniques used for solving an assignment problem?
6. State and discuss the methods of solving an assignment problem. How is the Hungarian method better than other methods for solving an assignment problem?
7. (a) Give an algorithm to solve an assignment problem.
   (b) Show that an assignment problem is a special case of a transportation problem.
8. Explain how an assignment problem can be solved by using the transportation approach?

SELF PRACTICE PROBLEMS A

1. Five men are available to do five different jobs. From past records, the time (in hours) that each man takes to do each job is known and is given in the following table:

		Jobs				
		I	*II*	*III*	*IV*	*V*
Men	*A*	2	9	2	7	1
	B	6	8	7	6	1
	C	4	6	5	3	1
	D	4	2	7	3	1
	E	5	3	9	5	1

Find out how men should be assigned the jobs in way that will minimize the total time taken.

2. A pharmaceutical company producing a single product sold it through five agencies situated in different cities. All of a sudden, there rouse a demand for the product in another five cities that didn't any agency of the company. The company is now facing the problem of deciding on how to assign the existing agencies in order to despatch the product to needy cities in such a way that the travelling distance is minimized. The distance between the surplus and deficit cities (in km) is given in the following table.

		Deficit cities				
		a	*b*	*c*	*d*	*e*
Surplus Cities	*A*	160	130	115	190	200
	B	135	120	130	160	175
	C	140	110	125	170	185
	D	50	50	80	80	110
	E	55	35	80	80	105

Determine the optimum assignment schedule.

3. A national truck rental service has a surplus of one truck in each of the cities, 1, 2, 3, 4, 5 and 6; and a deficit of one truck in each of the cities 7, 8, 9, 10, 11 and 12. The distances (in km) between the cities with a surplus and cities with deficit are displayed in the table below:

		To					
		7	*8*	*9*	*10*	*11*	*12*
From	*1*	31	62	29	42	15	41
	2	12	19	39	55	71	40
	3	17	29	50	41	22	22
	4	35	40	38	42	27	33
	5	19	30	29	16	20	23
	6	72	30	30	50	41	20

How should the trucks be displayed so as to minimize the total distance travelled?

4. An air freight company picks up and delivers freight where customers want. The company has two types of aircraft, *X* and *Y*, with equal loading capacities but different operations costs. These are shown in the following Table.

Type of Aircraft	*Operating Costs (Rs)*	
	Empty	*Loaded*
X	1.00	2.00
Y	1.50	3.00

The present four locations of the aircraft that the company has are; *J* – *X*; *K* – *Y*; *L* – *Y*, and *M* – *X*. Four customers of the company located at A, B, C and D want to transport nearly the same size of load to their final destinations. The final destinations are 600, 300, 1,000 and 500 km from the loading points A, B, C and D, respectively.

The distances (in km) between the aircraft and the loading points are as follows:

		Loading Point			
		A	*B*	*C*	*D*
Aircraft Location	*J*	200	200	400	100
	K	300	100	300	300
	L	400	100	100	500
	M	200	200	400	200

Determine the allocations which minimize the total cost of transportation.

5. A departmental head has four subordinates and four tasks to be performed. The subordinates differ in efficiency and the tasks differ in their intrinsic difficulty. His estimates of the times that each man would take to perform each task is given in the matrix below:

		Tasks			
		I	*II*	*III*	*IV*
Subordinates	*A*	8	26	17	11
	B	13	28	4	26
	C	38	19	18	15
	D	19	26	24	10

How should the tasks be allocated to subordinates so as to minimize the total man-hours?

6. An automobile dealer wishes to put four repairmen to four different jobs. The repairmen have somewhat different kinds of skills and they exhibit different levels of efficiency from one job to another. The dealer has estimated the number of man-hours that would be required for each job-man combination. This is given in matrix form in the following table:

		Jobs			
		A	*B*	*C*	*D*
Men	*1*	5	3	2	8
	2	7	9	2	6
	3	6	4	5	7
	4	5	7	7	8

Find the optimal assignment that will result in minimum man-hours needed.

7. A lead draftsman has five drafting tasks to accomplish and five idle draftsmen. Each draftsman is estimated to require the following number of hours for each task.

		Tasks				
		A	*B*	*C*	*D*	*E*
Draftsmen	*1*	60	50	100	85	95
	2	65	45	100	75	90
	3	70	60	110	97	85
	4	70	55	105	90	93
	5	60	40	120	85	97

If each draftsman costs the company Rs 15.80 per hour, including overhead, find the assignment of draftsmen to tasks that will result in the minimum total cost. What would be the total cost?

8. A construction company has requested bids for subcontracts on five different projects. Five companies have responded. Their bids are represented below.

		Bid Amounts ('000s Rs)				
		I	*II*	*III*	*IV*	*V*
Bidders	*1*	41	72	39	52	25
	2	22	29	49	65	81
	3	27	39	60	51	40
	4	45	50	48	52	37
	5	29	40	45	26	30

Determine the minimum cost assignment of subcontracts to bidders, assuming that each bidder can receive only one contract.

9. A shipbuilding company has been awarded a big contract for the construction of five cargo vessels. The contract stipulates that the company must subcontract a portion of the total work to at least five small ancillary companies. The company has invited bids from the small ancillary companies (A_1, A_2, A_3, A_4, and A_5) to take care of the subcontract work in five fields – materials testing, fabrication, assembly, scrap removal and painting. The bids received from the ancillary companies are given in the table.

Subcontract Bids (Rs)

Ancillary Companies	*Materials Testing*	*Fabrication*	*Assembly*	*Scrap Removal*	*Painting*
A_1	2,50,000	3,00,000	3,80,000	5,00,000	1,50,000
A_2	2,80,000	2,60,000	3,50,000	5,00,000	2,00,000
A_3	3,00,000	3,50,000	4,00,000	5,50,000	1,80,000
A_4	1,50,000	2,50,000	3,00,000	4,80,000	1,20,000
A_5	3,00,000	2,70,000	3,20,000	4,80,000	1,60,000

Which bids should the company accept in order to complete the contract at minimum cost? What is the total cost of the subcontracts?

10. In a textile sales emporium, four salesmen A, B, C and D are available to four counters *W*, *X*, *Y* and *Z*. Each salesman can handle any counter. The service (in hours) of each counter when manned by each salesman is given below:

		Salesmen			
		A	*B*	*C*	*D*
Counters	*W*	41	72	39	52
	X	22	29	49	65
	Y	27	39	60	51
	Z	45	50	48	52

How should the salesmen be allocated to appropriate counters so that the service time is minimized? Each salesman must handle only one counter.

11. A hospital wants to purchase three different types of medical equipments and five manufacturers have come forward to supply one or all the three machines. However, the hospital's policy is not to accept more than one machine from any one of the manufacturers. The data relating to the price (in thousand of rupees) quoted by the different manufacturers is given below:

		Machines		
		1	*2*	*3*
Manufacturers	*A*	30	31	27
	B	28	29	26
	C	29	30	28
	D	28	31	27
	E	31	29	26

Determine how best the hospital can purchase the three machines. [*Delhi Univ., MBA* (*HCA*)*, 2008*]

12. The secretary of a school is taking bids on the city's four school bus routes. Four companies have made the bids (in Rs), as detailed in the following table:

		Route 1	*Route 2*	*Route 3*	*Route 4*
Bus	*1*	4,000	5,000	–	–
	2	–	4,000	–	4,000
	3	3,000	–	2,000	–
	4	–	–	4,000	5,000

Suppose each bidder can be assigned only one route. Use the assignment model to minimize the school's cost of running the four bus routes.

13. A large oil company operating a number of drilling platforms in the North Sea is forming a high speed rescue unit in order to cope with emergency situations that may occur. The rescue unit comprises 6 personnel who, for reasons of flexibility, undergo the same comprehensive training programme. The six personnel are assessed as to their suitability for various specialist tasks and the marks they received in the training programme are given in the following table:

	Trainee Number					
Specialist Task	*I*	*II*	*III*	*IV*	*V*	*VI*
Unit Leader	21	5	21	15	15	28
Helicopter Pilot	30	11	16	8	16	4
First Aid	28	2	11	16	25	25
Drilling Technology	19	16	17	15	19	8
Firefighting	26	21	22	28	29	24
Communications	3	21	21	11	26	26

Based on the marks awarded, what role should each of the trainees be given in the rescue unit?

14. The personnel manager of ABC Company wants to assign Mr *X*, Mr *Y* and Mr *Z* to regional offices. But the firm also has an opening in its Chennai office and would send one of the three to that branch if it were more economical than a move to Delhi, Mumbai or Kolkata. It will cost Rs 2,000 to relocate Mr *X* to Chennai, Rs 1,600 to reallocate Mr *Y* there, and Rs 3,000 to move Mr *Z*. What is the optimal assignment of personnel to offices?

		Office		
		Delhi	*Mumbai*	*Kolkata*
Personnel	*Mr X*	1,600	2,200	2,400
	Mr Y	1,000	3,200	2,600
	Mr Z	1,000	2,000	4,600

HINTS AND ANSWERS

1. $A-III$, $B-V$, $C-I$, $D-IV$, $E-II$; Optimal value = 13 hours.
2. $A-e$, $B-c$, $C-b$, $D-a$, $E-d$; Minimum distance = 570 km.
4. 1 – 11, 2 – 8, 3 – 7, 4 – 9, 5 – 10, 6 – 12; Minimum distance = 125 km.
5. $A-I$, $B-III$, $C-II$, $D-IV$; Total man-hours = 41 hours.
6. $1-B$, $2-C$, $3-D$, $4-A$; $1-C$, $2-D$, $3-B$, $4-A$; Total man-hours = 17 hours.
7. $1-A$, $2-D$, $3-E$, $4-C$, $5-B$; Minimum cost = Rs 365 × 15.8
8. $1-V$, $2-II$, $3-I$, $4-III$, $5-IV$; Minimum cost = Rs 155.
9. A_1 – scrap, A_2 – Fabrication, A_3 – Painting, A_4 – Testing, A_5 – Assembly; Minimum cost = Rs 14,10,000
10. $W-C$, $X-B$, $Y-A$, $Z-D$; Optimal value = 147 hours.

10.4 VARIATIONS OF THE ASSIGNMENT PROBLEM

10.4.1 Multiple Optimal Solutions

While making an assignment in the reduced assignment matrix, it is possible to have two or more ways to strike off a certain number of zeros. Such a situation indicates that there are multiple optimal solutions with the same optimal value of objective function.

10.4.2 Maximization Case in Assignment Problem

If instead of cost matrix, a profit (or revenue) matrix is given, then assignments are mode in such a way that total profit is maximized. The profit maximization assignment problems are solved by converting them into a cost minimization problem in either of the following two ways:

(i) Put a negative sign before each of the elements in the profit matrix in order to convert the profit values into cost values.

(ii) Locate the largest element in the profit matrix and then subtract all the elements of the matrix from the largest element including itself.

The transformed assignment problem can be solved by using usual Hungarian method.

Example 10.4 A company operates in four territories, and four salesmen available for an assignment. The territories are not equally rich in their sales potential. It is estimated that a typical salesman operating in each territory would bring in the following annual sales:

Territory	:	*I*	*II*	*III*	*IV*
Annual sales (Rs)	:	1,26,000	1,05,000	84,000	63,000

The four salesmen also differ in their ability. It is estimated that, working under the same conditions, their yearly sales would be proportionately as follows:

Salesmen	:	*A*	*B*	*C*	*D*
Proportion	:	7	5	5	4

If the criterion is maximum expected total sales, the intuitive answer is to assign the best salesman to the richest territory, the next best salesman to the second richest, and so on; verify this answer by the assignment technique. [*Delhi Univ., MBA, 2004*]

Solution *Construct the Effectiveness Matrix:* To avoid the fractional values of annual sales of each salesman in each territory, for convenience, consider their yearly sales as 21 (i.e. the sum of sales proportions), taking Rs 1,000 as one unit. Now divide the individual sales in each territory by 21 in order to obtain the required annual sales by each salesman. The maximum sales matrix so obtained is given in Table 10.10.

		Territory				
		I	*II*	*III*	*IV*	*Sales Proportion*
Salesman	*A*	42	35	28	21	7
	B	30	25	20	15	5
	C	30	25	20	15	5
	D	24	20	16	12	4
Sales (*in '000 Rs*)		6	5	4	3	

Table 10.10 Effectiveness Matrix

Converting Maximization Problem into Minimization Problem: The given maximization assignment problem (Table 10.10) can be converted into a minimization assignment problem by subtracting from the highest element (i.e. 42) all the elements of the given table. The new data so obtained is given in Table 10.11(a).

(a)

	I	*II*	*III*	*IV*
A	0	7	14	21
B	12	17	22	27
C	12	17	22	27
D	18	22	26	30

(b)

	I	*II*	*III*	*IV*
A	0	3	6	9
B	0	1	2	3
C	0	1	2	3
D	0	0	0	0

Table 10.11 Equivalent Cost Matrix

Apply Hungarian Method to get Optimal Solution: Apply Step 2 of the Hungarian method in order to get the reduced matrix that has at least one zero, either in a row or column, as shown in Table 10.11(b).

The assignment is made in row *A*. All zeros in the assigned column *I* are crossed as shown in Table 10.12. Column *II* has only one zero, in cell (*D*, *II*). Assignment is made in this column, and other zeros are crossed in row *D*. Now all zeros are either assigned or crossed off as shown in Table 10.12.

	I	*II*	*III*	*IV*	
A	[0]	3	6	9	✓
B	~~0~~	1	2	3	✓
C	~~0~~	1	2	3	✓
D	~~0~~	[0]	~~0~~	~~0~~	
	✓				

Table 10.12

The solution shown in Table 10.12 is not optimal since only two assignments are made. Draw lines through marked columns and unmarked rows to cover zeros as shown in Table 10.12.

Develop the revised cost matrix by selecting the minimum element (= 1) among all uncovered elements by the lines. Subtract 1 from each uncovered element, including itself, and add it to the elements at the intersection of two lines. A revised cost table so obtained is shown in Table 10.13.

Table 10.13

	I	*II*	*III*	*IV*	
A	0	2	5	8	✓
B	~~0~~	0	1	2	✓
C	~~0~~	~~0~~	1	2	✓
D	1	~~0~~	0	~~0~~	
	✓	✓			

Repeat Steps 1 to 3 to mark the assignments in Table 10.13. Two alternative optimal assignments are shown in Tables 10.14(a) and (b).

Table 10.14 Alternative Optimal Solutions

(a)

	I	*II*	*III*	*IV*
A	0	2	4	7
B	~~0~~	~~0~~	0	1
C	~~0~~	0	~~0~~	1
D	2	1	~~0~~	0

(b)

	I	*II*	*III*	*IV*
A	0	2	4	7
B	~~0~~	0	~~0~~	1
C	~~0~~	~~0~~	0	1
D	2	1	~~0~~	0

The pattern of two alternative optimal assignments among territories and salesmen with their respective sales volume (in Rs 1,000) is given in the table.

Assignment Set I

Salesman	*Territory*	*Sales* (*Rs*)
A	I	42
B	III	20
C	II	25
D	IV	12
	Total	99

Assignment Set II

Salesman	*Territory*	*Sales* (*Rs*)
A	I	42
B	II	25
C	III	20
D	IV	12
	Total	99

Example 10.5 A marketing manager has five salesmen and five sales districts. Considering the capabilities of the salesmen and the nature of districts, the marketing manager estimates that the sales per month (in hundred rupees) for each salesman in each district would be as follows:

		Districts				
		A	*B*	*C*	*D*	*E*
Salesmen	*1*	32	38	40	28	40
	2	40	24	28	21	36
	3	41	27	33	30	37
	4	22	38	41	36	36
	5	29	33	40	35	39

Find the assignment of salesmen to districts that will result in maximum sales.

Solution The given maximization problem can be converted into a minimization problem by subtracting from the largest element (i.e. 41) all the elements of the given table. The new cost data so obtained is given in Table 10.15.

Table 10.15 Equivalent Cost Data

(a)

	A	*B*	*C*	*D*	*E*
1	9	3	1	13	1
2	1	17	13	20	5
3	0	14	8	11	4
4	19	3	0	5	5
5	12	8	1	6	2

(b)

	A	*B*	*C*	*D*	*E*
1	8	0	0	7	0
2	0	14	12	14	4
3	0	12	8	6	4
4	19	1	0	0	5
5	11	5	0	0	1

Apply Step 2 of the Hungarian method to get the opportunity cost table as shown in Table 10.15(b).

Make assignments in Table 10.15(b) by applying Hungarian method as shown in Table 10.16(a).

(a)

	A	*B*	*C*	*D*	*E*	
1	~~8~~	[0]	~~0~~	7	~~0~~	
2	[0]	14	12	14	4	✓
3	~~0~~	12	8	6	4	✓
4	~~19~~	1	~~0~~	[0]	5	
5	~~11~~	5	[0]	~~0~~	1	
	✓					

(b)

	A	*B*	*C*	*D*	*E*
1	12	0	0	7	0
2	0	10	8	10	0
3	0	8	4	2	0
4	23	1	0	0	5
5	15	5	0	0	1

Table 10.16

The solution shown in Table 10.16(a) is not optimal since only four assignments are made. Cover the zeros with the minimum number of lines (= 4) as shown in Table 10.16(a).

Develop the revised cost matrix by selecting the minimum element (= 4) among all uncovered elements by the lines. Subtract 4 from all uncovered elements, including itself, and add it to the element at the intersection of the lines. A revised cost table, so obtained, is shown in Table 10.16(b).

Repeat the above procedure again to make the assignments in the reduced Table 10.16(b). The two alternative assignments are shown in Tables 10.17(a) and (b). Two more alternative solutions exist due to presence of zero element in cells (4, *C*), (4, *D*) and cells (5, *C*), (5, *D*).

(a)

	A	*B*	*C*	*D*	*E*
1	12	[0]	~~0~~	7	~~0~~
2	[0]	12	10	12	~~0~~
3	~~0~~	10	4	2	[0]
4	23	1	[0]	~~0~~	5
5	15	5	~~0~~	[0]	1

(b)

	A	*B*	*C*	*D*	*E*
1	12	[0]	~~0~~	7	~~0~~
2	~~0~~	12	10	12	[0]
3	[0]	10	4	2	~~0~~
4	23	1	[0]	~~0~~	5
5	15	5	~~0~~	[0]	1

Table 10.17 Alternative Optimal Solutions

Two alternative optimal assignments are as follows:

Assignment Set I

Salesman	*District*	*Sales* (*in '000 Rs*)
1	*B*	38
2	*A*	40
3	*E*	37
4	*C*	41
5	*D*	35
	Total	191

Assignment Set II

Salesman	*District*	*Sales* (*in '000 Rs*)
1	*B*	38
2	*E*	36
3	*A*	41
4	*C*	41
5	*D*	35
	Total	191

10.4.3 Unbalanced Assignment Problem

The Hungarian method for solving an assignment problem requires that the number of columns and rows in the assignment matrix should be equal. However, when the given cost matrix is not a square matrix, the assignment problem is called an *unbalanced problem.* In such cases before applying Hungarian method, dummy row(s) or column(s) are added in the matrix (with zeros as the cost elements) in order to make it a square matrix.

10.4.4 Restrictions on Assignments

Sometimes it may so happen that a particular resource (say a man or machine) cannot be assigned to a particular activity (say territory or job). In such cases, the cost of performing that particular activity by a

particular resource is considered to be very large (written as M or ∞) so as to prohibit the entry of this pair of resource-activity into the final solution.

Example 10.6 In the modification of a plant layout of a factory four new machines M_1, M_2, M_3, and M_4 are to be installed in a machine shop. There are five vacant places *A, B, C, D* and *E* available. Because of limited space, machine M_2 cannot be placed at *C* and M_3 cannot be placed at *A*. The cost of locating a machine at a place (in hundred rupees) is as follows.

		Location				
		A	*B*	*C*	*D*	*E*
Machine	M_1	9	11	15	10	11
	M_2	12	9	–	10	9
	M_3	–	11	14	11	7
	M_4	14	8	12	7	8

Find the optimal assignment schedule. [*Delhi Univ., MBA, 2004, 2006*]

Solution Since cost matrix is not balanced, add one dummy row (machine) with a zero cost elements in that row. Also assign a high cost, denoted by *M,* to the pair (M_2, *C*) and (M_3, *A*). The cost matrix so obtained is given in Table 10.18(a).

Apply Hungarian method for solving this problem. An optimal assignment is shown in Table 10.18(b).

Table 10.18

(a)

	A	*B*	*C*	*D*	*E*
M_1	9	11	15	10	11
M_2	12	9	*M*	10	9
M_3	*M*	11	14	11	7
M_4	14	8	12	7	8
M_5	0	0	0	0	0

(b)

	A	*B*	*C*	*D*	*E*
M_1	[0]	2	6	1	2
M_2	3	[0]	*M*	1	~~0~~
M_3	*M*	4	7	4	[0]
M_4	7	1	5	[0]	1
M_5	~~0~~	~~0~~	[0]	~~0~~	~~0~~

The total minimum cost (Rs) and optimal assignments made are as follows:

Machine	*Location*	*Cost (in '000 Rs)*
M_1	*A*	9
M_2	*B*	9
M_3	*E*	7
M_4	*D*	7
M_5 (*dummy*)	*C*	0
	Total	32

Example 10.7 An airline company has drawn up a new flight schedule that involves five flights. To assist in allocating five pilots to the flights, it has asked them to state their preference scores by giving each flight a number out of 10. The higher the number, the greater is the preference. A few of these flights are unsuitable to some pilots, owing to domestic reasons. These have been marked with '×'.

		Flight Number				
		1	*2*	*3*	*4*	*5*
Pilot	*A*	8	2	×	5	4
	B	10	9	2	8	4
	C	5	4	9	6	×
	D	3	6	2	8	7
	E	5	6	10	4	3

What should be the allocation of the pilots to flights in order to meet as many preferences as possible? [*AIMA* (*Dip. in Mgt.*), *2005*]

Solution Since the problem is to maximize the total preference score, in order to apply the Hungarian method to solve this assignment problem, the equivalent cost matrix is required. This is obtained by subtracting all the elements of the given matrix from the largest element (= 10) including itself as shown in Table 10.19.

	1	*2*	*3*	*4*	*5*
A	2	8	*M*	5	6
B	0	1	8	2	6
C	5	6	1	4	*M*
D	7	4	8	2	3
E	5	4	0	6	7

Table 10.19 Equivalent Cost Matrix

Perform the Hungarian method on Table 10.19 to make assignments as shown in Table 10.20.

	1	*2*	*3*	*4*	*5*	
A	[0]	5	*M*	3	3	
B	~~0~~	[0]	8	2	5	
C	4	4	[0]	3	*M*	✓
D	5	1	6	[0]	~~0~~	
E	5	3	~~0~~	6	6	✓
			✓			

Table 10.20 Opportunity Cost Table

The solution shown in Table 10.20 is not the optimal solution because there is no assignment in row E. Draw minimum number of lines to cover all zeros in the table and then subtract the smallest element (= 3) from all uncovered elements including itself and add it to the element at the intersection of two lines. The new table so obtained is shown in Table 10.21.

	1	*2*	*3*	*4*	*5*
A	[0]	5	*M*	3	3
B	~~0~~	[0]	11	2	5
C	1	1	~~0~~	[0]	*M*
D	5	1	9	~~0~~	[0]
E	2	~~0~~	[0]	3	3

Table 10.21

Repeat the Hungarian method to make assignments in Table 10.21. Since the number of assignments in Table 10.21 is equal to the number of rows or columns, this solution is the optimal solution. The optimal assignment is as follows:

Pilot	*Flight Number*	*Preference Score*
A	1	8
B	2	9
C	4	6
D	5	7
E	3	10
	Total	40

Example 10.8 A city corporation has decided to carry out road repairs on four main arteries of the city. The government has agreed to make a special grant of Rs 50 lakh towards the cost with a condition that the repairs be done at the lowest cost and quickest time. If the conditions warrant, a supplementary token grant will also be considered favourably. The corporation has floated tenders and five contractors have sent in their bids. In order to expedite work, one road will be awarded to only one contractor.

		Cost of Repairs (Rs in lakh)			
		R_1	R_2	R_3	R_4
Contractors/Road	C_1	9	14	19	15
	C_2	7	17	20	19
	C_3	9	18	21	18
	C_4	10	12	18	19
	C_5	10	15	21	16

(a) Find the best way of assigning the repair work to the contractors and the costs.
(b) If it is necessary to seek supplementary grants, what should be the amount sought?
(c) Which of the five contractors will be unsuccessful in his bid? [*AMIE 2005*]

Solution (a) Since cost matrix is not balanced, therefore add one dummy column (road, R_5) with a zero cost elements. The revised cost matrix is given in Table 10.22.

Table 10.22
Cost Matrix

	R_1	R_2	R_3	R_4	R_5
C_1	9	14	19	15	0
C_2	7	17	20	19	0
C_3	9	18	21	18	0
C_4	10	12	18	19	0
C_5	10	15	21	16	0

Apply the Hungarian method to solve this problem. This is left as an exercise for the reader. An optimal assignment is shown in Table 10.23.

Table 10.23
Optimal Solution

	R_1	R_2	R_3	R_4	R_5
C_1	1	1	[0]	~~0~~	1
C_2	[0]	5	2	5	2
C_3	~~0~~	4	1	2	[0]
C_4	3	[0]	~~0~~	5	2
C_5	1	1	1	[0]	~~0~~

The total minimum cost (in rupees) and optimal assignment made are as follows:

Road	*Contractor*	*Cost (Rs in lakh)*
R_1	C_2	7
R_2	C_4	12
R_3	C_1	19
R_4	C_5	16
R_5	C_3	0
		Total 54

(b) Since the total cost exceeds 50 lakh, the excess amount of Rs 4 lakh (= 54 – 50) is to be sought as supplementary grant.
(c) Contractor C_3 who has been assigned to dummy row, R_5 (roads) loses out in the bid.

CONCEPTUAL QUESTIONS B

1. Can there be multiple optimal solutions to an assignment problem? How would you identify the existence of multiple solutions, if any?
2. How would you deal with the assignment problems, where (a) the objective function is to be maximized? (b) some assignments are prohibited?
3. Explain how can one modify an effectiveness matrix in an assignment problem, if a particular assignment is prohibited.
4. What is an unbalanced assignment problem? How is the Hungarian method applied for obtaining a solution if the matrix is rectangular?

SELF PRACTICE PROBLEMS B

1. A project work consists of four major jobs for which an equal number of contractors have submitted tenders. The tender amount quoted (in lakh of rupees) is given in the matrix.

		Job			
		a	*b*	*c*	*d*
Contractor	*1*	10	24	30	15
	2	16	22	28	12
	3	12	20	32	10
	4	9	26	34	16

Find the assignment which minimizes the total cost of the project when each contractor has to be assigned at least one job.

2. Alpha Corporation has four plants, each of which can manufacture any one of four products *A*, *B*, *C* or *D*. Production costs differ from one plant to another and so do the sales revenue. The revenue and the cost data are given below. Determine which product should each plant produce in order to maximize profit.

Sales Revenue (in '000 Rs)

		Plant			
		1	*2*	*3*	*4*
Product	*A*	50	68	49	62
	B	60	70	51	74
	C	52	62	49	68
	D	55	64	48	66

Production Cost (in '000 Rs)

		Plant			
		1	*2*	*3*	*4*
Product	*A*	49	60	45	61
	B	55	63	45	49
	C	55	67	53	70
	D	58	65	54	68

3. A company has four machines that are to be used for three jobs. Each job can be assigned to one and only one machine. The cost of each job on each machine is given in the following table.

		Machines			
		W	*X*	*Y*	*Z*
Jobs	*A*	18	24	28	32
	B	8	13	17	18
	C	10	15	19	22

What are the job-assignment pairs that shall minimize the cost? [*Gauhati, MCA, 2001*]

4. Five workers are available to work with the machines and the respective costs (in rupees) associated with each worker-machine assignment are given below. A sixth machine is available to replace one of the existing ones and the associated of that machine costs are also given below.

		Machines					
		M_1	M_2	M_3	M_4	M_5	M_6
Workers	W_1	12	3	6	–	5	9
	W_2	4	11	–	5	–	8
	W_3	8	2	10	9	7	5
	W_4	–	7	8	6	12	10
	W_5	5	8	9	4	6	1

(a) Determine whether the new machine can be accepted.
(b) Also determine the optimal assignment and the associated saving in cost.

5. A fast-food chain wants to build four stores. In the past, the chain has used six different construction companies, and having been satisfied with each, has invited each to bid on each job. The final bids (in lakh of rupees) are shown in the following table:

	Construction Companies					
	1	*2*	*3*	*4*	*5*	*6*
Store *1*	85.3	88.0	87.5	82.4	89.1	86.7
Store *2*	78.9	77.4	77.4	76.5	79.3	78.3
Store *3*	82.0	81.3	82.4	80.6	83.5	81.7
Store *4*	84.3	84.6	86.2	83.3	84.4	85.5

Since the fast-food chain wants to have each of the new stores ready as quickly as possible, it will award at the most one job to a construction company. What assignment would result in minimum total cost to the fast-food chain?

[*Delhi Univ., MBA, 2001, 2003*]

6. A methods engineer wants to assign four new methods to three work centres. The assignment of the new methods will increase production. The methods are given below.

Increase in Production (unit)

		Work Centres		
		A	*B*	*C*
Method	*1*	10	7	8
	2	8	9	7
	3	7	12	6
	4	10	10	8

If only one method can be assigned to a work centre, determine the optimum assignment.

7. Consider a problem of assigning four clerks to four tasks. The time (hours) required to complete the task is given below:

		Tasks			
		A	*B*	*C*	*D*
Clerks	*1*	4	7	5	6
	2	–	8	7	4
	3	3	–	5	3
	4	6	6	4	2

Clerk 2 cannot be assigned task A and clerk 3 cannot be assigned task B. Find all the optimum assignment schedules.

8. The marketing director of a multi-unit company is faced with a problem of assigning 5 senior managers to six zones. From past experience he knows that the efficiency percentage judged by sales, operating costs, etc., depends on the manager-zone combination. The efficiency of different managers is given below:

		Zones					
		I	*II*	*III*	*IV*	*V*	*VI*
Manager	*A*	73	91	87	82	78	80
	B	81	85	69	76	74	85
	C	75	72	83	84	78	91
	D	93	96	86	91	83	82
	E	90	91	79	89	69	76

Find out which zone should be managed by a junior manager due to the non-availability of a senior manager.

9. A head of department in a college has the problem of assigning courses to teachers with a view to maximize educational quality in his department. He has available to him one professor, two associate professors, and one teaching assistant (TA). Four courses must be offered. After appropriate evaluation, he has arrived at the following relative ratings (100 = best rating) regarding the ability of each instructor to teach each of the four courses.

	Course 1	*Course 2*	*Course 3*	*Course 4*
Prof. 1	60	40	60	70
Prof. 2	20	60	50	70
Prof. 3	20	30	40	60
TA	30	10	20	40

How should he assign his staff to the courses in order to realize his objective? [*Delhi Univ., MBA* (*HCA*)*, 1999*]

10. At the end of a cycle of schedules, a transport company has a surplus of one truck in each of the cities 1, 2, 3, 4, 5 and a deficit of one truck in each of the cities A, B, C, D, E and F. The distance (in kilometres) between the cities with a surplus, and cities with a deficit, is given below:

		To City					
		A	*B*	*C*	*D*	*E*	*F*
From City	*1*	80	140	80	100	56	98
	2	48	64	94	126	170	100
	3	56	80	120	100	70	64
	4	99	100	1,100	104	80	90
	5	64	80	90	60	60	70

How should the trucks be despatched so as to minimize the total distance travelled? Which city will not receive a truck? [*Madras, MBA, 2000*]

11. A company is considering expanding into five new sales territories. The company has recruited four new salesmen. Based on the salesmen's experience and personality traits, the sales manager has assigned ratings to each of the salesmen for each of the sales territories. The ratings are as follows:

		Territory				
		1	*2*	*3*	*4*	*5*
Salesmen	*A*	75	80	85	70	90
	B	91	71	82	75	85
	C	78	90	85	80	80
	D	65	75	88	85	90

Suggest optimal assignment of the salesmen. If for certain reasons, salesman D cannot be assigned to territory 3, will the optimal assignment be different? If so, what would be the new assignment schedule? [*Delhi Univ., MCom, 2000*]

12. The personnel manager of a medium-sized company has decided to recruit two employees D and E in a particular section of the organization. The section has five fairly defined tasks 1, 2, 3, 4 and 5; and three employees A, B and C are already employed in the section. Considering the specialized nature of task 3 and the special qualifications of the recruit D for task 3, the manager has decided to assign task 3 to employee D and then assign the remaining tasks to remaining employees so as to maximize the total effectiveness. The index of effectiveness of each employee of different tasks is as under.

		Tasks				
		1	*2*	*3*	*4*	*5*
Employees	*A*	25	55	60	45	30
	B	45	65	55	35	40
	C	10	35	45	55	65
	D	40	30	70	40	60
	E	55	45	40	55	10

Assign the tasks for maximizing total effectiveness. Critically examine whether the decision of the manager to assign task 3 to employee D was correct. [*Delhi Univ., MBA, 2003*]

13. The casualty medical officer of a hospital has received four requests for Ambulance van facility. Currently, six vans are available for assignment and their estimated response time (in minutes) are shown in the table below:

		Van					
		1	*2*	*3*	*4*	*5*	*6*
Incident	*1*	16	15	13	14	15	18
	2	18	16	12	13	17	16
	3	14	14	17	16	15	15
	4	13	17	19	18	14	17

Determine which van should respond, and what will be the average response time. [*Delhi Univ., MBA* (*HCA*)*, 1990, 96, 98*]

14. To stimulate interest and provide an atmosphere for intellectual discussion a finance faculty in a management school decides to hold special seminars on four contemporary topics: Leasing, portfolio management, private mutual funds, swaps and options. Such seminars are to be held be held once par week in the afternoons. However, scheduling these seminars (one for each topic, and not more than one seminar per afternoon) has to be done carefully so that the number of students unable to attend is kept to a minimum. A careful study indicates that the number of students who cannot attend a particular seminar on a specific day is as follows:

	Leasing	*Portfolio Management*	*Private Mutual Funds*	*Swaps and Options*
Monday	50	40	60	20
Tuesday	40	30	40	30
Wednesday	60	20	30	20
Thursday	30	30	20	30
Friday	10	20	10	30

Find an optimal schedule of the seminars. Also find out the total number of students who will be missing at least one seminar.

15. Five swimmers are eligible to compete in a relay team that is to consist of four swimmers swimming four different swimming styles. The styles are – back stroke, breast stroke, free style and butterfly. The time taken by the five swimmers – Anand, Bhasker, Chandru, Dorai and Easwar, to cover a distance of 100 metres in various swimming styles is given below, in minutes, seconds.
    - Anand swims the back stroke in 1 : 09, the breast stroke in 1 : 15 and has never competed in the free style or butterfly.
    - Bhasker is a free style specialist averaging 1 : 01 for the 100 metres but can also swim the breast stroke in 1 : 16 and butterfly in 1 : 20.
    - Chandru swims all styles: back 1 : 10, butterfly 1 : 12, free style 1 : 05 and breast stroke 1 : 20.
    - Dorai swims only the butterfly 1 : 11, while Easwar swims the back stroke 1 : 20, breast stroke 1 : 16, free style 1 : 06 and the butterfly 1 : 10.

    Which swimmer should be assigned which swimming style? Who will not be in the relay?

16. (*a*) At the end of a cycle of schedules, a trucking company has a surplus of one vehicle in each of the cities – 1, 2, 3, 4 and 5 and a deficit of one vehicle in each of the cities A, B, C, D, E and F. The cost (in rupees) of transportation and handling between the cities with a surplus and the cities with deficits are shown in the following table:

		To City					
		A	*B*	*C*	*D*	*E*	*F*
From City	*1*	134	116	167	233	164	97
	2	114	195	260	166	178	130
	3	129	117	48	94	66	101
	4	71	156	92	143	114	136
	5	97	134	125	83	142	118

Find the assignment of surplus vehicles to deficit cities that will result in a minimum total cost? Which city will not receive a vehicle? [*Delhi Univ., MBA, 2002*]

HINTS AND ANSWERS

1. (i) 1 – *b,* 2 – *c,* 3 – *d,* 4 – *a,*
   (ii) 1 – *c,* 2 – *b,* 3 – *d,* 4 – *a;* Minimum cost = Rs 71,00,000.
   (iii) 1 – *c,* 2 – *d,* 3 – *b,* 4 – *a,*
   (iv) 1 – *b,* 2 – *c,* 3 – *d,* 4 – *a*
2. Construct the profit matrix using the relationship: Profit = Revenue – Cost
   A – 2, *B* – 4, *C* – 1, *D* – 3 and Maximum profit = Rs 42,000
3. (i) *A – W, B – X, C – Y,*
   (ii) *A – W, B – Y, C – X.* Minimum cost = Rs 50.
4. (a) $W_1 - M_5$, $W_2 - M_6$, $W_3 - M_2$, $W_4 - M_4$, $W_5 - M_1$ and W_6 (dummy) – M_3; Minimum cost = Rs 21
   (b) $W_1 - M_5$, $W_2 - M_1$, $W_3 - M_2$, $W_4 - M_3$ and $W_5 - M_4$, Minimum cost = Rs 23.
   The sixth machine should be accepted because saving in cost is Rs (23 – 21) = Rs 2
5. $S_1 - 4$, $S_2 - 3$, $S_3 - 2$, $S_4 - 1$; Cost = Rs 325.4
6. (i) 1 – A, 2 – dummy, 3 – B, 4 – C
   (ii) 1 – C, 2 – dummy, 3 – B, 4 – A. Total production = 30 units
7. (i) 1 – B, 2 – D, 3 – A, 4 – C
   (ii) 1 – C, 2 – D, 3 – A, 4 – B. Minimum hours = 18.
8. A – III, B – II, C – III, D – I, E – IV, dummy – V.
9. (i) Prof. 1 – Course 3, Prof. 2 – Course 2, Prof. 3 – Course 4, T.A. – Course 1
   (ii) Prof. 1 – Course 1, Prof. 2 – Course 2, Prof. 3 – Course 4, T.A. – Course 3. Maximum educational quality = 210.
10. 1 – E, 2 – B, 3 – A, 4 – F, 5 – D; Cost = Rs 326.
11. A – 5, B – 1, C – 2, D – 3; 359.
12. A – 4, B – 2, C – 5, D – 3, E – 1; 300.
15. The assignment matrix with time expressed in seconds and adding a dummy style to balance it is given by

	Back Stroke	*Breast Stroke*	*Free Style*	*Butterfly*
Anand	69	75	–	–
Bhasker	–	76	61	80
Chandru	70	80	65	72
Dorai	–	–	–	71
Easwar	80	76	66	70

10.5 A TYPICAL ASSIGNMENT PROBLEM

Example 10.9 A small airline company that owns five planes operates on all the seven days of a week. Flights between three cities A, B and C, according to the schedule, is given below. The layover cost per stop is roughly proportional to the square of the layover time:

Flight No.	*From*	*Departure Time (in hrs)*	*To*	*Arrival Time (in hrs)*
1	*A*	09.00	*B*	12.00
2	*A*	10.00	*B*	13.00
3	*A*	15.00	*B*	18.00
4	*A*	20.00	*C*	Midnight
5	*A*	22.00	*C*	02.00
6	*B*	04.00	*A*	07.00
7	*B*	11.00	*A*	14.00
8	*B*	15.00	*A*	18.00
9	*C*	07.00	*A*	11.00
10	*C*	15.00	*A*	19.00

Find out how the planes should be assigned to the flights so as to minimize the total layover cost. If you have made any particular assumptions, state them clearly.

Solution We have made the following assumptions:

(i) A plane cannot make more than two trips (to and fro).

(ii) A plane flying from a particular city must be back within 24 hours for the next scheduled trip from the city.

From the given data, it is clear that there is no route between city B and C. Therefore, the present problem can be divided into two subproblems: (i) routes between A and C, and (ii) routes between A and B.

Let us now construct the cost matrix for the routes connecting cities A and C. This is given in Table 10.24.

Table 10.24

Flight Number	*9*	*10*
4	130	226
5	146	178

The elements in the Table 10.24 are interpreted as follows: For route 4 – 9 a plane taking flight number 4 from A to C and number 9 from C to A, would have a layover time of 9 hours (11.00 to 20.00) at city A and 7 hours (midnight – 7.00) at city C. Thus, the layover cost for the route 4 – 9 would be $(9)^2 + (7)^2$ = 130 units. Similarly, the other route costs elements are also obtained.

The optimal assignment can now be obtained by applying the Hungarian method of assignment. The optimal solution is given in Table 10.25.

Table 10.25

Flight Number	*9*	*10*
4	[0]	64
5	~~0~~	[0]

Optimal assignment : 4 – 9, 5 – 10

Total cost : 308 units.

Similarly, cost matrix for the routes connecting cities A and B can be constructed as shown in Table 10.26.

Table 10.26

Flight Number	*6*	*7*	*8*
1	260	*M*	234
2	234	*M*	260
3	164	290	*M*

The elements in Table 10.26 are interpreted as follows: For route 1–6, a plane taking flight number 1 from A to B and number 6 from B to A would have a total layover time of 18 hours (16 + 2) with an associated layover cost of $(16)^2 + (2)^2 = 260$ units. Since a plane taking flight number 1 cannot return for flight number 7 because of assumption (ii), the layover cost associated with this flight is considered very high, say M. Similarly, the other route cost elements are also obtained.

The optimal assignment can now be obtained by applying the Hungarian method of assignment. The optimal solution is given in Table 10.27.

Flight Number	*6*	*7*	*8*
1	26	*M*	[0]
2	[0]	*M*	26
3	~~0~~	[0]	*M*

Table 10.27

Optimal assignment : 1 – 8, 2 – 6, 3 – 7
Total cost : 758 units.

Hence, from the two solutions obtained above, the complete flight schedule of planes is arrived at and is given in Table 10.28.

Plane Number	*Flight Number*	*Departure*		*Arrival*	
		City	*Time (hrs)*	*City*	*Time (hrs)*
1	1	*A*	09.00	*B*	12.00
	8	*B*	15.00	*A*	18.00
2	2	*A*	10.00	*B*	13.00
	6	*B*	04.00	*A*	07.00
3	3	*A*	15.00	*B*	18.00
	7	*B*	11.00	*A*	14.00
4	4	*A*	20.00	*C*	Midnight
	9	*C*	07.00	*A*	11.00
5	5	*A*	22.00	C	0.200
	10	*C*	15.00	*A*	19.00

Table 10.28 Optimal Flight Schedule

The total minimum layover cost is 308 + 758 = 1,066 units.

10.6 TRAVELLING SALESMAN PROBLEM

The travelling salesman problem may be solved as an assignment problem, with two additional conditions on the choice of assignment. That is, how should a travelling salesman travel starting from his home city (the city from where he started), visiting each city only once and returning to his home city, so that the total distance (cost or time) covered is minimum. For example, given n cities and distances d_{ij} (cost c_{ij} or time t_{ij}) from city i to city j, the salesman starts from city 1, then any permutation of 2, 3, . . ., n represents the number of possible ways of his tour. Thus, there are $(n - 1)!$ possible ways of his tour. Now the problem is to select an optimal route that is able to achieve the objective of the salesman.

To formulate and solve this problem, let us define:

$$x_{ij} = \begin{cases} 1, & \text{if salesman travels from city } i \text{ to city } j \\ 0, & \text{otherwise} \end{cases}$$

Since each city can be visited only once, we have

$$\sum_{i=1}^{n-1} x_{ij} = 1, \quad j = 1, 2, \ldots, n; \quad i \neq j$$

Again, since the salesman has to leave each city except city n, we have

$$\sum_{j=1}^{n} x_{ij} = 1, \quad i = 1, 2, \ldots, n-1; \quad i \neq j$$

A travelling salesman starts from his home city and returns back visiting each city, so that the total distance (cost or time) covered is the minimum.

The objective function is then

$$\text{Minimize } Z = \sum_{i=1}^{n-1} \sum_{j=1}^{n} d_{ij}\, x_{ij}$$

Since $d_{ji} = d_{ij}$ is not required, therefore $d_{ij} = \infty$ for $i = j$. However, all d_{ij}s must be non-negative, i.e. $d_{ij} \geq 0$ and $d_{ij} + d_{jk} \geq d_{jk}$ for all i, j, k.

Example 10.10 A travelling salesman has to visit five cities. He wishes to start from a particular city, visit each city once and then return to his starting point. The travelling cost (in '000 Rs) of each city from a particular city is given below:

		To City				
		A	*B*	*C*	*D*	*E*
From City	*A*	∞	2	5	7	1
	B	6	∞	3	8	2
	C	8	7	∞	4	7
	D	12	4	6	∞	5
	E	1	3	2	8	∞

What should be the sequence of visit of the salesman so that the cost is minimum? [*Delhi Univ., MBA, 2004*]

Solution Solving the given travelling salesman problem as an assignment problem, by Hungarian method of assignment, an optimal solution is shown in Table 10.29. However, this solution is not the solution to the travelling salesman problem as it gives the sequence A – *E* – *A*. This violates the condition that salesmen can visit each city only once.

Table 10.29 Optimal Solution

		To City				
		A	*B*	*C*	*D*	*E*
From City	*A*	∞	1	3	6	[0]
	B	4	∞	[0]	6	~~0~~
	C	4	3	∞	[0]	3
	D	8	[0]	1	∞	1
	E	[0]	2	~~0~~	7	∞

The 'next best' solution to the problem that also satisfies the extra condition of unbroken sequence of visit to all cities, can be obtained by bringing the next (non-zero) minimum element, i.e. 1, into the solution. In Table 10.29, the cost element 1 occurs at three different places. Therefore, consider all three different cases one by one until the acceptable solution is reached.

Case 1: Make the unit assignment in the cell (*A*, *B*) instead of zero assignment in the cell (*A*, *E*) and delete row *A* and column *B* so as to eliminate the possibility of any other assignment in row *A* and column *B*. Now make the assignments in the usual manner. The resulting assignments are shown in Table 10.30.

Table 10.30

		To City				
		A	*B*	*C*	*D*	*E*
From City	*A*	~~∞~~	[1]	~~3~~	~~6~~	~~0~~
	B	4	~~∞~~	[0]	6	~~0~~
	C	4	~~3~~	∞	[0]	3
	D	8	~~0~~	~~1~~	∞	[1]
	E	[0]	2	~~0~~	7	∞

The solution given in Table 10.30 gives the sequence: $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$, $E \rightarrow A$. The cost corresponding to this feasible solution is Rs 15,000.

Case 2: If we make the assignment in the cell (D, C) instead of (D, E), then no feasible solution is obtained in terms of zeros or in terms of element 1 which may give cost less than Rs 15,000.

Hence, the best solution is: $A - B - C - D - E - A$, and the total cost associated with this solution is Rs 15,000.

Example 10.12 ABC Ice Cream Company has a distribution depot in Greater Kailash Part I for distributing ice-cream in South Delhi. There are four vendors located in different parts of South Delhi (call them A, B, C and D) who have to be supplied ice-cream everyday. The following matrix displays the distances (in kilometres) between the depot and the four vendors:

		To				
		Depot	*Vendor A*	*Vendor B*	*Vendor C*	*Vendor D*
From	*Depot*	–	3.5	3	4	2
	Vendor A	3.5	–	4	2.5	3
	Vendor B	3	4	–	4.5	3.5
	Vendor C	4	2.5	4.5	–	4
	Vendor D	2	3	3.5	4	–

What route should the company van follow so that the total distance travelled is minimized?

[*Delhi Univ., MBA, 2003, 2004, 2006*]

Solution Solving the given travelling salesman problem as an assignment problem, by using the Hungarian method of assignment, an optimal solution is shown in Table 10.31.

		To				
		Depot	*Vendor A*	*Vendor B*	*Vendor C*	*Vendor D*
From	*Depot*	–	1.5	~~0~~	2	[0]
	Vendor A	1.5	–	0.5	[0]	0.5
	Vendor B	[0]	0.5	–	1	~~0~~
	Vendor C	2	[0]	1	–	1.5
	Vendor D	~~0~~	0.5	[0]	1.5	–

Table 10.31
Optimal Solution

The solution shown in Table 10.31 is an optimal solution to an assignment problem, but it is not the optional solution to the travelling salesman problem. This is because it gives the sequence: Depot $\rightarrow$ Vendor $D \rightarrow$ Vendor $B \rightarrow$ Depot. This violates the condition of the travelling salesman problem.

The 'next best' solution to the problem that also satisfies travelling salesman condition of unbroken sequence, can be obtained by bringing next (non-zero) minimum element, i.e. 0.5, into the solution. In Table 10.31, the element 0.5 occurs at four different places. Therefore, consider all four different cases, one by one, until an acceptable solution is reached.

All four cases of possible solution to travelling salesman problem tried with element 0.5 as well as zero element do not provide a desired solution. Thus, we look for the 'next best' solution by bringing the next (non-zero) element 1 along with 0.5 and zero elements into the solution.

Make assignment in cell (C, B) and delete row 4 and column 3. Then make assignments in the usual manner using elements 1, 0.5 and 0 in the cells as shown in Table 10.32.

Table 10.32

		To				
		Depot	*Vendor A*	*Vendor B*	*Vendor C*	*Vendor D*
From	*Depot*	–	1.5	~~0~~	2	0
	Vendor A	1.5	–	0.5	0	~~0.5~~
	Vendor B	0	~~0.5~~	–	~~1~~	~~0~~
	Vendor C	2	~~0~~	1	—	1.5
	Vendor D	~~0~~	0.5	0	1.5	–

The set of assignments given in Table 10.38 is a feasible solution to the travelling salesman problem. The route for the salesman is: Vendor $C \rightarrow$ Vendor $B \rightarrow$ Depot $\rightarrow$ Vendor $D \rightarrow$ Vendor $A \rightarrow$ Vendor C. The total distance (in km) to be covered in this sequence is 15 km.

The reader may try other cases. This is left as an exercise.

SELF PRACTICE PROBLEMS C

1. An airline that operates seven days a week has the timetable which is given below. The crew must have a minimum layover time of five hours between flights.

Flight No.	*Delhi-Jaipur*		*Flight No.*	*Jaipur-Delhi*	
	Dep.	*Arrival*		*Dep.*	*Arrival*
101	7.00 am	8.00 am	201	8.00 am	9.15 am
102	8.00 am	9.00 am	202	8.30 am	9.45 am
103	1.30 pm	2.30 pm	203	12.00 noon	1.15 pm
104	6.30 pm	7.30 pm	204	5.30 pm	6.45 pm

Obtain the pairing of flights that minimizes layover time away from home. For any given pairing, the crew will be based at the city that results in the smaller layover. For each pair also mention the town where the crew should be based.

2. A machine operator processes four types of items on his machine and he must choose a sequence for them. The set-up cost per change depends on the items currently on machine and the set-up to be made according to the following table:

		To			
		A	*B*	*C*	*D*
From Item	*A*	–	4	7	3
	B	4	–	6	3
	C	7	6	–	7
	D	3	3	7	–

If he processes each of the item once and only once each week, then how should he sequence the item on his machine? Use the method for the problem of travelling salesman.
[*Delhi Univ., MBA, 2001*]

3. A salesman has to visit five cities A, B, C, D and E. The distances (in hundred kilometres) between the five cities are as follows:

		To City				
		A	*B*	*C*	*D*	*E*
From City	*A*	–	1	6	8	4
	B	7	–	8	5	6
	C	6	8	–	9	7
	D	8	5	9	–	8
	E	4	6	7	8	–

If the salesman starts from city A and has to come back to city A, which route should he select so that the total distance travelled is minimum?

4. A salesman must travel from city to city to maintain his accounts. This week he has to leave his home base and visit other cities and the return home. The table shows the distances (in km) between the various cities. His home city is city A.

		To City				
		A	*B*	*C*	*D*	*E*
From City	*A*	–	375	600	150	190
	B	375	–	300	350	175
	C	600	300	–	350	500
	D	160	350	350	–	300
	E	190	175	500	300	–

Use the assignment method to determine the tour that will minimize the total distance of visiting all cities and then returning home. [*Delhi Univ., MBA, 2002*]

5. A salesman travels from one place to another; he cannot, however, travel from one place and back. The distances (in km) between pairs of cities are given below:

		To City			
		P	*Q*	*R*	*S*
From City	*P*	–	15	25	20
	Q	22	–	45	55
	R	40	30	–	25
	S	20	26	38	–

The problem is to chalk out a route which enables him to visit each of the cities only once, so that the total distance covered by him is minimum. [*Delhi Univ., MBA, 2003*]

6. Products 1, 2, 3, 4 and 5 are to be processed on a machine. The set-up costs in rupees per change depend upon the product presently on the machine and the set-up to be made. These are given by the following data:

$C_{12} = 16,\ C_{13} = 4,\ C_{14} = 12,\ C_{23} = 6,\ C_{24} = 5,\ C_{25} = 8,$
$C_{35} = 6,\ C_{45} = 20;\ C_{ij} = C_{ji},\ C_{ij} = \infty$ for $i = j$

for all values of i and j not given in the data. Find the optimum sequence of products in order to minimize the total set-up cost.

7. A salesman has to visit five cities A, B, C, D and E. The distances (in hundred km) between the five cities are as follows:

		To City A	B	C	D	E
	A	–	17	16	18	14
	B	17	–	18	15	16
From City	C	16	18	–	19	17
	D	18	15	19	–	18
	E	14	16	17	18	–

If the salesman starts from city A and has to come back to city A, which route should he select so that total distance travelled by him is minimized? [*Delhi Univ., MBA, 2004*]

8. The expected times required to be taken by a salesman in travelling from one city to another are as follows:

		To City C_1	C_2	C_3	C_4	C_5
	C_1	–	10	13	11	–
	C_2	10	–	12	10	12
From City	C_3	14	13	–	13	11
	C_4	11	10	14	–	10
	C_5	12	11	12	10	–

How should the salesman plan his trip so that he covers each of these cities no more than once, and completes his trip in minimum possible time required for travelling?

9. Solve the travelling salesman problem given by the following data

$C_{12} = 20$, $C_{13} = 4$, $C_{14} = 10$, $C_{23} = 5$, $C_{24} = 6$. $C_{25} = 10$, $C_{35} = 6$, $C_{45} = 20$, where $C_{ij} = C_{ji}$

and there is no route between cities i and j if the value for C_{ij} is not shown.

HINTS AND ANSWERS

1. Calculate layover times when (i) the crew is based in Delhi and (ii) when the crew is based in Jaipur, by considering 15 minutes = 1 unit.
   (i) 103 – 201; 104 – 202, 101 – 203; 102 – 204 (ii) 103 – 201; 101 – 202; 104 – 203; 102 – 204.
   Total minimum layover time is 52 hours and 30 minutes.
2. (i) $A - D$, $D - B$, $B - C$, $C - A$; (ii) $A - C$, $C - B$, $B - D$, $D - A$. Total minimum cost = Rs 19.
3. $A \to C \to D \to B \to E \to A$; 30 km
4. $A \to D \to C \to B \to E \to A$; 1,165 km
5. $P \to R \to S \to Q \to P$; 98 km
6. $1 \to 2 \to 5 \to 3 \to 4 \to 1$. Total cost = Rs 47.

CHAPTER SUMMARY

Given a set of tasks to be performed and a set of assignees who are available to perform these tasks (one assignee per task), an assignment problem deals with the allocation of an assignee to a particular task so as to minimize the total cost of performing all the tasks. The assignees can be people, machines, vehicles, plants and so on. The formulation of an assignment problem requires constructing a cost matrix for each possible pair of an assignee to a task.

The aim of this chapter is to enable readers to recognize whether a problem can be formulated and analyzed as an assignment problem or as a variant of one of these problem types.

CHAPTER CONCEPTS QUIZ

True or False

1. In assignment problem, an optimal assignment requires that the maximum number of lines which can be drawn through squares with zero opportunity cost be equal to the number of rows.
2. In an assignment problem, if a constant is added or subtracted from every element of any row/column in the given cost matrix, then an assignment that minimizes the total cost in one matrix also maximizes the total cost in other matrix.
3. The travelling salesman problem can not be solved as an assignment problem.
4. Multiple zeros in columns and rows are all indicative of multiple optimal solutions.
5. Assignment problem deals with assignment of persons to jobs that they can perform with varying efficiency.
6. If all entries in a row of the cost matrix are increased by a constant, then it will affect the optimal solution to the problem.
7. All dummy rows or columns in the assignment problem are assumed to be non-zero.
8. The minimum number of lines required to cover all zeroes can not be more than the number of columns/rows in the problem.
9. In an assignment problem involving assignment of salesmen to different zones to exploit their sales potential fully, if a salesman cannot be assigned to a particular zone, then the relevant cell value would be replaced by M for solving the problem.
10. The relevant cost element is replaced by a zero in case a certain work is not to be assigned to a particular job.

Fill in the Blanks

11. An assignment problem that has an objective of maximizing profit is solved as a minimization problem after converting all cost value into _____ values.
12. When applying Hungarian method, we subtract the smallest element from all elements in each column. Then we _____ the smallest element in each row from all elements in the row and the remaining elements represent _____.

13. An optimal solution can be found from an opportunity cost table by drawing lines covering all zero squares when the minimum number of the lines possible equals the number of _____.
14. In the assignment problem, the number of allocations in each row and column are _____.
15. The _____ method provides an efficient method of finding the optimal solution without making a direct comparison of every solution.
16. If in the assignment problem, there is no assignment in a _____, then it implies that the total number of assignments are _____ the number of rows/columns in the square matrix.
17. In the _____ method, a list of possible assignments among the given assignees and activities is prepared.
18. The problem of degeneracy makes the transportation method _____ for solving an assignment problem.
19. The _____ problem is a special case of the balanced transportation problem where all rim requirements equals _____.
20. In the assignment problem, the _____ cost is the difference between the best possible assignment of an assignee to an activity and the best possible assignment is an assignment with opportunity cost equal to _____.

Multiple Choice

21. An assignment problem is considered as a particular case of a transportation problem because
    (a) the number of rows equals columns
    (b) all $x_{ij} = 0$ or 1
    (c) all rim conditions are 1
    (d) all of the above
22. An optimal assignment requires that the maximum number of lines that can be drawn through squares with zero opportunity cost be equal to the number of
    (a) rows or columns
    (b) rows and columns
    (c) rows + columns – 1
    (d) none of the above
23. While solving an assignment problem, an activity is assigned to a resource through a square with zero opportunity cost because the objective is to
    (a) minimize total cost of assignment
    (b) reduce the cost of assignment to zero
    (c) reduce the cost of that particular assignment to zero
    (d) all of the above
24. The method used for solving an assignment problem is called
    (a) reduced matrix method
    (b) MODI method
    (c) Hungarian method
    (d) none of the above
25. The purpose of a dummy row or column in an assignment problem is to
    (a) obtain balance between total activities and total resources
    (b) prevent a solution from becoming degenerate
    (c) provide a means of representing a dummy problem
    (d) none of the above
26. Maximization assignment problem is transformed into a minimization problem by
    (a) adding each entry in a column from the maximum value in that column
    (b) subtracting each entry in a column from the maximum value in that column
    (c) subtracting each entry in the table from the maximum value in that table
    (d) any one of the above
27. If there were n workers and n jobs there would be
    (a) $n!$ solutions
    (b) $(n-1)!$ solutions
    (c) $(n!)^n$ solutions
    (d) n solutions
28. An assignment problem can be solved by
    (a) simplex method
    (b) transportation method
    (c) both (a) and (b)
    (d) none of the above
29. For a salesman who has to visit n cities which of the following are the ways of his tour plan
    (a) $n!$
    (b) $(n+1)!$
    (c) $(n-1)!$
    (d) n
30. The assignment problem
    (a) requires that only one activity be assigned to each resource
    (b) is a special case of transportation problem
    (c) can be used to maximize resources
    (d) all of the above
31. An assignment problem is a special case of transportation problem, where
    (a) number of rows equals number of columns
    (b) all rim conditions are 1
    (c) values of each decision variable is either 0 or 1
    (d) all of the above
32. Every basic feasible solution of a general assignment problem, having a square pay-off matrix of order, n should have assignments equal to
    (a) $2n+1$ (b) $2n-1$
    (c) $m+n-1$ (d) $m+n$
33. To proceed with the MODI algorithm for solving an assignment problem, the number of dummy allocations need to be added are
    (a) n (b) $2n$
    (c) $n-1$ (d) $2n-1$
34. The Hungarian method for solving an assignment problem can also be used to solve
    (a) a transportation problem
    (b) a travelling salesman problem
    (c) both (a) and (b)
    (d) only (b)
35. An optimal solution of an assignment problem can be obtained only if
    (a) each row and column has only one zero element
    (b) each row and column has at least one zero element
    (c) the data are arrangement in a square matrix
    (d) none of the above

Answers to Quiz

1. F 2. F 3. F 4. T 5. T 6. F 7. F 8. F 9. T 10. T
11. regret 12. subtract; opportunity cost 13. rows or columns 14. equal 15. Hungarian
16. row or columns; less than 17. enumeration 18. inefficient 19. assignment; one
20. job opportunity; zero. 21. (d) 22. (a) 23. (a) 24. (c) 25. (a) 26. (c) 27. (a) 28. (c)
29. (c) 30. (d) 31. (d) 32. (b) 33. (c) 34. (b) 35. (d)

CASE STUDY

Case 10.1: Shreya & Sons

Shreya & Sons are planning to develop three products 1, 2 and 3 in its three plants *A*, *B* and *C*. Only a single product is decided to be introduced in each of the plants. The unit cost of producing one product in a plant, is given in the following matrix.

		Plant		
		A	*B*	*C*
Product	*1*	8	12	–
	2	10	6	4
	3	7	6	6

The management is interested to understand how should the product be assigned so that the total unit cost is minimized? Also

(a) If the quantity of different products to be produced is as follows, then what assignment shall minimize the aggregate production cost?

Product	*Quantity* (*in units*)
1	2,000
2	2,000
3	10,000

(b) If the three products were to be produced in equal quantities, then what is consequence?

It is expected that the selling prices of the products produced by different plants would be different. The prices are shown in the following table:

		Plant		
		A	*B*	*C*
Product	*1*	15	18	–
	2	18	16	10
	3	12	10	8

Assuming the quantities mentioned in (a) above would be produced and sold, how should the products be assigned to the plants in order to obtain maximum profits?

Case 10.2: City Corporation

A city corporation has decided to carry out repairs on four main dispensaries of the city. The government has agreed to make a special grant of Rs 50 lakh towards the costs, with a condition that the repairs must be done at the lowest cost and the quickest time. If conditions warrant, then a supplementary token grant will also be considered favourably. The corporation has floated tenders and five contractors have sent in their bids. In order to expedite work, one dispensary will be awarded to only one contractor.

Cost of Repairs (*Rs in lakh*)

Contractor	*Dispensary*			
	D_1	D_2	D_3	D_4
C_1	9	14	19	15
C_2	7	17	20	19
C_3	9	18	21	18
C_4	10	12	18	19
C_5	10	15	21	16

You as a consultant suggest to the corporate the best way of assigning the repair work to the contractors. Also

(a) If it is necessary to seek supplementary grants, then what would be the amount sought?

(b) Which of the five contractors will be unsuccessful in his bid?

Case 10.3: Kamal Transport

A trip from Chennai to Bangalore takes six hours by bus. A typical time-table of the bus service prepared by the transport authorities in both directions is given below:

Departure from Chennai	*Route Number*	*Arrival at Bangalore*	*Arrival at Chennai*	*Route Number*	*Departure from Bangalore*
06.00	*a*	12.00	11.30	1	05.30
07.30	*b*	13.30	15.00	2	09.00
11.30	*c*	17.30	21.00	3	15.00
19.00	*d*	01.00	00.30	4	18.30
00.30	*e*	06.30	06.00	5	00.00

The cost of providing this service by the transport company depends upon the time spent by the bus crew (driver and conductor) away from their places in addition to service time. There are five crew. There is, however, a constraint that every crew should be provided with more than 4 hours of rest before the return trip again and should not wait for more than 24 hours for the return trip. The company has residential facilities for the crew at Chennai as well as at Bangalore. Suggest how crew should be assigned with which line of service or which service line should be connected with which other line, so as to reduce the waiting time to the minimum.

APPENDIX: IMPORTANT RESULTS AND THEOREMS

Theorem 10.1 In an assignment problem, if a constant is added to or subtracted from every element of any row or column in the given cost matrix, then an assignment that minimizes the total cost in one matrix also minimizes the total cost in the other matrix.

Alternately Let $x_{ij} = x_{ij}^*$, Minimize $Z = \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij}\, x_{ij}$ such that $\sum_{i=1}^{n} x_{ij} = \sum_{j=1}^{n} x_{ij} = 1$, for all $x_{ij} = 0$ or 1, then x_{ij}^* also Minimize $Z^* = \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij}^*\, x_{ij}$, where $c_{ij}^* = c_{ij} - u_i - v_j$, for $i, j = 1, 2, \ldots, n$ and where u_i and v_j are some real numbers.

Proof Given that the assignment $x_{ij} = x_{ij}^*$ minimizes the total cost,

$$Z^* = \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij}^*\, x_{ij} = \sum_{i=1}^{n} \sum_{j=1}^{n} (c_{ij} - u_i - v_j)\, x_{ij}$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij}\, x_{ij} - \sum_{i=1}^{n} u_i \sum_{j=1}^{n} x_{ij} - \sum_{j=1}^{n} v_j \sum_{i=1}^{n} x_{ij}$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij}\, x_{ij} - \sum_{i=1}^{n} u_i - \sum_{j=1}^{n} v_j = Z - \sum_{i=1}^{n} u_i - \sum_{j=1}^{n} v_j$$

As the terms $\sum_{i=1}^{n} u_i$ and $\sum_{j=1}^{n} v_j$, which are subtracted from Z to give Z^* are independent of x_{ij}, therefore Z^* is also minimum for $x_{ij} = x_{ij}^*$.

Theorem 10.2 If all $c_{ij} > 0$ and it is possible to find a set $x_{ij} = x_{ij}^*$ so that $\sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij}\, x_{ij} = 0$, then this assignment is optimal.

Proof Left to the reader as an exercise. A state of nature can be a state of economy (e.g. inflation), a weather condition, a political development, etc. The states of nature are usually not determined by the action of an individual or an organization. These are the result of an 'act of God' or result of many situations pushing in various directions.

The most relevant states of nature may be identified through some technique such as *scenario analysis*, i.e. there may be certain possible states of nature that may not have a serious impact on the decision, while others could be quite serious. In scenario analysis, various knowledgeable section of people are interviewed – stakeholders, long-time managers, etc., in order to determine the most relevant states of nature that affect the decision.